THE TEACHER ABOVE AI

INTERNATIONAL STEM SKILLS ROUND TABLE PHASE III

The Teacher Above AI

STEM Education, Human Judgment, and the New Learning Ecosystem

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ABOUT THIS BOOK

This book is a synthesis of the International STEM Skills Round Table Phase III, written by a Round Table participant. It is not an official publication of the Samuel Neaman Institute or of any other organisation represented in the discussions, and it does not speak on behalf of the participants named in it.

The manuscript was drafted with the assistance of generative AI tools working from the collected Round Table materials, and every part of it was reviewed and edited by the author, who is responsible for the final text. Where the sources named speakers and summarised their contributions, the book refers to them by name and role rather than by direct quotation.

Comments, corrections, and suggestions are welcome at avi.salmon@gmail.com.

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TABLE OF CONTENTS

LIST OF FIGURES

Editorial Note

This book is built from the collected Round Table materials. It is written as a synthesis, not as meeting minutes. Where the source material names speakers and summarizes their contributions, the book refers to them by name and role. Where a source was available only as a meeting outline or steering summary, the book treats it as a reported contribution and avoids direct quotation.

This book is not an official publication of the Samuel Neaman Institute or of any organization represented in the discussions, and it does not speak on behalf of the participants named in it. The text was developed with assistance from generative AI tools working from the collected Round Table materials, then reviewed and edited by the author, who is responsible for the final text.

The author was himself a participant in the Round Table and presented at the sixth meeting. His own contributions are cited on the same basis as those of every other participant, and readers should weigh them in that light. The Contributors section that follows names the people whose work this book rests on.

The evidence base is uneven across the book. Meetings one, two, four, and six left detailed written summaries, and most of them also left presentation decks. Meeting three left a discussion document prepared before the session rather than a record of it, so the chapters drawn from that meeting report the questions put to the room more faithfully than the answers given in it. Meeting five left only steering-team summaries and outlines, with no primary record of the session itself. The Meeting 1 summary and the Phase I and Phase II reports reached the author late in the preparation of this edition, and the affected chapters were revised to use them. Chapters that rest mainly on retrospective summaries say so in their sources, and the reader should treat their attributions as reported rather than verbatim. The Meeting 6 summary reached the author as a draft circulated for participant comment, and is used here on that basis.

This book uses external references that appear in the collected materials, including the OECD Teaching Compass, OECD Learning Compass 2030, UNESCO AI competency frameworks, TALIS 2024 material, and research references listed in the Meeting 3 Round Table discussion document. Some working sources are internal Round Table or steering-team documents and are cited by document type and date when no public link is available.

Source citations that point to `../materials/` refer to the project's internal working library. Those files are third-party documents that the project does not hold publication rights to, so they are not distributed with the book or the public repository. They are listed so that the chain of evidence behind each chapter is visible.

Contributors

One name appears on the cover of this book, and that is a statement about editorial responsibility, not about authorship of the ideas. Nothing in these pages was invented at a desk. The International STEM Skills Round Table Phase III was a shared inquiry conducted over six meetings and a year of steering discussions, and the substance of this book belongs to the people who were in that room.

The people who made Phase III happen. Dr. Eli Eisenberg initiated and led the Round Table. He opened the meetings, introduced each dilemma, moderated the discussion, and insisted that the process be an active professional inquiry rather than a documentation exercise. Prof. Arnon Bentur gave the inquiry its conceptual spine. The sequence of dilemmas he set out at the second meeting – educator profile, possession versus cultivation of competencies, the teacher-student-AI relationship, professional development, learning environments – is the sequence this book follows chapter by chapter. Dr. Avigdor Zonnenshain brought the systems and engineering-education perspective and summarized the sessions. Tamar Dayan organized the meetings and built the source library on which every citation in this book depends; without that record there would be nothing to synthesize. Dr. Yael Granot-Bein and Dr. Revital Duek moderated discussions and shaped the treatment of competency development, repeated practice, teamwork, and visible thinking. Inna Zertser supported the Round Table process throughout Phase III.

Each of these six people was closely involved in the work that produced this book, and an earlier draft carried their names on the cover. They are credited here instead because the drafting decisions, the structure, the interpretations, and any errors that follow from them are the editor's own, and it would be unfair to attach their names to judgments they did not make.

The people whose arguments the chapters are built from. Jan Morrison of TIES and Dr. Gabi Shafat of Afeka Academic College of Engineering presented at Meeting 2. Danielle Eisenberg of Ignite/Ed and ETS and Dr. Iris Pinto of the Institute for Futures Research in Education presented at Meeting 4. Prof. Russell Tytler of Deakin University presented at Meeting 5. Emlen Metz of the OECD and Berkeley presented at Meeting 6. Their material is the primary evidence behind large parts of this book.

Positions, objections, and questions that shaped the argument came from Dr. Marwa Maklada, Dr. Noa Ragonis, Prof. Sigal Tifferet, Dr. Tzur Karelitz, Dr. Einat Sprinzak, Prof. Gil Noam, and Dr. Olena Bekh. Several of them disagreed with directions the group was taking, and Chapter 24 exists because they did. Dr. Rinat Itzhaki and Orly Rauch took part in the sessions.

Appendix A records each contributor's role and background in full. The attributions throughout the book are drawn from meeting summaries, presentation files, discussion documents, and steering-team records. Where a source reported a contribution rather than recording it verbatim, the book says so. Any misattribution is unintended, and corrections are welcome.

Preface

The International STEM Skills Round Table Phase III took place when education systems were already feeling the pressure of generative artificial intelligence. AI tools had moved from novelty to everyday practice. Students could use them to explain a concept, solve a problem, write code, summarize a paper, generate a draft, translate a text, or rehearse an argument. Teachers and lecturers faced a difficult question: if AI can perform many routine intellectual tasks, what should human education now do?

The Round Table did not treat AI as a narrow technical issue. Its central concern was broader and more demanding: the profile of the teacher and lecturer in the age of generative AI, especially in STEM education. The group asked how educators should cultivate skills, competencies, ethical judgment, critical thinking, curiosity, teamwork, and real understanding when machines can produce fluent answers but cannot take human responsibility for learning.

This book captures the arc of that discussion. It follows the six Phase III meetings and the steering discussions around them. It draws on presentations, summaries, discussion documents, background papers, and the voices of participants from education, higher education, policy, research, and industry.

The book's central claim is simple: AI does not remove the need for teachers. It raises the bar for teaching. The educator is no longer only a transmitter of content. The educator becomes a designer of learning environments, a mediator of judgment, a cultivator of human competencies, a connector between knowledge and practice, and a guardian of the principle that the human must remain above the machine.

Methodology

This book is based on the materials of the International STEM Skills Round Table Phase III: meeting summaries, steering-team summaries, presentation files, discussion documents, program outlines, correspondence, and background papers.

The main Phase III meetings addressed the following sequence:

1. Introduction and skills in the era of artificial intelligence.
2. Whether the profile of lecturers and teachers should be uniform or diverse.
3. Whether teachers and lecturers must themselves possess the competencies they are expected to cultivate.
4. Whether the educational process should be understood as a teacher-student-AI triangle.
5. How professional development of teachers and lecturers can be ensured over time.
6. How physical, virtual, workplace, and industry-connected environments should support future STEM skills.

This book follows that sequence and organizes the material into a repeated professional pattern:

- The challenge.
- The dilemma.
- The main positions.
- Pros and cons.

- Points of view from different sectors.
- Round Table voices.
- Synthesis.
- Recommendation in the spirit of the discussion.

This approach is intended to preserve the complexity of the Round Table. The goal is not to force consensus where the discussion remained open. The goal is to present a coherent professional compass that reflects the direction, tensions, and insights of the group.

Chapters one to twenty each carry their own source list. The closing chapters draw on the whole body of material rather than on any single meeting, so they carry no separate source list; their evidence is the twenty chapters that precede them.

Chapter 24 departs from this pattern deliberately. It collects the strongest objections to the book's own argument, including objections raised inside the Round Table itself and objections from research literature that the Round Table did not resolve.

PART I

Why STEM Education Must Change Now

CHAPTER 1

The New Pressure on STEM Education

The Challenge

STEM education was built on the belief that disciplined knowledge matters. Mathematics, physics, chemistry, engineering, computer science, and biology all require sequence, practice, abstraction, symbolic fluency, and conceptual control. A learner cannot reason responsibly about a circuit, a model, a biological process, a data set, or an engineering system without foundations.

Generative AI does not cancel that truth. It makes the truth harder to protect.

A student can now ask an AI system to explain a concept, solve a calculation, write code, summarize a paper, produce a design idea, draft a laboratory report, or generate a polished answer. The output may be helpful. It may also be shallow, biased, wrong, or disconnected from understanding. This changes the basic evidence problem in education. Producing an answer no longer proves that a student has learned. Producing a fluent answer no longer proves that a student can judge, transfer, explain, or take responsibility for the result.

The second Round Table meeting placed this pressure inside a broader historical shift. Prof. Arnon Bentur described education as moving away from a model centered mainly on transmission of disciplinary knowledge toward a model in which teachers and lecturers must also cultivate cross-disciplinary competencies such as critical thinking, teamwork, curiosity, reflection, and AI literacy. That was the working vocabulary of the second meeting. Chapter 3 sets out the five competencies that this book uses from that point onward, and curiosity, creativity, communication, and reflection are treated there as ways those five show themselves in practice rather than as separate items on the list. What matters here is the direction of the shift: disciplinary knowledge does not become less important, but it must now be taught as something learners can use, question, connect, and apply in uncertain situations.

The Meeting 3 discussion document sharpened the point further. It set out five competencies for an AI-rich world. Four came from the Phase I Round Table consensus; the fifth, ethical judgment, was added because AI raises questions of integrity, intellectual ownership, fairness, transparency, and accountability that cut across all the others. The document's guiding principle was clear: keep the human above the machine. Across all competencies, human judgment and moral responsibility must remain the final authority over any tool or system.

This is the new pressure on STEM education. The system must still teach foundations, but it can no longer be satisfied when students reproduce procedures or submit correct-looking products. It must cultivate learners who can ask whether an answer makes sense, whether a tool should be trusted, whether a model fits the situation, whether a method is fair, and whether a decision can be defended.

The Dilemma

Should STEM education continue to focus mainly on mastery of disciplinary content, or should it reorganize around deeper human competencies that remain valuable when AI can perform many routine tasks?

This question is often misunderstood as a choice between content and skills. The Round Table materials point to a more serious dilemma. STEM education must preserve rigorous disciplinary learning, but it must also change what counts as successful learning. A student who can repeat a procedure but cannot evaluate an AI-generated solution is not prepared. A student who can speak generally about critical thinking but lacks mathematical, scientific, or engineering foundations is also not prepared.

The deeper dilemma is therefore this: how can education preserve the slow work of conceptual formation while preparing students for a world in which many technical outputs can be produced quickly by machines?

Main Positions

The first position is content-first. This view argues that students still need strong disciplinary foundations before they can use AI responsibly. Without mathematics, physics, scientific method, engineering reasoning, and computational thinking, students cannot reliably evaluate the outputs of AI systems. They may become dependent on confident answers that they do not understand.

The second position is competency-first. This view argues that the future value of education lies in human capacities for which no machine can take responsibility: moral responsibility, judgment, persistence, accountability, and the ability to decide when a tool should not be trusted. It also treats critical thinking, collaboration, creativity, communication, curiosity, and reflection as abilities that AI may imitate in output but cannot own in educational responsibility. From this perspective, education must prepare students not only to know, but to act wisely with what they know.

The third position, which best reflects the spirit of the Round Table, is integrated. STEM education must preserve disciplinary depth, but it must use disciplinary learning as the arena in which students develop human competencies. Students should not learn critical thinking as a slogan. They should learn it by testing claims, comparing models, identifying failure modes, explaining assumptions, and reflecting on evidence inside real STEM work.

Jan Morrison's presentation from TIES supported this integrated view from the education-system perspective. She argued that success requires transdisciplinary teaching, career-connected learning, real-world experiences, team-based teaching, and systems designed for iteration and improvement. Her examples showed that problem-based and transdisciplinary learning fail when they are placed inside systems that were never designed to support them. Time for teacher collaboration, cross-sector training, and structured teamwork are not decorations. They are conditions for success.

Dr. Gabi Shafat's presentation from Afeka brought the same issue into higher education. The shift from "teaching engineering" to "engineering education" means starting not only from course content, but also from the desired graduate profile. AI can provide assistance, summaries, explanations, and feedback, but it lacks accountability and human interaction, and it can be confidently wrong. Therefore, students must learn to use AI while understanding its limits.

Pros and Cons

The content-first approach protects rigor. It reminds educators that AI fluency without knowledge is fragile. Students cannot check a mathematical derivation, debug code, critique a model, or recognize a flawed engineering assumption if they lack the underlying concepts. Its weakness is that it can preserve a model of education designed for an earlier evidence environment, where producing the answer was more tightly connected to doing the thinking.

The competency-first approach is future-facing. It recognizes that employers and societies need people who can think, judge, create, cooperate, and keep learning. Its weakness is that competencies can become vague if they are separated from disciplinary practice. A student does not become a critical thinker in general. A student learns to think critically through repeated work with specific materials, methods, claims, tools, and consequences.

The integrated approach is strongest, but hardest. It asks teachers and lecturers to design learning tasks in which content and competencies develop together. It asks institutions to change assessment, scheduling, professional development, class size, collaboration time, and learning environments. It also asks policy makers to recognize that competency development is not free. It requires investment.

Point of View Map

From an industry perspective, Avi Salmon emphasized that employers, especially in high technology, are becoming less interested in routine technical skills that AI can easily perform and more interested in deep conceptual understanding, curiosity, and genuine engagement with the profession. His point was not that technical skill no longer matters. It was that technical skill must be connected to judgment, meaning, and the ability to keep learning. Schools should help develop love of the profession. Higher education should deepen conceptual understanding beyond curriculum coverage.

From a school perspective, the challenge is implementation. Teachers face prescribed curricula, external exams, mixed student motivation, class size, limited time, and different expectations across age groups. The Meeting 2 discussion made clear that middle school, upper secondary school, and higher education cannot be treated as identical settings.

From a higher education perspective, the challenge is to protect foundations while redesigning programs around graduate profiles and competencies. Prof. Arnon Bentur noted that the first years of higher education, especially in mathematics, physics, and chemistry foundations, often resemble secondary education in their constraints. Later years may allow more autonomy, differentiation, and specialization.

From a policy perspective, the challenge is to define skills clearly enough to support teacher development, curriculum design, assessment, and institutional investment. Dr. Marwa Maklada's contribution points to the importance of validating competency-based tasks in formal assessment systems, including matriculation processes.

From an international education-system perspective, Jan Morrison's examples suggest that isolated innovation is not enough. Systems need cross-sector communication, industry involvement, teacher externships, flexible professional pathways, and measures used for improvement rather than accountability alone.

Round Table Voices

Dr. Eli Eisenberg opened the first meeting by naming the shift that defines Phase III. Phase I had described the generic STEM competencies. Phase II had produced practical tools for educators. Phase III moves the question from the profile of the graduate to the profile of the teacher and the lecturer, and treats AI both as an educational tool and as a force that changes which skills are worth having.

Avi Salmon opened the substantive discussion with the industry view of what that pressure looks like from outside the school. Teachers, he argued, are frustrated, working with methods built for a different era, and often feel behind their own students. Some leave the profession. This is not a complaint about teachers. It is a description of a system whose tools have moved faster than its design.

Dr. Yael Granot-Bein named a gap that runs through the rest of this book. When educators talk about their own use of AI they are curious and often enthusiastic. When the conversation turns to students using AI, the tone changes to suspicion and to fear of cheating. Institutions then spend their energy controlling behavior rather than asking what AI now makes possible that was not possible before.

Dr. Einat Sprinzak argued that the first need is foundational AI literacy, because many people use these tools without understanding their limits, their biases, or their distortions. Telling students to use AI responsibly is not enough. Learning environments have to be designed so that reflection and discernment are unavoidable.

Dr. Olena Bekh widened the frame. Learning now happens across blurred boundaries between formal and informal education, digital and physical environments, and cognitive and emotional development. She also raised the cost side: digital saturation can erode empathy, so educators have to nurture emotional awareness deliberately rather than assume it survives on its own.

Danielle Eisenberg raised the objection that returns in Chapter 24. Expectations of educators keep rising while professional development, institutional support, and systemic guidance often do not. Teachers cannot carry a transformation of this size alone, and a system that asks them to will not sustain it.

Dr. Avigdor Zonnenshain closed the first meeting with the insight that framed the sessions that followed: the pressing need is to understand students better, in their diverse abilities, interests, and ways of learning, and to move away from one-size-fits-all teaching.

Dr. Eli Eisenberg opened Meeting 2 by emphasizing that the initiative's value lies not only in documentation but also in active participation and brainstorming. This matters because the dilemmas are not administrative. They concern the future purpose of education.

Prof. Arnon Bentur framed Phase III as a sequence of connected dilemmas: the educator profile, possession versus cultivation of competencies, the teacher-student-AI relationship, professional development, and the learning environment. This framing prevents the discussion from treating AI as a stand-alone technology issue. AI is part of a larger redesign of educational purpose, roles, environments, and evidence.

Jan Morrison argued that teaching must become a team-driven, cross-sector profession connected to real-world experience. Her warning is important: transdisciplinary and problem-based learning cannot succeed when inserted into systems built for isolated individual teaching.

Dr. Gabi Shafat showed how engineering education can respond through graduate profiles, skill mapping, AI integration, personalized learning, flipped classrooms, project-based learning, continuous feedback, and industry visits. His examples make the abstract debate concrete.

Avi Salmon brought the industry perspective into the conversation. His contribution was not that education should become narrow job training. It was that STEM education must cultivate the kind of thinking, curiosity, and meaning that remain valuable when routine production is automated.

Emlen Metz and Prof. Russell Tytler added a warning that belongs already in the opening chapter. AI can support learning, but it can also reduce independent thinking if students rely on it passively. Decision-making and responsibility cannot be delegated to machines. Long-term habits of independent, reflective STEM learning must be deliberately cultivated.

Synthesis

The Round Table points toward a new understanding of STEM rigor. Rigor is not only the ability to produce a correct answer. Rigor is the ability to understand why an answer is correct, judge whether a method is appropriate, see the limits of a tool, connect knowledge to context, and take responsibility for decisions.

In the AI era, rigor must include visible thinking. Students should be asked to show how they framed the problem, what assumptions they made, how they used tools, how they checked outputs, where they found uncertainty, what alternatives they considered, and why they trust their conclusion. The polished final product is no longer enough.

This also changes the teacher's role. The teacher or lecturer is not only a source of content. The educator becomes a designer of learning situations in which students must think with tools, against tools, and beyond tools. The educator makes judgment visible. The educator protects the human work of learning.

Recommendation

STEM education should preserve disciplinary rigor, but redesign learning so that every significant task also develops human competencies. Students should not only learn content. They should learn how to use content to think, judge, create, collaborate, and act responsibly in an AI-rich world.

The practical recommendation is to redesign STEM learning around three layers:

1. Foundations: disciplinary knowledge, methods, concepts, and technical fluency.
2. Competencies: critical thinking, self-directed learning, teamwork and interpersonal communication, solving complex problems, and ethical judgment and responsible use, with AI literacy running across all five.
3. Evidence of learning: visible reasoning, tool critique, explanation, reflection, application, and responsibility.

This is the first principle of the book: the future of STEM education is not less knowledge. It is knowledge placed under stronger human responsibility.

Chapter 23 sets out the practical form this takes in a classroom: a sequence in which students meet the phenomenon, commit to an explanation, and gather evidence before AI enters the room. Chapter 24 sets out the strongest arguments against the whole approach, including the objection that its evidence base is thinner than its confidence suggests.

Practical Recommendation for Teachers

Act as a designer of visible thinking, not only as a presenter of content. In every substantial STEM task, ask students to show the path behind the answer: assumptions, tool use, checks, errors, corrections, and final responsibility.

In practice:

1. Add a short "how I know" component to assignments.
2. Ask students to compare their reasoning with an AI-generated explanation.
3. Require students to mark what they accepted, rejected, or changed after using AI.
4. Use class discussion to examine one wrong or biased AI answer.
5. Keep disciplinary foundations strong, but assess whether students can explain and defend them.

Classroom activity examples:

1. AI answer audit. Give students a familiar STEM problem and one AI-generated solution. Students mark correct steps, weak assumptions, missing explanations, and possible errors, then write a short statement explaining what they trust and why.
2. Reasoning trail submission. Students solve a problem and submit the final answer together with a reasoning trail: what they tried first, where they checked themselves, what evidence supports the answer, and what remains uncertain.
3. Human responsibility debate. Present two polished answers, one student-made and one AI-assisted. Students discuss which answer shows stronger learning and define what evidence would prove ownership.

Evaluation method:

Evaluate success through evidence of visible thinking, not only the correctness of the final answer. The student should be able to explain assumptions, identify tool use, verify claims, correct errors, and take responsibility for the conclusion. A strong performance includes a defensible answer, a clear reasoning trail, and at least one explicit check of reliability.

For your own development, practice the same task yourself. Take one familiar lesson, use AI to solve or explain it in several ways, identify where the tool is strong and weak, and redesign the lesson so students must think beyond the output. This follows the Round Table spirit expressed by Avi Salmon, Emlen Metz, Prof. Russell Tytler, Jan Morrison, and Dr. Gabi Shafat: AI use must deepen judgment, not replace it.

SOURCES

Meeting 1 summary, Meeting 2 presentation, Meeting 3 discussion document, OECD Teaching Compass, TALIS 2024 background.

CHAPTER 2

From Knowledge Transmission to Competency Formation

The Challenge

Many education systems now speak the language of skills and competencies. They mention critical thinking, teamwork, creativity, communication, adaptability, AI literacy, ethical judgment, and lifelong learning. Yet many classrooms, courses, timetables, examinations, and promotion systems remain organized around content delivery, individual performance, and final answers.

This gap between language and structure was one of the recurring concerns in the Round Table. It is easy to say that students need skills. It is harder to design courses, programs, assessment, teacher development, and institutional routines that actually cultivate them.

The OECD Teaching Compass helps explain why the gap is so difficult. It argues that modern teaching requires more than subject knowledge and technical skill. Teachers need professional identity, agency, competencies for complexity, well-being, and participation in a learning ecosystem. In other words, competencies cannot be demanded only from students. The entire teaching system must become capable of supporting them.

The TALIS 2024 material reinforces this point from the teacher side. Teachers work under job demands such as workload, work pressure, student behavior, diverse learning needs, uncertainty, and change. They also need resources such as learning opportunities, agency, leadership, support, and professional status. A system that asks teachers to cultivate complex competencies while leaving their work environment unchanged creates a contradiction.

The challenge, therefore, is not whether competencies are desirable. The challenge is how to move them from policy language into daily educational practice without weakening disciplinary rigor or overloading teachers.

The Dilemma

Should competencies be taught as separate skills, or embedded inside disciplinary courses and real learning tasks?

A separate skills course can make a skill visible. It can give students vocabulary, tools, and dedicated practice. But it can also signal that the skill is external to the discipline. Students may treat it as secondary, optional, or disconnected from real STEM work.

An embedded model makes the skill authentic. It asks students to practice critical thinking inside a physics problem, teamwork inside an engineering project, ethical judgment inside an AI-supported design task, and reflection inside a real learning process. But embedding competencies is demanding. It requires every teacher and lecturer to understand how to design for the competency, make it visible, support it, and assess it.

The Round Table discussion made the dilemma sharper by asking whether competencies should be taught in individual courses, across whole programs, or through a wider ecosystem of teachers, lecturers, industry partners, and institutional structures.

Main Positions

The separate-skills position argues that competencies should be taught directly. Students need explicit language and practice. If teamwork, communication, or AI literacy matters, then it deserves dedicated time.

The embedded-skills position argues that competencies only become meaningful when used in context. Critical thinking in physics is not identical to critical thinking in history. Collaboration in engineering is shaped by technical constraints, design decisions, and shared responsibility.

The repeated-practice position, raised in the discussion through Dr. Yael Granot-Bein's reference to European evidence, argues that competencies are internalized when three conditions are present: they are linked to a discipline, reinforced over time, and supported by structured reflection. A one-time activity is not enough.

The program-level mapping position argues that individual courses are not enough. Institutions should define a graduate profile and then map where each competency is introduced, practiced, assessed, and deepened across the program.

The ecosystem position goes one step further. It argues that competencies require teachers, lecturers, school leaders, teaching centers, assessment bodies, industry partners, and policy makers to work in alignment. A teacher cannot create a competency-based system alone.

Pros and Cons

Separate skills courses provide clarity. They can be easy to schedule and communicate. Their risk is that students may treat them as secondary or fail to transfer them into real disciplinary work.

Embedded skills feel more authentic. They connect competencies to actual problems, repeated practice, and disciplinary identity. Their risk is that implementation depends heavily on teacher preparation. If teachers are not supported, the term "embedded" may mean only that the competency is mentioned in the syllabus but never deliberately taught.

Program-level mapping creates coherence. It prevents skills from being left to chance. Its cost is institutional coordination. Departments, teachers, lecturers, and assessment systems must work together.

The ecosystem approach has the greatest long-term potential, but it is also the most demanding. It requires policy, leadership, professional development, data, time, collaboration, and investment. This is why the Afeka presentation is important. It did not present competency formation as a slogan. It presented it as course design, graduate profile work, dynamic learning processes, feedback, student presentations, flipped learning, AI-supported problem solving, project-based learning, industry visits, and evaluation.

Point of View Map

Higher education needs graduate profiles and program-level mapping. The Afeka example, presented by Dr. Gabi Shafat, shows how engineering education can move from a collection of courses toward a coherent profile of the graduate. The presentation described the future lecturer profile, the Afeka learning experience, the limits of the traditional lecturer model, and the need for team-based structures that combine teaching, project-based learning, industry engagement, AI, and faculty development.

Schools need competencies that fit curriculum, assessment, and student development. Dr. Marwa Maklada's contribution pointed to work in STEM subjects where competency-based tasks can be connected to assessment processes.

Students need to see why competencies matter. If they experience skills as general slogans, they will not internalize them. If they use them to solve real problems, compare approaches, expose mistakes, and reflect on their decisions, competencies become part of learning.

Assessment bodies need ways to evaluate process, reasoning, reflection, and growth, not only final products.

Teachers and lecturers need agency and support. The OECD Teaching Compass explicitly connects teacher agency with curriculum change. If educators are expected to lead modern learning, they need autonomy, confidence, capabilities, professional identity, and well-being. A competency agenda that ignores teacher conditions will remain rhetorical.

Industry needs graduates who can perform in uncertain and collaborative settings. Avi Salmon's contribution connects directly to this point: future readiness is not only mastery of routine technical skills, but deep understanding, curiosity, and professional engagement.

Round Table Voices

Dr. Yael Granot-Bein brought evidence from European contexts suggesting that competencies are internalized more effectively when they are linked to a discipline, reinforced over time, and accompanied by reflection.

Prof. Sigal Tifferet raised an important question: is embedding competencies inside disciplinary courses evidence-based? This question is central because education systems must avoid adopting attractive language without testing its effectiveness.

Dr. Gabi Shafat described Afeka's practice of defining skills and competencies for courses and evaluating whether students achieved them. His presentation adds operational depth. In the Advanced Digital Signal Processing course, drivers of change included teamwork, flipped classroom, personalized learning, students teaching, AI as a tool for improving learning processes, and project-based or problem-based learning. The course design included pre-assessment questionnaires, problems at the beginning of lectures, student presentations, pre-class preparation, immediate formative feedback through Delta-Plus, discussions to bridge knowledge gaps, industry site visits, and guest experts.

Dr. Marwa Maklada described progress in integrating skills in STEM subjects and connecting competency-based tasks to assessment.

Dr. Avigdor Zonnenshain emphasized that education must go beyond knowledge transmission and foster a love of the profession and lifelong learning. He connected this to team-driven education, industry realities, project-based learning, active learning, meaningful assessment, and student motivation.

Danielle Eisenberg's comment about institutions mapping competencies across curricula added another layer. The issue is not only whether one lecturer teaches one skill. The issue is whether an institution knows where its core competencies are developed, where gaps exist, and how professors are supported to teach competencies as intentionally as they teach content.

Synthesis

Competencies become real when they are practiced, reflected on, and assessed in context. They cannot remain posters on a wall or terms in a policy document.

The Round Table material points to a discipline-linked model of competency formation. It has six parts:

1. Name the competency clearly.
2. Anchor it in a disciplinary or professional context.

3. Give students repeated opportunities to practice it.
4. Make the process visible through reflection and feedback.
5. Assess evidence of reasoning, contribution, and growth.
6. Map the competency across courses, years, and institutional structures.

This model also changes what teachers and lecturers need. They need to know how to design learning processes, not only how to deliver knowledge. They need to manage AI as a learning tool, not only as a threat. They need to work in teams, because one individual educator cannot carry the full complexity of modern STEM learning alone.

Recommendation

The book recommends a program-level embedded model. Competencies should be explicitly named, repeatedly practiced, connected to disciplinary work, supported by reflection, and assessed through evidence of process as well as outcomes.

For schools, this means connecting competencies to subject learning, assessment tasks, classroom routines, and age-appropriate development.

For higher education, this means defining a graduate profile and mapping competencies across the program rather than leaving them to individual lecturer preference.

For teacher and lecturer development, this means giving educators lived experience with the learning processes they are expected to design for students.

For policy, this means supporting time, structures, assessment redesign, and investment. Competency formation is not a low-cost add-on to traditional education. It is a different design logic.

Practical Recommendation for Teachers

Turn one unit from content coverage into competency formation. Do not add competencies as a slogan at the end of the lesson. Build them into the task itself.

In practice:

1. Choose one disciplinary topic already in the curriculum.

2. Name one competency that naturally belongs inside it, such as critical thinking, teamwork, reflection, communication, or ethical judgment.
3. Design a task where students must use the competency to solve the disciplinary problem.
4. Add structured reflection, so students explain how the competency helped them work.
5. Repeat the same competency later in the course, so it develops over time.

Classroom activity examples:

1. Competency inside content. During a disciplinary lesson, assign a task that requires both content and one named competency, such as explaining a physics result to a nonexpert or justifying a design choice under constraints.
2. Reflection loop. At the end of a project, ask students to write where they used critical thinking, teamwork, or ethical judgment and how that changed the technical result.
3. Repeated competency tracker. Return to the same competency three times across a unit. Students record how their performance changed from first attempt to final task.

Evaluation method:

Evaluate success by looking for transfer of the competency inside the discipline. The student should show accurate content knowledge and also demonstrate the named competency in action: explaining, collaborating, questioning, reflecting, or making a justified decision. Use a rubric with two equal parts: disciplinary quality and competency evidence.

For your own development, map one course or semester and ask: where is each competency introduced, practiced, assessed, and revisited? This follows Dr. Yael Granot-Bein's point that competencies are internalized when linked to discipline, repeated over time, and supported by reflection, and it follows Dr. Gabi Shafat's Afeka material on defining and evaluating skills inside courses.

SOURCES

Meeting 2 presentation, Dr. Gabi Shafat STEM material, OECD Teaching Compass, TALIS 2024 background.

CHAPTER 3

The Five Competencies in an AI-Rich World

The Challenge

The Round Table built on earlier work identifying core STEM competencies. Phase III added a new pressure: generative AI raises questions of integrity, fairness, responsibility, authorship, trust, dependence, bias, and human agency. These questions cannot be placed outside the competency framework.

The Meeting 3 discussion document, prepared by Dr. Eli Eisenberg, Prof. Arnon Bentur, and Avi Salmon with AI assistance, made this point directly. It stated that the first four competencies had been selected through the Phase I Round Table consensus process, and that ethical judgment and responsible use were added as a fifth competency because AI raises questions of integrity, fairness, transparency, accountability, and responsibility that cut across all the others.

This move is important. It means that AI does not merely add one more technical requirement to STEM education. It changes the conditions under which every competency must be understood. Critical thinking now includes judging AI output. Self-directed learning now includes knowing when to continue learning beyond an automated answer. Teamwork and interpersonal communication now include collaboration in human-machine environments without hiding human contribution. Solving complex problems now includes framing the problem that AI cannot frame for the learner. Ethical judgment and responsible use become cross-cutting conditions for all four.

The five competencies used in this chapter are therefore: critical thinking; self-directed learning; teamwork and interpersonal communication; solving complex problems; and ethical judgment and responsible use. The chapter also treats AI literacy as a necessary operating layer across all five rather than as a replacement for them.

The Dilemma

Are existing STEM skills sufficient, or does the AI era require adding ethical judgment and stronger AI literacy to the core profile?

The question is not only whether a fifth competency should be added. The harder question is whether AI changes the meaning of every competency already named. A student may demonstrate critical thinking in a traditional task, but fail to evaluate an AI-generated explanation. A student may collaborate well with classmates, but use AI in a way that hides contribution and responsibility. A student may show creativity in an assignment, but rely on AI to generate ideas without understanding where they came from. A student may be fluent with tools, but unable to explain when not to use them.

The dilemma is therefore structural. Should AI-era education update the list of competencies, or reinterpret the whole framework through the principle of human responsibility?

Main Positions

One position is continuity. The competencies identified in earlier phases remain valid. Education systems should avoid constantly changing frameworks.

Another position is expansion. AI creates new conditions, so education must add explicit AI literacy and ethical judgment.

A third position is integration. Ethical judgment and AI literacy should not be treated only as separate skills. They should become cross-cutting dimensions of every competency.

A fourth position, implied by the OECD Teaching Compass, is professional mirroring. If students are expected to develop agency, responsibility, and competencies for complexity, teachers must also be supported as agents of curriculum change. The student competency framework and the educator competency framework cannot be separated.

Pros and Cons

Continuity offers stability. Teachers and institutions need frameworks they can implement. But continuity can understate the specific risks created by AI.

Expansion addresses the new reality directly. But if every new technology adds another competency, teachers and curricula become overloaded.

Integration is stronger. It says that ethical judgment and responsible AI use should be present in critical thinking, self-directed learning, teamwork and interpersonal communication, and solving complex problems. Its challenge is assessment. Cross-cutting competencies are harder to measure.

Professional mirroring is essential but demanding. It prevents the system from asking teachers to cultivate capacities that they are not supported to practice themselves. But it requires time, professional development, leadership, trust, and attention to teacher well-being.

Point of View Map

Teachers need definitions they can teach. A vague call for "future skills" is not enough. Educators need to know what each competency looks like in a lesson, in a project, in feedback, in assessment, and in AI-supported work.

Students need habits, not only rules. They need repeated practice in asking: What did the tool do? What did I do? What do I understand? What do I still need to verify? What are the consequences if this answer is wrong?

Policy makers need a framework that can be adapted locally. The OECD Teaching Compass is useful here because it connects competencies to professional identity, agency, well-being, and learning ecosystems. It also makes clear that transformative student learning cannot happen unless teachers are empowered and supported.

Industry needs graduates who can work responsibly with powerful tools. Avi Salmon's Meeting 2 contribution and the AI-era discussion documents point to a future in which routine output matters less than the ability to understand, judge, question, collaborate, and continue learning.

The UNESCO AI competency framework, referenced in the collected materials, similarly supports the Round Table's direction. It frames AI-era teaching through human-centered mindsets, ethics, foundations, pedagogy, and professional learning. That structure reinforces a central idea: AI competence is not just tool use. It is responsible practice.

Round Table Voices

Dr. Eli Eisenberg, Prof. Arnon Bentur, and Avi Salmon framed the Meeting 3 question around the role of the teacher in cultivating competencies. The discussion document emphasized a guiding principle: the human must remain above the machine.

This phrase is more than a slogan. It means that AI can support analysis, explanation, production, and feedback, but responsibility remains human. The learner must remain responsible. The teacher must remain responsible. The institution must remain responsible.

Emlen Metz's Meeting 2 contribution is directly relevant to this chapter. The warning was that students must use AI thoughtfully and critically, and that deep understanding, critical thinking, and writing require students to think and produce work independently, not merely edit AI-generated text. Decision-making and responsibility cannot be delegated to machines.

Prof. Russell Tytler added a related concern: as AI becomes more sophisticated, students may relinquish control over their thinking processes. This risk makes AI literacy and critical thinking not optional enrichments, but long-term habits that STEM education must deliberately cultivate.

Dr. Gabi Shafat's Afeka material gives this principle a practical form. Students should confront problems where AI tools fail, produce biased output, or provide incomplete solutions. The educational value lies not only in using AI, but in learning to evaluate it.

Dr. Marwa Maklada's assessment perspective also matters. If competencies include ethical and critical AI use, assessment systems must be able to recognize more than final correctness. They must evaluate reasoning, process, judgment, and evidence of independent thought.

Synthesis

AI does not only add a technology skill. It changes the moral and intellectual conditions of learning. Students must learn not only how to use AI, but also when to question it, when to reject it, when to disclose it, when to seek human help, and when to rely on their own judgment.

An AI-era competency is therefore not a simple skill. It has at least four layers:

1. Capability: the student can perform or participate in the activity.
2. Understanding: the student knows why the action or answer makes sense.
3. Judgment: the student can evaluate quality, limits, risks, and alternatives.
4. Responsibility: the student can explain and defend the decision to use, revise, reject, or disclose AI support.

This four-layer model helps prevent a common mistake. AI literacy should not be reduced to knowing how to prompt a system. Prompting may be useful, but it is not enough. The deeper educational question is whether the student remains intellectually present and morally responsible.

The same applies to teachers. Teachers need enough AI literacy to design meaningful learning, enough ethical judgment to set boundaries, enough pedagogical confidence to make thinking visible, and enough professional agency to adapt as tools change.

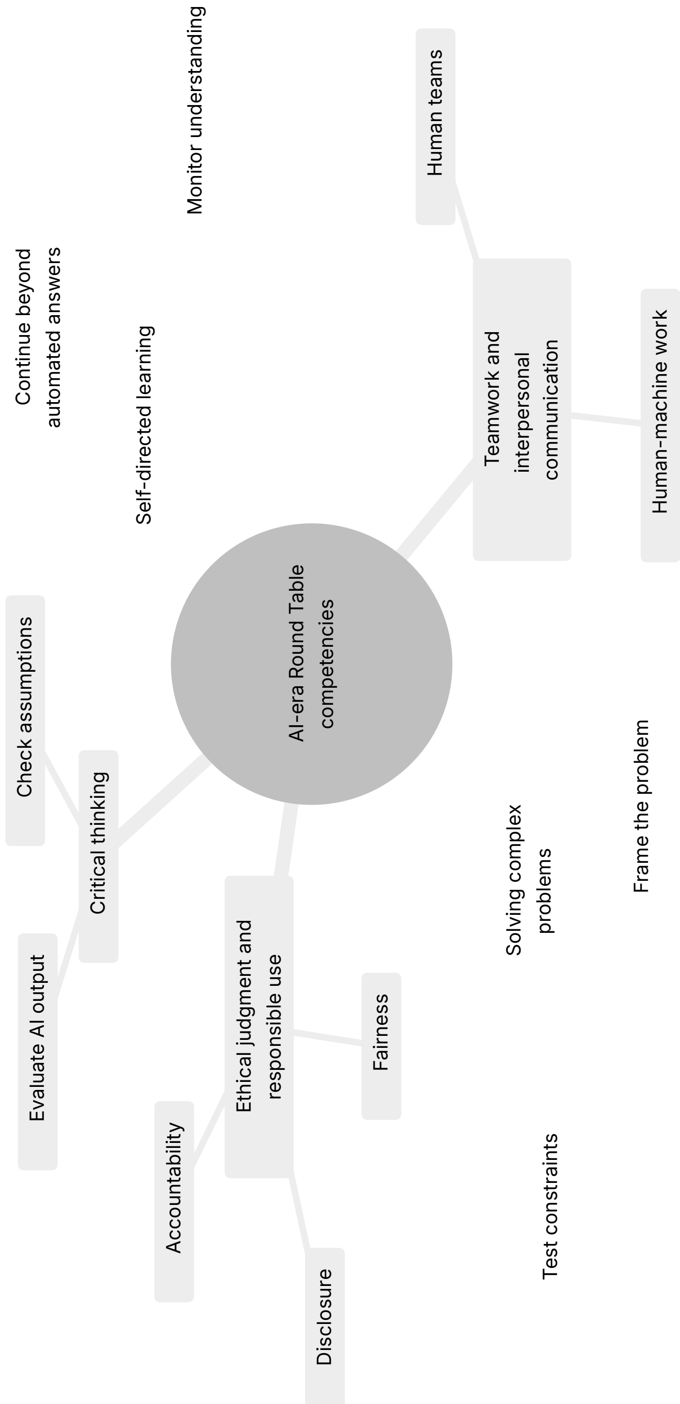

Figure 1. The five Round Table competencies in an AI-rich world.

Recommendation

The book recommends an AI-era competency architecture rather than only a list. The architecture preserves the competencies developed in the earlier Round Table work, adds ethical judgment and responsible use explicitly, and embeds AI literacy across the whole framework.

The guiding principle should be: keep the human above the machine.

In practice, this means that every STEM learning task involving AI should ask students to provide evidence of:

1. What they asked the AI to do.
2. What they accepted, rejected, or changed.
3. How they checked the output.
4. What they learned independently.
5. What ethical or practical risks they considered.
6. Why the final answer is their responsibility.

This principle should guide curriculum, assessment, professional development, and institutional policy. It should also guide the language of the book. The goal is not anti-AI education. The goal is human-centered AI education.

Practical Recommendation for Teachers

Teach AI literacy as responsibility, not only as tool use. A student who knows how to prompt AI but cannot verify, question, or own the result has not developed the needed competency.

In practice:

1. Require an AI-use note when students use a tool.
2. Ask students to identify one risk in the AI output: error, bias, missing context, weak source, or ethical concern.
3. Make students explain what remains their own responsibility.
4. Use short classroom routines for verification: source check, calculation check, assumption check, and consequence check.

5. Include ethical judgment in STEM tasks whenever AI affects fairness, privacy, accountability, or safety.

Classroom activity examples:

1. AI use declaration. Students complete an assignment with a short declaration: what AI tool was used, for what purpose, what was checked, and what remained their own work.
2. Bias and fairness check. Give students an AI answer to a STEM or technology scenario and ask them to identify possible bias, missing stakeholders, or fairness concerns.
3. Responsibility card. Before submitting AI-assisted work, each student writes one sentence beginning, "I am responsible for this answer because..." and supports it with one verification step.

Evaluation method:

Evaluate success through responsible AI behavior. The student should disclose assistance, distinguish personal thinking from tool output, identify possible ethical or reliability risks, and show how the final work was checked. The grade should reward transparency and judgment, not hidden automation or polished language alone.

For your own development, build a personal AI competency journal. Once a week, test one AI tool on a real teaching task, record what worked, what failed, and what boundary you would set for students. This turns the Meeting 3 principle, keeping the human above the machine, into a teacher habit.

SOURCES

Meeting 3 discussion document, OECD Teaching Compass, UNESCO AI competency frameworks, AI Literacy Framework.

PART II

The Future Profile of the Teacher and Lecturer

CHAPTER 4

One Profile or Many?

The Challenge

Teachers and lecturers are now expected to do many things at once. They must teach content, cultivate competencies, understand AI, design meaningful tasks, support student motivation, assess process and product, work with diverse learners, collaborate with peers, and connect education to society and work.

No single person can be equally strong in all these areas. This creates the central dilemma of Meeting 2: should the future profile of teachers and lecturers be uniform, or should education systems build diversity of profiles within a coordinated ecosystem?

The question is not technical. It goes to the identity of the profession. If the system defines one ideal teacher profile, it may create clarity but also overload. If the system accepts many profiles, it may gain realism but lose coherence. If it builds a common foundation with differentiated roles, it may support both equity and excellence, but only if institutions are designed to coordinate those differences.

Jan Morrison's TIES presentation framed the issue as a shift from "no alternative" toward a strategic approach to defining the future teacher or lecturer. Her proposal included openness to recruiting individuals from diverse backgrounds, including industry, not as a temporary solution but as part of a systemic strategy. It also included team-based models in which teachers and lecturers with different strengths complement one another rather than all being expected to fit a single mold.

The Dilemma

Should there be a uniform future profile for all teachers and lecturers, or a diversity of profiles within a coordinated educational ecosystem?

The dilemma contains several smaller questions. Are school teachers and university lecturers similar enough to share one profile? Does the answer change between middle school, upper secondary school, first-year higher education, and advanced university courses? Should all educators be expected to embed the same competencies in the same way? Can individual educators be retrained fast enough, or should systems create complementary teams? How much of the future profile belongs to the individual educator, and how much belongs to the institution?

Main Positions

The uniform-profile position argues for a common expected profile. Every educator should share a clear foundation. This supports equity, policy, teacher education, and public expectations.

The diverse-profile position argues that educators should bring different strengths. Some may be strong in disciplinary depth, others in project-based learning, AI integration, mentoring, assessment, industry connection, or student support.

The common-foundation-plus-specialization position combines both. Every educator needs a baseline, but not every educator needs the same depth in every competency.

The ecosystem position goes further. It argues that the future profile should not be imagined only as the profile of one person. It should be imagined as the profile of a school, department, program, or learning system. The question becomes: what capabilities must exist around the learner, and how should they be distributed among teachers, lecturers, mentors, AI tools, industry partners, teaching centers, and institutional leaders?

Pros and Cons

A uniform profile creates clarity. It can help teacher education and professional standards. But it can become unrealistic and may ignore differences between school, higher education, discipline, student age, and institutional mission.

Diverse profiles are more realistic. They allow teams to combine strengths. But diversity without coordination becomes fragmentation.

A common foundation plus specialization is the strongest model. It supports shared expectations and differentiated roles. But it requires institutional design.

The ecosystem model is even stronger when implemented well. It recognizes that STEM education in the AI era requires disciplinary expertise, pedagogy, technology use, assessment design, project design, industry relevance, mentoring, and student support. These capabilities can be distributed. The risk is that distributed responsibility can become no one's responsibility unless governance is clear.

The Round Table discussion suggests that this is the real tradeoff. Uniformity without realism produces overload. Diversity without structure produces incoherence. A coordinated ecosystem can produce both professional dignity and practical capacity.

Point of View Map

School teachers often work under prescribed curricula and closer regulation. University lecturers may have greater academic freedom, especially in advanced courses. Prof. Gil Noam warned against treating school teachers and university lecturers as identical. His concern is important because a shared vocabulary can hide very different professional conditions.

Prof. Arnon Bentur complicated the distinction. He noted that early higher education and secondary education may share important constraints, especially in large foundational courses in mathematics, physics, chemistry, and other basic subjects. In those settings, lecturers often work within predefined programs because later courses depend on those foundations. Greater academic freedom appears more strongly in later years.

Dr. Eli Eisenberg added a governance distinction. Schools in Israel operate under prescribed curricula and close supervision by the Ministry of Education, while universities usually grant more academic autonomy. This changes what can reasonably be expected from the individual educator.

Industry tends to value diverse teams. Complex problems are rarely solved by one ideal worker. This supports an ecosystem view of teaching.

Jan Morrison's TIES material adds a systems perspective. Teaching should become a team-driven, cross-sector endeavor. It should connect education to workforce and economic development without reducing education to training. Career-connected learning should be infused with real-world experience and expertise. Emerging technologies such as AI should not remain optional or peripheral to teaching competencies.

Dr. Gabi Shafat's Afeka material gives the higher education version of the same idea. The traditional lecturer model is no longer sufficient because the lecturer is asked to carry too many roles. His proposed principle is a shift from a single-lecturer model to a complementary team-based model with targeted integration of teaching, project-based learning, industry engagement, AI, and faculty development.

Round Table Voices

Prof. Gil Noam emphasized that the terms teacher and lecturer refer to different professional contexts. His warning matters because a book about educator profiles must not flatten those differences.

Prof. Arnon Bentur argued that some parts of higher education, especially foundational first-year courses, resemble secondary education in their constraints and challenges. He also described the shift from academic programs as collections of individual courses toward programs designed backward from a graduate profile.

Dr. Eli Eisenberg pointed to differences in academic freedom between schools and universities.

Jan Morrison emphasized team-based and system-based teaching. Her contribution supports the idea that the future profile may not be one individual profile but a designed professional ecosystem. She described the need to leave the traditional "yellow brick road" career path and adopt an "interstate road" model with ramps on and off, including industry, nonprofit, post-military, and other entry paths into teaching.

Dr. Marwa Maklada emphasized that educator profiles differ not only between schools and universities, but also across educational stages and subject areas. Middle school, upper secondary school, STEM subjects, and humanities do not pose identical pedagogical demands.

Dr. Einat Sprinzak added that teachers must experience learning processes themselves in order to mediate them. This supports a differentiated profile because not every educator will enter the transformation from the same starting point.

Avi Salmon's contribution connects the profile question to industry. If employers value deep understanding, curiosity, and meaning-making more than routine AI-replaceable skills, then teachers and lecturers need enough capacity to cultivate these qualities. But this does not imply that every educator must do so in the same way.

Synthesis

The Round Table points away from a single ideal educator. The better model is a shared baseline combined with diverse strengths, team structures, and institutional coherence.

The shared baseline should include disciplinary responsibility, pedagogical care, ethical judgment, AI awareness, commitment to student agency, and ability to participate in collaborative professional learning. These are not optional.

The differentiated layer should allow educators to specialize. Some may lead deep disciplinary foundations. Some may lead project-based learning. Some may design AI-supported learning. Some may connect with industry. Some may mentor students. Some may develop assessment. Some may coordinate interdisciplinary work.

The institutional layer must make these profiles work together. Without coordination, specialization becomes fragmentation. With coordination, specialization becomes capacity.

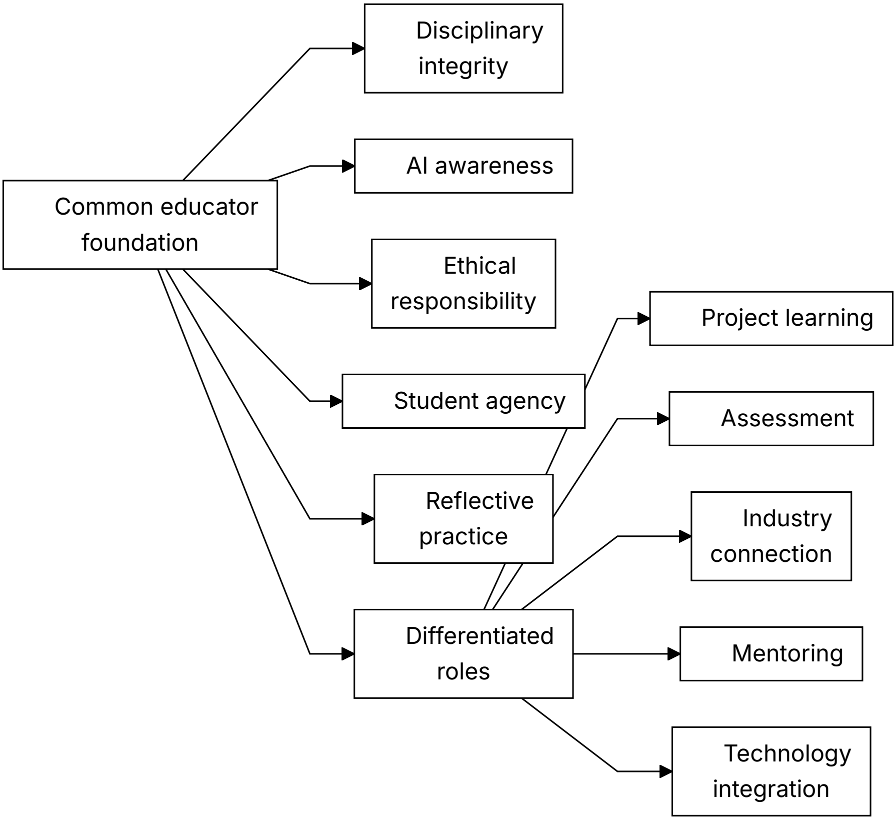

Figure 2. Common foundation plus diverse educator profiles.

Recommendation

Define a common AI-era educator foundation, then design differentiated roles within schools, departments, and programs. The future teacher or lecturer should not be alone. The educator should be part of a coordinated ecosystem.

The recommended model has three levels:

1. A common foundation for all educators: ethical responsibility, disciplinary integrity, AI awareness, student agency, and reflective practice.
2. Differentiated professional strengths: project learning, industry connection, assessment, mentoring, technology integration, foundational teaching, or interdisciplinary design.

3. Institutional coordination: graduate profiles, team teaching, shared planning, professional development, quality indicators, and leadership support.

This model protects equity without demanding an impossible teacher. It recognizes that excellence in AI-era STEM education is not a heroic individual trait. It is a designed professional system.

Practical Recommendation for Teachers

Stop trying to become every kind of future educator alone. Build a personal strength profile and connect it to colleagues with complementary strengths.

In practice:

1. Identify your current strongest contribution: foundations, project work, mentoring, assessment, AI use, industry connection, or interdisciplinary design.
2. Identify one gap that matters for your students this year.
3. Find a colleague, teaching center, or external partner who is stronger in that area.
4. Co-design one activity or assessment together.
5. Make the collaboration visible to students, so they see professional teamwork modeled.

Classroom activity examples:

1. Expert roles project. Students work in teams with differentiated roles such as concept lead, AI checker, evidence lead, ethics lead, and presentation lead. The class discusses why different strengths improved the outcome.
2. Teacher profile reflection. After a learning activity, students identify which teacher role mattered most: content expert, coach, assessor, AI guide, mentor, or industry connector.
3. Complementary teaching moment. Invite a colleague or external expert for one part of a lesson and explicitly show students how different professional profiles complement one another.

Evaluation method:

Evaluate success by checking whether students can explain how different roles contribute to learning and quality. In team work, assess both the shared product and each student's role evidence. A successful student understands that expertise can be distributed while responsibility remains clear.

For your own development, create a two-layer growth plan: maintain the common foundation required of every educator, then develop one specialized strength deeply. This reflects the Round Table's balanced answer to the one-profile-or-many dilemma: common responsibility plus differentiated expertise inside a coordinated ecosystem.

SOURCES

Meeting 2 presentation, TIES Jan Morrison presentation, Dr. Gabi Shafat STEM material, OECD Teaching Compass.

CHAPTER 5

Teaching as a Team Profession

The Challenge

Teaching is often organized as individual work. One teacher, one classroom, one course, one syllabus. But the competencies discussed by the Round Table, especially teamwork, creativity, problem solving, communication, industry relevance, and real-world learning, are difficult to cultivate through isolated practice.

This creates a contradiction. Education systems ask students to become collaborative, interdisciplinary, adaptive, and connected to real problems, but they often place teachers in structures that are solitary, fragmented, and disconnected from the world of practice. A teacher may be asked to cultivate teamwork while having little time to work in a team. A lecturer may be asked to connect learning to industry while having no structured connection to industry partners. A school may be asked to develop creativity while preserving schedules, spaces, and assessments designed for individual content delivery.

Jan Morrison's TIES presentation made this contradiction explicit. Future STEM teaching, in her framing, requires transdisciplinary instruction that is team-based and connected to real-life problems and issues. It also requires career-connected learning, externships, diverse recruitment, and cross-sector collaboration. Teaching must become a professional system, not a private act performed behind a classroom door.

The Dilemma

Should teachers remain primarily individual classroom actors, or should teaching become a team-based, cross-sector profession?

The dilemma is not whether individual teacher expertise matters. It does. Students still need teachers and lecturers who know their subjects, care about learning, and can build trust. The question is whether individual expertise is sufficient for AI-era STEM education.

If the future of STEM learning requires transdisciplinary problems, project-based learning, AI use, industry context, teamwork, ethical reasoning, and complex assessment, then the profession must decide whether one teacher should be expected to carry all of that alone.

Main Positions

The individual teacher model preserves autonomy and simplicity. The team-based model distributes expertise and models collaboration. The ecosystem model adds industry, mentors, communities, technology specialists, and institutional support.

The individual model sees teaching as a craft practiced by one educator with one group of students. It values responsibility, continuity, classroom intimacy, and professional autonomy.

The team model sees teaching as shared professional design. Teachers plan together, divide strengths, observe one another, co-assess, and build learning sequences that no one teacher could design alone.

The cross-sector ecosystem model connects educators with industry experts, community partners, university mentors, technology specialists, assessment professionals, and policy structures. This model does not remove the teacher. It surrounds the teacher with capacity.

Pros and Cons

The individual model is familiar. It is easy to schedule and govern. But it overloads teachers and limits the kinds of learning they can support.

The team model allows teachers to share expertise. It also models the collaborative behavior students need to learn. But it requires time, trust, scheduling, leadership, and recognition.

The ecosystem model connects school and higher education to real practice. But it requires governance. Without clear educational purpose, partnerships can become superficial.

Jan Morrison's criticism of failed transdisciplinary instruction is important here. Problem-based and transdisciplinary learning are often criticized when they do not work. But the failure may not be in the pedagogy itself. It may be in the system around it. Without time for faculty collaboration, cross-sector training, and structured teamwork, such approaches are likely to fail and then be dismissed as ineffective.

The Egyptian STEM schools example, as reported in the meeting summary, illustrates the opposite condition. When the educational structure itself is intentionally redesigned to support transdisciplinary teaching, industry and university involvement, and long-term iteration, the approach can become sustainable. The lesson is not that every country should copy one model. The lesson is that team-based STEM learning needs system design.

Point of View Map

From the TIES perspective, represented by Jan Morrison, future STEM teaching must be transdisciplinary, team-based, career-connected, and supported by systems. Industry can provide context, real problems, mentors, and externship opportunities.

From a teacher perspective, teamwork cannot simply be demanded. It must be supported by time, structures, and professional norms. Teachers need planning time, shared goals, protocols for collaboration, and recognition for work that happens outside direct classroom instruction.

From a policy perspective, collaboration must be funded and recognized.

From a student perspective, teamwork must be explicitly taught. Dr. Revital Duek asked how teamwork can be established as a foundational competency in teacher development and then in student learning. Jan Morrison responded by drawing on medicine, where team-based practice is non-negotiable and "learning to team" is taught intentionally during professional formation. STEM education can learn from that model.

From an industry perspective, team-based learning reflects how complex work actually happens. Dr. Avigdor Zonnenshain emphasized that education should reflect industry realities, especially in engineering, where teamwork is central. This does not mean importing industry uncritically into education. It means recognizing that isolated individual performance is not enough preparation for complex professional life.

From a recruitment perspective, Morrison's "interstate road" model matters. If teaching remains a narrow pathway in which people enter once and remain until burnout, the profession loses potential talent. A more flexible system can include industry workers, post-military professionals, nonprofit experts, and others who bring complementary strengths into education.

Round Table Voices

Jan Morrison described teaching as a profession that should connect to industry and be treated with a status beyond a calling or service role. She emphasized externships, flexible teacher pathways, systemic redesign, and international learning.

Dr. Revital Duek raised teamwork as a foundational competency. If students must learn teamwork, teachers must experience and model it.

Dr. Avigdor Zonnenshain summarized the importance of education that reflects industry realities, especially in engineering, where teamwork is central.

Danielle Eisenberg's question to Jan Morrison about whether change was happening only in innovative pockets or also in large traditional systems sharpened the implementation issue. Morrison's answer, pointing to system-level developments in districts such as Chicago and Anaheim and higher-education structures such as UTeach, suggested that team-based redesign is not only a boutique innovation. It can enter mainstream systems when trust, measurement, and political conditions support it.

Prof. Sigal Tifferet raised criticism of transdisciplinary instruction, especially the concern that problem-based learning may reduce explicit instruction. This is a serious point. Team-based and transdisciplinary approaches should not become excuses for weak teaching. They must protect disciplinary clarity while expanding the context of learning.

Jan Morrison's response was that criticism often reflects poor implementation in systems not designed for the pedagogy. This distinction is central. A good idea can fail when inserted into the wrong structure.

Synthesis

Teamwork is both an educational goal and a professional condition for achieving that goal. Students cannot learn modern STEM teamwork from systems that isolate teachers.

A team profession has at least four dimensions:

1. Team teaching: educators plan, teach, observe, and assess together.
2. Cross-disciplinary design: STEM problems are built across subject boundaries where appropriate.
3. Cross-sector partnership: industry, universities, communities, and mentors contribute authentic contexts.
4. Professional mobility: people can enter, leave, and re-enter teaching through flexible pathways without weakening standards.

This model also protects teachers. When the individual teacher is expected to be content expert, AI expert, project designer, mentor, assessor, social worker, industry connector, and innovation leader, burnout is predictable. Team structures distribute load and expertise.

Recommendation

Treat teaching as a team profession. Build time, shared planning, joint assessment, cross-disciplinary projects, and partnerships into the structure of schools and higher education.

A serious implementation model should include:

1. Scheduled collaboration time for educators.
2. Team-based project design and assessment.
3. Teacher externships or structured contact with industry.
4. Flexible professional pathways that bring diverse expertise into teaching.
5. Cross-sector advisory groups that meet regularly, not only at conferences.
6. Measurement used for improvement and iteration, not only accountability.
7. Explicit teaching of teamwork to students, beginning before advanced STEM specialization.

The recommendation is not to erase the individual teacher. It is to stop leaving the individual teacher alone with a system-level task.

Practical Recommendation for Teachers

Practice team teaching even before the institution formally redesigns around teams. Start small, but make the collaboration real.

In practice:

1. Select one complex topic that benefits from more than one perspective.
2. Invite a colleague, industry professional, advanced student, or community partner to help shape the task.
3. Ask students to work in teams with explicit roles, not only informal groups.
4. Assess both the product and the collaboration process.
5. Hold a short team debrief: what did each person contribute, where did coordination fail, and what should improve next time?

Classroom activity examples:

1. Cross-discipline design sprint. Pair two subjects, such as physics and design or biology and ethics. Students solve one problem that requires both perspectives and present how the disciplines changed each other.
2. Teamwork rubric lab. Students complete a short team task, then use a rubric to evaluate roles, communication, conflict, and contribution before discussing the technical result.

3. Industry-style standup. During a project week, students hold brief team standups: what was done, what is blocked, what help is needed, and what will be tested next.

Evaluation method:

Evaluate success through the quality of collaboration as well as the quality of the solution. Evidence should include role clarity, contribution records, response to feedback, and the team's ability to combine disciplinary perspectives. Individual assessment should include what each student added to the collective result.

For your own development, experience professional teamwork yourself. Join or create a small educator design group that meets regularly to build, test, and improve learning tasks. This follows Jan Morrison's TIES argument that career-connected, transdisciplinary learning requires a system shift, not isolated heroic teachers.

SOURCES

TIES Jan Morrison presentation, Meeting 2 presentation.

CHAPTER 6

The Engineering Education Case

The Challenge

Engineering education is a powerful test case for the Round Table's ideas. It requires rigorous foundations, but it also promises graduates who can design, solve, collaborate, innovate, and work with real systems. Traditional course structures do not always deliver that combination.

Dr. Gabi Shafat's presentation, "The Challenge in Engineering Higher Education," made this case concrete. Afeka Academic College of Engineering works with graduate profiles, educational models, and the Afeka learning experience across undergraduate and graduate programs. The challenge he presented was not a rejection of engineering rigor. It was a warning that the traditional lecturer model is no longer sufficient for the changing engineering landscape.

The pressure comes from several directions at once. Engineering knowledge is expanding. AI tools can explain, compute, summarize, and write code. Students expect more personalized learning. Industry expects graduates who can work in teams and solve real problems. Faculty workload is increasing. Traditional courses in mathematics, physics, and engineering foundations often remain poorly aligned with applied engineering practice. The result is role overload for engineering faculty and a gap between what engineering programs promise and what course structures can deliver.

The Dilemma

Should engineering programs mainly transmit disciplinary foundations, or redesign around graduate profiles and competency-based learning?

The dilemma is especially sharp in engineering because foundations are not optional. Mathematics, physics, signal processing, circuits, programming, and systems thinking are necessary. A weak foundation damages later learning.

At the same time, engineering graduates are not hired only to reproduce foundations. They are expected to design under constraints, evaluate tradeoffs, work with uncertainty, collaborate, communicate, use tools responsibly, and continue learning. AI raises the pressure further because some routine technical tasks can now be performed quickly by tools that may also be confidently wrong.

The question is not whether to keep foundations or move to projects. The question is how to connect foundations, projects, AI, assessment, industry, and graduate profiles into one coherent educational design.

Main Positions

The traditional-sequence position protects mathematics, physics, and engineering fundamentals. The graduate-profile position starts from the desired graduate and works backward. The hybrid position protects foundations while connecting them earlier to practice, projects, AI, and industry.

The Afeka model, as presented in the meeting, belongs to the hybrid and graduate-profile family. It shifts the frame from “teaching engineering” to “engineering education.” This shift matters. Teaching engineering can mean delivering the content of separate courses. Engineering education asks what kind of graduate the program is producing and how every course contributes to that profile.

The AI position inside this model is also balanced. AI is neither rejected nor worshipped. Dr. Shafat described AI almost as a “perfect lecturer” in certain narrow ways: available at all hours, patient, fast, and confident. But the same list exposes the problem. AI can be very confident when wrong. It is not responsible when things break in production. It does not provide human interaction. Therefore, engineering education must teach students to use AI while recognizing its strengths and limitations.

Pros and Cons

The traditional sequence is clear and rigorous. Its weakness is that students may not connect early foundations to engineering practice.

The graduate-profile approach creates coherence and relevance. Its weakness is that faculty buy-in and assessment redesign are difficult.

The hybrid approach is the most practical. It keeps fundamentals but asks every stage of learning to connect to purpose.

The Afeka example shows the cost of doing this seriously. It requires adaptive and dynamic course design, lecturer availability, pre-assessment, student preparation, active tasks, formative feedback, industry visits, and careful assessment of learning outcomes. It is not a matter of adding one AI assignment to a traditional course.

There are also limits. Dr. Shafat noted that AI-supported approaches do not work effectively in very large groups without investment. Post-hoc grading alone has little educational impact. Take-home assignments lose value when students can rely heavily on AI-generated solutions. Team-based projects raise fairness questions about individual contribution. These are not reasons to abandon redesign. They are reasons to treat redesign as serious professional work.

Point of View Map

Students need motivation and connection. In the Advanced Digital Signal Processing course described in the presentation, drivers of change included teamwork, flipped classroom, personalized learning, students teaching, AI as a tool to improve learning processes, and project-based or problem-based learning. Each lecture began with a problem to solve. Students prepared presentations. Pre-class preparation included videos or instructional content. Discussion was used to bridge knowledge gaps.

Faculty need professional support and evidence. Dr. Shafat's presentation described immediate formative feedback through Delta-Plus, assessment of learning outcomes, continuous improvement, and cross-departmental discussions about the strengths and limitations of AI tools and advanced teaching methods.

Industry needs graduates who can reason, collaborate, and judge. Industry site visits and guest experts were part of the course design, not external decoration. This connects engineering learning to professional practice.

Institutions need program-level design. Prof. Arnon Bentur emphasized that programs have traditionally been collections of individual courses, while the new direction requires defining a graduate profile and working backward. Afeka's mapping of competencies across courses provides a concrete example.

Policy makers need to understand that innovation requires investment. Dr. Shafat's policy message was that education requires more investment, not less. If institutions want project-based learning, AI integration, feedback, industry links, and meaningful assessment, they must fund the conditions that make them possible.

Round Table Voices

Dr. Gabi Shafat described Afeka's shift from teaching engineering to engineering education. This shift includes graduate profiles, skill mapping, AI integration, feedback, industry visits, flipped classrooms, and project-based learning.

Avi Salmon emphasized that deep understanding and genuine engagement with the profession matter more as routine technical work becomes automated.

Prof. Arnon Bentur pointed to the importance of defining the graduate profile and designing backward.

Dr. Avigdor Zonnenshain asked what Afeka was doing to demonstrate the effectiveness of these processes. This question is crucial because innovation must not be judged only by enthusiasm. It needs evidence, feedback, and evaluation.

Dr. Gabi Shafat responded that rapid change means there are no definitive answers yet, but that Afeka conducts ongoing cycles of feedback and evaluation through cross-departmental discussions. He also identified practical lessons: take-home assignments must be reconsidered when AI can produce solutions, in-class assessment remains important, large classes constrain innovation, and foundational subjects need stronger applied integration.

Avi Salmon later sharpened the faculty-development issue: the real question is how to get professors involved. The challenge is not only to redefine curriculum content, but to enable teachers and lecturers to personally experience new pedagogical and technological realities.

Synthesis

Engineering education shows that skills cannot be added only at the edges. They must be designed into courses, projects, assessment, and institutional culture.

The Afeka case suggests a concrete model:

1. Start from the graduate profile.
2. Map knowledge and competencies across courses.
3. Redesign courses around problems, preparation, feedback, and active learning.
4. Use AI as a learning tool, but deliberately expose its limits.
5. Connect learning to industry and professional practice.
6. Assess process, products, contribution, and understanding.
7. Support faculty through team models and institutional investment.

This is not a softening of engineering education. It is a more demanding form of rigor. It asks whether students can use foundations in conditions closer to the work they will actually do.

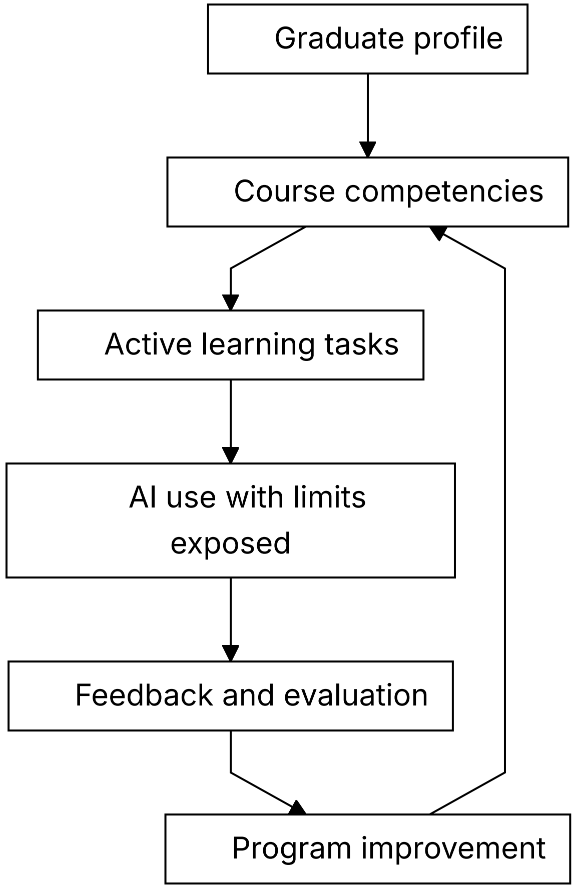

Figure 3. Engineering education as backward design from the graduate profile.

Recommendation

Use engineering education as a model for program-level redesign. Define the graduate profile, map competencies across the curriculum, connect foundations to practice, and use AI in ways that expose rather than hide student thinking.

Engineering programs should review every major course through five questions:

1. Which part of the graduate profile does this course develop?
2. Which competencies are practiced, not merely named?

3. How does the course make student thinking visible?
4. How does the course teach students to evaluate AI outputs and tool limits?
5. What evidence shows that students can transfer knowledge into real engineering judgment?

The deeper recommendation is structural. Higher education should move from the heroic individual lecturer model toward a complementary team-based model. Teaching, project-based learning, industry engagement, AI integration, assessment, and faculty development should be realigned so that the program, not only the lecturer, carries the responsibility for graduate readiness.

Practical Recommendation for Teachers

Use the engineering education case as a model even outside engineering: begin with the desired graduate or learner profile, then work backward to the lesson.

In practice:

1. Write one sentence describing what a successful learner should be able to do with the knowledge.
2. Design a task that requires application, not only recall.
3. Include a moment where students must evaluate an AI output or technical tool.
4. Add a real or realistic constraint, such as cost, safety, user need, data quality, or tradeoff.
5. Collect evidence from process, product, explanation, and reflection.

Classroom activity examples:

1. Constraint-based design task. Students design a solution under real constraints such as budget, safety, time, user need, or data quality, then explain the tradeoffs.
2. AI tool stress test. Students use AI on an engineering or STEM problem, compare its output to disciplinary principles, and document where the tool was useful or unreliable.

3. Graduate profile presentation. Students present a project by explaining which future professional capabilities it developed, not only what technical result they produced.

Evaluation method:

Evaluate success with a design-and-justification rubric. The student should show technical correctness, awareness of constraints, explanation of tradeoffs, and ability to connect the work to future professional capabilities. Strong work includes iteration, not only a final product.

For your own development, redesign one course artifact using the Afeka logic: graduate profile, course skills, active learning, AI as a learning tool, feedback, and evaluation. The aim is not to copy engineering education mechanically. It is to adopt its discipline of connecting foundations to practice and evidence.

SOURCES

Dr. Gabi Shafat STEM material, Meeting 2 presentation, TIES Jan Morrison presentation.

PART III

Must Teachers Possess the Skills They Teach?

CHAPTER 7

Possessing Versus Cultivating Competencies

The Challenge

Teachers are expected to cultivate competencies in students. But must they personally possess those competencies first? Can a teacher teach creativity without being creative? Can a lecturer cultivate teamwork without working in teams? Can an educator teach ethical AI use without deep AI experience?

The Meeting 3 discussion document framed this not as a binary question but as a continuum. Its purpose was not to settle the issue in advance, but to lay out the competencies, tensions, and arguments on each side as a basis for informed discussion.

The challenge is especially difficult because competencies are not like isolated facts. A teacher can teach a fact from a textbook with limited personal experience. But competencies such as judgment, collaboration, curiosity, ethical responsibility, and AI literacy are partly visible through practice. Students notice how educators think, respond, question, collaborate, and handle uncertainty.

At the same time, expecting every educator to be a full practitioner of every competency is unrealistic. It risks turning the future teacher profile into an impossible demand. The real question is not whether teachers must possess everything perfectly. The question is what level of lived understanding is necessary for credible cultivation.

The Dilemma

Should lecturers and teachers themselves possess the required competencies, or should their primary expertise lie in cultivating these competencies in students?

The Meeting 3 document translated this dilemma into precise subquestions:

1. What counts as sufficient mastery?
2. Can pedagogical expertise substitute for personal practice?
3. Do all competencies require the same depth of teacher experience?
4. How should limited professional development time be allocated?
5. What is the role of teacher vulnerability and not-knowing?
6. How can enough competency be assessed?
7. Does the answer change across age, discipline, institution, culture, or resource level?

These questions prevent an easy answer. The teacher's role differs by context. A primary school teacher building foundational literacy, a high school physics teacher, a university lecturer in signal processing, and an engineering project mentor do not need identical competency profiles. But all need enough understanding to protect student learning and responsibility.

Main Positions

The possession position argues that teachers need lived experience. Students learn not only from instruction but from modeling.

The cultivation position argues that teachers do not need to be expert practitioners of every competency. Their professional expertise is pedagogy.

The balanced position argues that teachers need enough lived experience to model, recognize, and judge the competency, but they do not need identical mastery in all domains.

The distributed-expertise position adds that some competencies can be held more deeply by specific members of a professional team. One educator may be strong in AI-supported learning, another in project design, another in assessment, another in industry connection, and another in student mentoring. The system must ensure that the full set of capabilities is available to learners.

The developmental position argues that teacher competency should be seen as growth, not as a static qualification. Educators can develop their own competencies while learning how to cultivate them in students.

Pros and Cons

Possession gives credibility. It allows teachers to model thinking and practice. Its risk is unrealistic expectations.

Cultivation focuses on pedagogy. It recognizes that teachers help students grow beyond the teacher's own experience. Its risk is hollow instruction if the teacher lacks any lived understanding.

The balanced approach is most workable. It requires defining levels of competence and support.

Distributed expertise reduces overload and reflects real professional life. Its risk is fragmentation if roles are not coordinated.

Developmental expectations support professional growth. Their risk is vagueness unless institutions define evidence of progress.

The professional-development tradeoff is real. Training time and resources are limited. If all time is spent building teachers' own competencies, pedagogy may suffer. If all time is spent on how to teach competencies, teachers may lack the lived experience to model them credibly. A strong system must do both, but not all at once and not at the same depth for every role.

Point of View Map

Teachers need expectations they can meet. If the profile becomes too idealized, teachers may experience it as accusation rather than professional direction.

Students need credible guidance. They must see adults who can ask good questions, use AI critically, collaborate, admit uncertainty, and take responsibility.

Teacher educators need priorities. They must decide what belongs in initial preparation, what belongs in induction, and what belongs in continuing professional development.

Policy makers need standards that support growth instead of creating impossible demands. Standards should define baseline competence, development pathways, and support structures.

Institutions need role design. If some competencies require deep specialization, then schools and departments should not pretend that every educator will hold the same level of expertise.

Industry and society need trust. If graduates are expected to work responsibly with powerful tools, the educators who prepare them must be able to recognize responsible and irresponsible use.

Round Table Voices

Dr. Eli Eisenberg, Prof. Arnon Bentur, and Avi Salmon framed this dilemma in the Meeting 3 document.

Emlen Metz warned against students delegating thinking and responsibility to machines. This implies that teachers must be able to guide responsible use.

Prof. Russell Tytler expressed concern that AI may reduce deep, sustained thinking. This strengthens the case for teachers who can model intellectual discipline.

Avi Salmon argued in Meeting 2 that teachers and lecturers must remain above AI. In this chapter, that claim becomes a competency requirement. The educator does not need to know more than every AI system. The educator does need enough judgment to keep the human process of learning, responsibility, and meaning above the machine.

Dr. Einat Sprinzak's Meeting 2 contribution also belongs here. Teachers who mediate skills and competencies should first experience these learning processes themselves as learners. This does not mean they must become perfect models. It means they need firsthand understanding of the student journey.

Dr. Gabi Shafat's Afeka material shows one way to make this practical: educators and students can use AI tools on challenging problems where the tools fail or show bias, then discuss the limits and produce a better solution. The teacher's role is not to be an all-knowing AI expert, but to design and guide the learning process responsibly.

Synthesis

Teachers do not need to be perfect examples of every competency. But they do need enough lived understanding to model learning, ask good questions, recognize quality, and guide students responsibly.

The Round Table materials support a levels model:

1. Awareness: every educator understands the competency and why it matters.
2. Personal practice: every educator has some lived experience using the competency.
3. Pedagogical design: educators can design learning situations that cultivate the competency.
4. Assessment judgment: educators can recognize evidence of student growth.
5. Specialist leadership: some educators develop deeper expertise and support colleagues.

This levels model avoids two mistakes. It avoids saying that teachers can teach competencies they have never practiced. It also avoids saying that every teacher must master every competency equally.

Recommendation

Define competency expectations in levels. All educators need baseline awareness and responsible practice. Some need deeper competence because of their discipline or role. Specialist expertise should be distributed across teams and institutions.

Professional development should be designed in two linked tracks:

1. Educator experience: teachers and lecturers personally practice AI use, critical questioning, teamwork, reflection, and ethical judgment.
2. Pedagogical translation: educators learn how to design tasks, feedback, assessment, and classroom routines that cultivate these competencies in students.

The recommendation is balanced: teachers must possess enough to model and judge, but they do not need to carry the whole future alone.

Practical Recommendation for Teachers

Build enough personal practice to model and judge the competency you want students to develop. Do not teach a competency only as vocabulary.

In practice:

1. Choose one competency you expect from students.
2. Practice it yourself in a real professional context.
3. Show students one example of your process, including difficulty or revision.
4. Design a student task where the competency is necessary, not decorative.
5. Use a simple rubric that separates awareness, responsible use, teaching translation, assessment judgment, and specialist leadership.

Classroom activity examples:

1. Teacher modeling session. The teacher demonstrates a real thinking process, including a mistake, revision, or uncertainty, then students identify the competency being modeled.
2. Competency transfer task. Students use one competency, such as critical questioning, in two different contexts and compare what changed.
3. Enough mastery discussion. Students examine a scenario where a teacher or expert guides a competency without being the top specialist, then discuss what level of mastery is enough to guide others responsibly.

Evaluation method:

Evaluate success by looking for student awareness of the competency and evidence of use in more than one setting. A student succeeds when they can name the competency, apply it, explain how it helped the work, and recognize what level of guidance or expertise was needed.

For your own development, choose one competency per semester for deliberate growth. For example, if the focus is ethical AI use, use AI yourself, test its limits, read one relevant framework, design one classroom routine, and discuss results with colleagues. This follows the Meeting 3 balance: teachers need enough lived practice to model and judge, while specialist depth can be distributed across teams.

SOURCES

Meeting 3 discussion document, Meeting 2 presentation.

CHAPTER 8

Teacher Vulnerability, Authority, and Not-Knowing

The Challenge

AI destabilizes traditional authority. A student may use a tool the teacher has not yet mastered. A system may generate an answer faster than the teacher can. Knowledge changes quickly. The teacher can no longer rely only on being the person who knows.

The Meeting 4 discussion document described the classical educational relationship as a teacher who knows and a student who learns. The teacher decides what to present, how to present it, when the student is ready, and whether understanding has been achieved. AI challenges this model because it can explain, tutor, quiz, adapt, and respond continuously. It does not replace the teacher, but it enters the space between teacher and student.

This creates a new professional pressure. If a student can get an instant answer, the teacher's authority cannot rest only on speed, access, or factual possession. The teacher's authority must rest on judgment, context, trust, care, interpretation, and the ability to guide inquiry.

The Dilemma

Should teachers preserve confident authority, or model uncertainty, inquiry, and learning in real time?

The Meeting 3 document named the question directly: what is the role of teacher vulnerability and not-knowing? Should teachers model uncertainty and learning in real time, or do students need confident authority?

This is not a soft psychological issue. It is a core pedagogical question. Too much certainty can become false confidence. Too much vulnerability can become loss of structure. The AI era requires a new balance: professional authority that is confident enough to lead and humble enough to investigate.

Main Positions

One position says students need confident authority. Without it, classrooms lose structure.

Another position says students need to see how experts learn, revise, verify, and admit uncertainty.

A third position is guided authority. Teachers remain responsible leaders, but they model inquiry and judgment openly.

A fourth position, grounded in cognitive apprenticeship, says that expert thinking should be made visible. Students learn not only from final answers, but from seeing how an expert frames a problem, tests a claim, notices uncertainty, checks evidence, and revises a conclusion.

Pros and Cons

Traditional authority gives clarity and trust. Its weakness is brittleness. If authority depends on always knowing, it fails in fast-changing contexts.

Vulnerability models lifelong learning. Its weakness is that it can confuse students if not framed with care.

Guided authority is strongest. The teacher does not pretend to know everything. The teacher demonstrates how responsible learners handle what they do not know.

Cognitive apprenticeship adds a practical method. The teacher can model thinking aloud, coach students through uncertainty, scaffold the task, and gradually transfer responsibility. In an AI-rich classroom, this may mean comparing an AI explanation with a textbook, asking students to identify unsupported claims, showing how to verify a result, or explaining why a fluent answer is still not trustworthy.

The danger is performance vulnerability. A teacher saying “I do not know” without showing the next professional move may weaken trust. A teacher saying “I do not know yet, so here is how we will find out” strengthens trust.

Point of View Map

Younger students may need stronger structure. Older students can benefit from explicit modeling of uncertainty. In AI-rich classrooms, all students need to see how claims are checked, compared, and judged.

From a teacher perspective, authority becomes interpretive rather than merely declarative. The teacher does not only state what is correct. The teacher helps students understand why, under what conditions, with what evidence, and with what limits.

From a student perspective, visible teacher inquiry can build self-efficacy. Students learn that not knowing is not failure if it leads to disciplined investigation.

From a policy perspective, teacher vulnerability must not be confused with lowered standards. Professional learning systems should help teachers develop confidence in uncertainty, not abandon expertise.

From the OECD Teaching Compass perspective, teachers need agency to navigate complexity and uncertainty. The Compass describes teachers as agents of change who adapt, innovate, and act with professional judgment. That is exactly the authority needed here.

From the AI design perspective, the Meeting 4 and Danielle Eisenberg materials point to the teacher as the interpretive layer. AI can propose, surface evidence, or generate signals. The teacher who knows the student and the context decides.

Round Table Voices

The Meeting 3 document explicitly asked about the role of teacher vulnerability and not-knowing.

Avi Salmon argued that teachers and lecturers must remain above AI. That does not mean they must know more facts than every tool. It means they must lead the human process of judgment.

Emlen Metz emphasized that decision-making and responsibility cannot be delegated to machines.

The Meeting 4 discussion document asked whether AI enhances or undermines teacher authority. It suggested that the teacher's role may evolve from knowledge authority toward a guide who teaches judgment, context, and critical evaluation of AI-generated content.

Danielle Eisenberg's presentation added a design formulation: AI proposes, teachers decide. AI may surface skill evidence, but the teacher who knows the student remains the interpretive layer.

The OECD Teaching Compass reinforces this stance through teacher agency, professional identity, ethical values, and the capacity to act under uncertainty.

Synthesis

Authority in the AI era shifts from having all answers to leading the process of responsible inquiry.

Professional authority now has four parts:

1. Knowledge authority: the educator understands the field and its standards.
2. Process authority: the educator knows how to investigate and verify.
3. Ethical authority: the educator keeps responsibility human.
4. Relational authority: the educator knows the student, the context, and the learning process.

AI can challenge the first if authority is understood narrowly as information possession. It cannot replace the other three. That is where the teacher's future authority becomes stronger, not weaker.

Recommendation

Adopt transparent professional authority. Teachers should be able to say, "I do not know yet," while showing students how to investigate, evaluate, decide, and take responsibility.

In practice, this means teachers should model professional inquiry in front of students:

1. State what is known.
2. Name what is uncertain.
3. Check sources and assumptions.
4. Compare human reasoning with AI output.
5. Decide what can be trusted.
6. Explain why the final judgment remains human.

The teacher above AI is not the teacher who always knows first. It is the teacher who knows how responsible knowing is done.

Practical Recommendation for Teachers

Model disciplined not-knowing. Do not hide uncertainty, and do not perform uncertainty without structure. Show students the professional moves that follow uncertainty.

In practice:

1. When a question is uncertain, name what is known and what is not yet known.
2. Ask students to propose how the class should verify the issue.
3. Compare at least two sources or explanations, including AI when relevant.
4. Explain why one answer is more trustworthy than another.
5. End with a clear human judgment and the evidence behind it.

Classroom activity examples:

1. I do not know yet protocol. When a difficult question arises, the class records what is known, what is uncertain, how to investigate, and what evidence would settle the issue.
2. Competing explanations. Students compare a textbook explanation, an AI explanation, and a peer explanation, then decide which is strongest and why.
3. Trust ladder. Students rank sources or answers from least to most trustworthy and explain what evidence would move each answer up or down.

Evaluation method:

Evaluate success through epistemic responsibility: how students handle uncertainty, evidence, and source trust. The student should be able to say what is known, what is unknown, what evidence is needed, and why one explanation is stronger than another. Reward careful reasoning even when the answer is not immediate.

For your own development, rehearse transparent authority. Before class, identify one place in the lesson where uncertainty, competing explanations, or AI output could be examined openly. This develops the authority described in the Round Table: not authority based only on having answers, but authority based on judgment, process, ethics, and relationship.

SOURCES

Meeting 3 discussion document, Meeting 2 presentation.

CHAPTER 9

Assessing Enough Competency

The Challenge

If teachers are expected to possess or cultivate competencies, systems need ways to judge readiness. But competencies such as ethical judgment, critical thinking, teamwork, reflection, AI literacy, and professional judgment cannot be reduced to simple checklists.

The assessment challenge appears at three levels at once. First, students must be assessed for competencies, not only final answers. Second, teachers and lecturers must be assessed or supported in their ability to cultivate those competencies. Third, institutions and programs must be evaluated to see whether their reforms actually work.

The Round Table materials repeatedly warn against confusing measurement with learning. Assessment must make practice visible. It must support improvement. It must be credible enough for policy and institutions, but humane enough not to punish educators for engaging in complex change.

The Dilemma

Can these competencies be assessed meaningfully for teachers and lecturers?

The Meeting 3 document asked: if possession matters, what does evaluation look like for something personal and contextual? This question is sharper than it first appears. A teacher's ethical judgment, AI literacy, ability to model uncertainty, or capacity to cultivate teamwork cannot be captured by a single written test.

At the same time, avoiding assessment entirely is not acceptable. If systems cannot tell whether teachers, courses, or programs are developing competencies, reforms remain aspirational. The dilemma is how to assess deeply without reducing complex professional practice to shallow indicators.

Main Positions

Formal standards create clarity. Portfolios capture practice. Peer review brings professional judgment. Support-first approaches encourage growth. Each has value and each has limits.

The standards position says that systems need explicit expectations. Teachers and lecturers should know what is required. Institutions and policy makers need benchmarks.

The portfolio position says that competencies require evidence over time: lesson designs, student artifacts, reflections, feedback cycles, AI-use policies, assessment rubrics, peer observations, and examples of changed practice.

The peer-review position says that professional judgment should be evaluated by professionals, not only by external metrics. Colleagues can observe teaching, review course design, and discuss evidence.

The program-evaluation position, raised strongly by Avi Salmon, says that reforms themselves should be evaluated professionally and with data. A program should not be declared successful because it sounds modern.

The support-first position says that the purpose of assessment should be growth. Especially in a period of rapid change, teachers need accompaniment and feedback, not only accountability.

Pros and Cons

Formal standards support accountability but can become bureaucratic. Portfolios show evidence but require time. Peer review is contextual but can be inconsistent. Support-first models reduce fear but may lack accountability.

Program evaluation can reveal whether reforms work, but it requires careful design. Bad metrics may reward superficial compliance. Good metrics must look at student learning processes, teacher practice, institutional support, and long-term outcomes.

AI-supported assessment can help surface patterns, but it introduces bias, privacy, trust, and interpretation problems. The Meeting 4 materials are relevant here: AI can propose analysis and feedback, but the teacher who knows the student and context must remain the interpretive layer.

The deepest risk is that assessment becomes another burden on teachers without improving learning. The Meeting 5 steering-team materials warned that professional development cannot be one-time training. It requires implementation coaching, implementation support, institutional infrastructure, resources, evaluation, and policy backing. The same is true of assessment.

Point of View Map

Teachers need assessment that helps them improve. Evidence should point to next steps: what to redesign, what to practice, what support is needed, what student evidence shows.

Institutions need quality assurance. They must know whether competency-based learning is actually happening across courses and programs.

Policy makers need mechanisms that can scale. But scalable mechanisms must not flatten practice into simplistic compliance.

Students need confidence that educators can guide them responsibly. If AI is part of learning, students need teachers who can interpret evidence, detect shallow work, guide ethical use, and evaluate process.

Assessment bodies need new tools. Dr. Marwa Maklada's comments about validating competency-based tasks in matriculation processes show that formal systems can evolve, but they must be careful and evidence-based.

Higher education needs course-level and program-level evidence. Dr. Gabi Shafat's Afeka material included pre-assessment questionnaires, assessment of learning outcomes, immediate formative feedback, class discussions, evaluation of AI-supported work, and cross-departmental review.

Round Table Voices

Dr. Gabi Shafat described assessing students' ability to think critically around AI, including cases where AI tools fail or produce biased output.

Avi Salmon argued for professional, data-based evaluation of programs.

Dr. Marwa Maklada described work on validating competency-based tasks and assessment processes.

The Meeting 5 steering-team materials add that there is no single actor with clear responsibility for sustained professional development. This matters for assessment because no assessment system can work if no one owns the follow-up. Evidence without support becomes judgment without improvement.

Danielle Eisenberg's Meeting 4 material argued that traditional assessment measures outputs and cannot detect shallow AI use, equity gaps, or whether AI-supported learning is actually producing understanding. It proposed visibility into process, not only products.

Dr. Tzur Karelitz's question in Meeting 4, asking why the teacher should mediate between AI and students in evaluation, sharpened the trust issue. The answer from the discussion was not that AI has no role. It was that AI evidence still needs human interpretation because students, contexts, bias, and relationships matter.

Synthesis

Assessment should be formative, evidence-based, and tied to practice. It should not become a single test of teacher worth.

Four assessment objects must remain distinct:

1. Student competency: evidence of reasoning, collaboration, reflection, AI critique, and responsibility.
2. Teacher practice: evidence that the educator can design, guide, assess, and improve competency-rich learning.
3. Course design: evidence that competencies are embedded in tasks, feedback, and assessment.
4. Program or system effectiveness: evidence that reforms improve learning over time and are supported institutionally.

Each object requires different evidence. A student portfolio is not the same as a teacher portfolio. Course redesign evidence is not the same as program evaluation. Confusing these layers creates weak assessment.

Kind of evidence	What the student produces	What it can show	What it cannot show
Product	Final answer, report, artefact, design, code	Quality of the outcome against a standard	Whether the student, or a tool, did the thinking
Process	Drafts, prompts, verification steps, rejected suggestions, team logs	Reasoning, AI use, collaboration, and where judgment was applied	Whether the student can transfer the competency elsewhere
Reflection	Written or spoken account of what changed and why	Growth, ownership, responsibility, and transfer	Technical accuracy on its own

Figure 4. Three kinds of assessment evidence for competency-rich learning.

Recommendation

Use layered evidence: reflective portfolios, peer observation, student artifacts, course redesign evidence, professional learning records, and institutional data.

A credible assessment model should include:

1. Formative evidence: feedback cycles, reflections, drafts, revisions, and student explanations.
2. Performance evidence: presentations, projects, problem-solving tasks, oral defense, and team contribution.
3. AI-use evidence: prompts, tool outputs, verification steps, rejected suggestions, and student justification.
4. Teacher-practice evidence: lesson designs, assessment rubrics, peer observation, and reflective analysis.
5. Program evidence: competency mapping, student progression data, graduate readiness indicators, and external review.
6. Support evidence: professional development participation, implementation coaching, and institutional resources.

The principle is simple: assess what the system claims to value. If the system values judgment, collaboration, responsibility, and learning with AI, assessment must make those processes visible.

Practical Recommendation for Teachers

Use assessment as a learning mirror. Do not assess only the finished answer. Assess the evidence that shows whether the student actually developed the competency.

In practice:

1. Add a small portfolio element to major tasks: draft, feedback, revision, reflection, and final product.
2. Ask students to explain their contribution in team work.
3. Require students to document AI use, verification, and rejected suggestions.
4. Include one oral or written defense of reasoning.
5. Review student evidence with a colleague once per unit or semester.

Classroom activity examples:

1. Mini portfolio task. Students submit a final answer with one draft, one feedback note, one revision, and one reflection explaining how their thinking changed.
2. Oral defense circle. Small groups present a solution and answer peer questions about reasoning, AI use, evidence, and responsibility.
3. Team contribution evidence. In a group project, each student submits one artifact that proves their contribution and one reflection on how the team improved the result.

Evaluation method:

Evaluate success through a portfolio and short defense, not only a final score. The student should show growth, revision, evidence of contribution, and ability to explain decisions under questioning. A successful portfolio contains artifacts from the process and a reflection that connects effort to improved understanding.

For your own development, build a teacher-practice portfolio as well. Save one redesigned task, one rubric, one student evidence sample, one reflection, and one improvement decision. This follows Dr. Gabi Shafat's course-evaluation approach, Danielle Eisenberg's evidence-design logic, Avi Salmon's call for data-based evaluation, and Dr. Tzur Karelitz's concern that AI evidence needs human interpretation.

SOURCES

Meeting 2 presentation, Meeting 3 discussion document, Meeting 4 discussion document, Meeting 4 presentation, Dr. Gabi Shafat STEM material.

PART IV

The Teacher, the Student, and AI

CHAPTER 10

AI Enters the Triangle

The Challenge

AI has entered the learning relationship. It is no longer outside the classroom. Students can use it as tutor, explainer, editor, evaluator, simulator, translator, and collaborator.

The Meeting 4 discussion document, prepared by Avi Salmon, Dr. Eli Eisenberg, and Prof. Arnon Bentur, stated the challenge clearly. For centuries, education rested on a direct relationship between teacher and student: a teacher who knows and a student who learns. The teacher decided what to present, how to present it, when the student was ready, and whether understanding had been achieved. AI now challenges this model at its core because it can explain, tutor, quiz, adapt, and respond continuously, patiently, and at scale.

The document made an important distinction. AI does not simply replace the teacher. It inserts itself into the space between teacher and student in ways that cannot be ignored. This makes AI different from a textbook, calculator, or static digital resource. It can engage the student in dialogue, respond to questions, assess understanding, and adapt in real time.

The challenge is therefore structural. Education must decide whether AI remains an optional tool selected by the teacher, becomes a recognized participant in a teacher-student-AI triangle, or becomes a primary student learning partner with the teacher serving as mentor, validator, or safety net.

The Dilemma

Should AI be treated as a tool controlled by the teacher, or as a third actor in a redesigned teacher-student-AI triangle?

The dilemma is not only about technology adoption. It asks whether the structure of education itself should change. If AI can teach, quiz, explain, and respond, then the old didactic relation among teacher, student, and subject matter must be reconsidered.

The Meeting 4 document placed one principle above all configurations: the teacher's professional judgment must remain the final authority over the educational process. The question is not whether to eliminate the teacher. The question is how much of the instructional relationship should be shared with, or mediated by, AI.

Main Positions

The tool model treats AI as an aid selected by the teacher. The student-AI direct model recognizes independent student use. The educational triangle model defines roles for teacher, student, and AI.

The teacher-led model keeps the teacher as the sole orchestrator of learning. AI is a tool the teacher may choose to deploy. The educational relationship remains teacher-student, and the teacher decides if, when, and how AI enters the process.

The triangle model recognizes AI as a co-participant in the educational process. The student interacts directly with AI as a learning partner, while the teacher designs, moderates, and oversees the dynamic among teacher, student, and AI.

The student-AI direct model recognizes that students may engage AI independently as a primary learning resource. In this configuration, learning becomes more student-driven and AI becomes an always-available guide, while the teacher serves as mentor, validator, safety net, and interpreter.

The Round Table did not treat these models as slogans. It used them to ask hard questions about authority, mediation, relationships, equity, assessment, accountability, and context.

Pros and Cons

The tool model is safer and easier to govern, but ignores much student use. The direct model supports independence, but risks shallow learning and hidden dependency. The triangle model is more realistic, but requires new pedagogy and assessment.

The teacher-led model protects accountability and gives teachers control. Its weakness is that it may not match reality. Students already use AI outside formal teacher control, and pretending otherwise leaves educators blind to actual learning processes.

The student-AI direct model supports agency, personalization, and access. It carries the promise of something like personal tutoring at scale. But it also risks cognitive offloading, misinformation, social isolation, shallow understanding, and unequal use.

The triangle model is the most realistic and most demanding. It recognizes that AI is already active, but insists that teacher judgment, student responsibility, and institutional design must shape its role. It requires teachers to understand AI, students to disclose and reflect on use, and institutions to redesign assessment and policy.

Point of View Map

Teachers need role clarity. If AI can explain and assess, teachers must know what remains uniquely human: judgment, relationships, motivation, values, context, and the design of meaningful learning.

Students need agency and guardrails. They should learn to use AI as part of modern life, but not in ways that replace thinking, curiosity, effort, or responsibility.

Institutions need policy. They must define acceptable use, disclosure expectations, assessment redesign, data privacy, tool procurement, teacher preparation, and accountability.

Families and society need trust. AI-supported learning raises questions about who is responsible when AI gives wrong, biased, or harmful guidance.

Equity must be central. The Meeting 4 document asked whether AI will equalize learning or deepen inequality. AI could offer personal tutoring to students who lack support. It could also create a new gap in which advantaged students use AI to augment thinking while disadvantaged students use it to replace thinking.

Discipline and age matter. A primary school teacher shaping literacy, a high school STEM teacher, and a university lecturer running an advanced seminar may need different AI relationships. The triangle should be locally designed, not imposed as one uniform model.

Round Table Voices

Dr. Eli Eisenberg framed the Meeting 4 dilemma around whether the teacher alone should lead the process or whether AI becomes part of the educational relationship.

Avi Salmon, Dr. Eli Eisenberg, and Prof. Arnon Bentur prepared the Meeting 4 discussion document that explored the new triangle.

UNESCO sources cited in the materials also refer to AI transforming the traditional teacher-student relationship into a teacher-AI-student dynamic.

Dr. Iris Pinto's Meeting 4 lecture connected the triangle to a wider futures perspective. In a volatile, uncertain, complex, and ambiguous world, education must prepare learners for multiple possible futures. Her summary presented the new model as teacher-student-AI collaboration: teachers lead pedagogy, students take active responsibility for learning, and AI supports both as a tool.

Danielle Eisenberg's presentation complicated the triangle further by arguing that AI in education is not a simple tool but a role-fluid system. It already acts as content, tutor, evaluator, and collaborator. Student-AI interactions occur whether or not schools design them. The issue is not whether AI will enter the system, but whether the system will design its role.

Avi Salmon argued during the discussion that AI is already an active participant in education. The real question is what the teacher's role should be inside the triangle. He emphasized that teachers must gain hands-on experience with AI, understand its capabilities and limits, think critically about it, and remain educational leaders.

He had set out the underlying model at Meeting 1, and it gives this book its title. There are three levels of AI knowledge. The first is the user level, where the skills that matter are not technical: constructing a real dialogue with the tool, judging outputs for truthfulness and bias, and continuing to learn faster than institutions can absorb new knowledge. The second is the practitioner level, where professionals use AI to accelerate work they already understand, as engineers do when they treat a model as a coding partner rather than a substitute for engineering judgment. Using a tool without understanding the principles beneath it does not survive contact with a hard problem. The third is genuine technical knowledge of how models are built and trained. His argument was that education depends on teachers reaching the second level: able to generate material with AI, able to stay above it, and therefore able to mentor students through it. That cannot be delegated, because no one can teach an experience they have not had.

Dr. Marwa Maklada stated that AI should not design learning or control knowledge. The teacher must remain the central architect, responsible for guiding learning, preserving deep thinking, and ensuring values-based education.

Dr. Revital Duek asked what must remain fundamentally human in the evolving model. Her question captures the chapter's central concern: the triangle is not complete until the human role is clearly defined.

Synthesis

The triangle already exists in practice. The only question is whether education systems design it intentionally.

The Round Table's materials suggest that the teacher-student-AI triangle should not be understood as three equal points. It is a structured relationship:

1. The teacher leads pedagogy, judgment, values, trust, and learning design.
2. The student takes responsibility for active learning, disclosure, reflection, and independent thought.
3. AI supports explanation, practice, feedback, simulation, diagnosis, and exploration, but does not carry moral or educational responsibility.

The triangle becomes dangerous when AI replaces thinking, hides inequity, weakens relationships, or shifts accountability away from humans. It becomes powerful when it makes learning more visible, more personalized, more reflective, and more connected to human judgment.

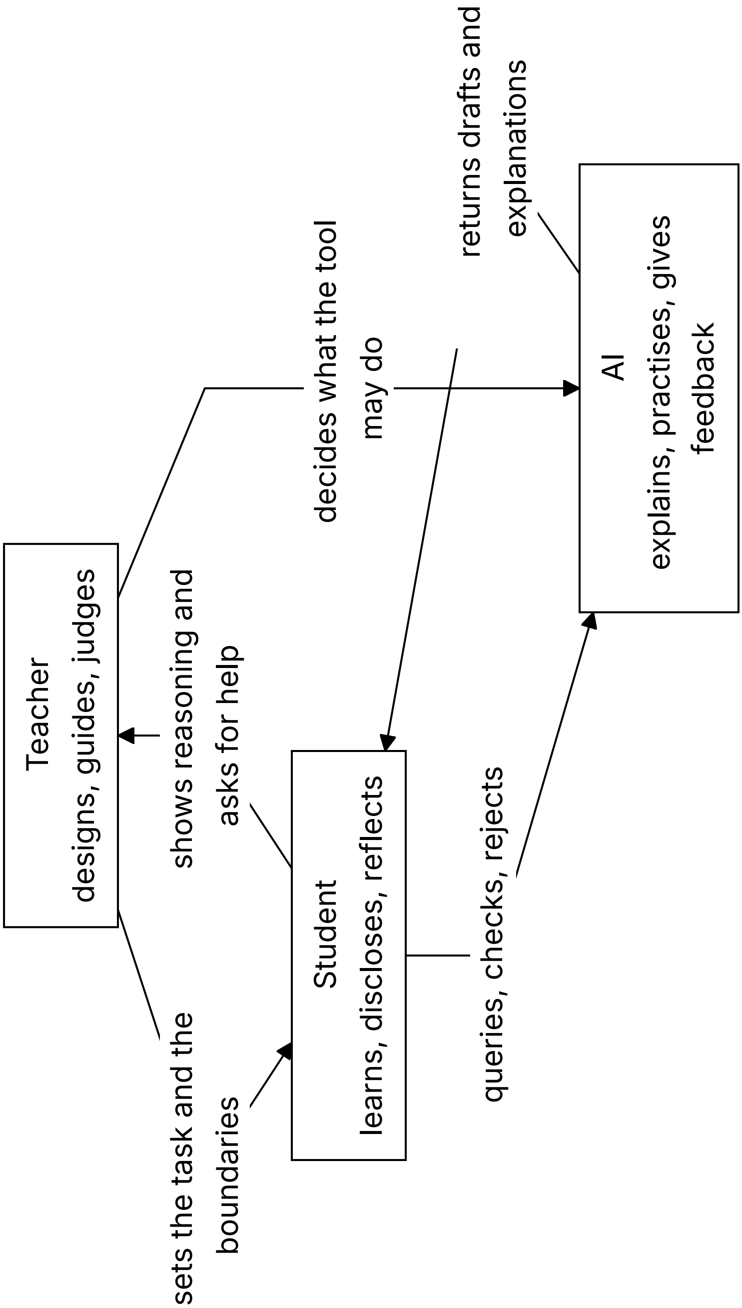

Figure 5. The teacher–student–AI triangle. Responsibility runs along the teacher–student edge; the tool is bounded on both sides.

Recommendation

Adopt the teacher-student-AI triangle as a design model. Teachers lead pedagogy, students take active responsibility, and AI serves as a powerful but bounded support.

Every institution should define its triangle model explicitly:

1. What may students use AI for?
2. What must students disclose?
3. What remains the student's own responsibility?
4. What decisions remain with the teacher or lecturer?
5. How will AI-supported learning be assessed?
6. How will equity, privacy, bias, and tool quality be handled?
7. How will teachers gain hands-on experience before being asked to lead students?

The recommendation is not to let AI into education without boundaries. It is to admit that AI is already there and design the human structure around it.

Practical Recommendation for Teachers

Create a clear teacher-student-AI classroom agreement. Students should know what AI may do, what they must do themselves, what they must disclose, and how their work will be judged.

In practice:

1. Define allowed AI uses for the topic: explanation, practice, brainstorming, feedback, simulation, or checking.
2. Define forbidden or limited uses, especially where AI would replace the student's own reasoning.
3. Require students to disclose meaningful AI use.
4. Build one activity where students compare AI help with teacher guidance or peer explanation.
5. End the activity by asking students what remained their own responsibility.

Classroom activity examples:

1. Triangle roles simulation. Assign three roles: teacher, student, and AI. Students analyze who decides the goal, who gives support, who verifies, and who is responsible.
2. AI boundary workshop. Students sort classroom AI uses into allowed, limited, and not allowed, then justify each decision using learning and integrity criteria.
3. Student responsibility contract. Before an AI-supported task, students write what AI may help with and what they must prove independently.

Evaluation method:

Evaluate success by checking whether the student understands the roles in the learning triangle. The student should know when AI is a helper, when the teacher is the designer and evaluator, and when the student must act independently. Evidence can include an AI-use contract, annotated drafts, and a short explanation of personal responsibility.

For your own development, use AI hands-on before setting policy. Test the tool on the same assignment students will face. Notice what it explains well, where it misleads, and what judgment students will need. This follows Avi Salmon's point that teachers must gain practical AI experience and Dr. Marwa Maklada's insistence that the teacher remains the architect of learning.

SOURCES

Meeting 1 summary, Meeting 4 discussion document, Meeting 4 presentation, UNESCO AI competency frameworks.

CHAPTER 11

Default or Design

The Challenge

AI use is already happening. Students use it whether institutions are ready or not. If systems do not design AI use, AI will shape learning by default.

Danielle Eisenberg's Meeting 4 presentation, "Default or Design," named this as a system architecture problem. AI is not simply a tool inserted into an existing classroom. It is a role-fluid agent that can act as content, tutor, evaluator, and collaborator at the same time. It enters several layers of the system, often without formal permission.

The challenge is that schools and universities can see that students are using AI, but they cannot yet reliably see the thinking around that use. A student may use AI to deepen understanding, test alternatives, and improve reasoning. Another student may use the same tool to avoid thinking altogether. The visible product may look similar. The learning underneath may be completely different.

This is why default integration is dangerous. Without design, AI creates hidden processes, hidden dependency, hidden inequity, and hidden erosion of student thinking.

The Dilemma

Will education systems allow AI integration to happen by default, or will they design roles, boundaries, assessment, and equity protections deliberately?

The dilemma is not between AI and no AI. That choice is already unrealistic. The real dilemma is whether AI use will be governed by educational purpose or by convenience, market pressure, student habit, and tool design.

Danielle Eisenberg framed three central questions for the discussion:

1. What does it look like for a teacher to govern the student-AI layer, not just permit or prohibit it?
2. What would it take in assessment, policy, and practice to make student thinking visible?
3. If privileged students augment with AI while disadvantaged students replace thinking with AI, who is responsible for equity, and what authority do they have?

Main Positions

Restriction tries to block use. Permission gives broad principles. Design creates structured roles and visible learning processes.

Restriction says that AI is too risky and should be blocked or minimized. This position may protect some forms of academic integrity in the short term, but it ignores actual student behavior.

Permission says that students may use AI within broad guidelines. This is more realistic, but often too weak. It can leave teachers without tools to evaluate process or prevent shallow use.

Design says that AI use must be built into the educational architecture. It defines what AI may do, what students must disclose, how teachers interpret evidence, how systems track patterns, and how equity is protected.

The design position is the only one that treats AI as a system condition rather than a classroom accessory.

Pros and Cons

Restriction may protect integrity briefly, but it is unrealistic. Permission is flexible, but often too vague. Design is harder, but it is the only durable approach.

Design is harder because it requires infrastructure. Danielle Eisenberg's presentation argued for assessment infrastructure that sees inside learning, funding the teacher's role as system designer, defining AI as a policy lever with equity guardrails, and training teachers in evidence design rather than only AI tools.

Design also requires product discipline. Human-in-the-loop should be core architecture, not compliance language. Tools should be built for the margins first. Systems should separate demonstration from confidence level.

Design requires educational courage. It forces institutions to ask what evidence of learning is defensible in an AI-rich environment. It also forces them to admit that a single artifact, such as one essay or one homework submission, is a weak signal.

Point of View Map

Students use AI unevenly. Advantaged students may use it to extend and refine thinking. Disadvantaged students may be more vulnerable to using it as a replacement for thinking. The same tool can produce opposite outcomes.

Teachers need guidance. They must learn to govern the student-AI layer, not only react to it. They need time for contextual judgment and support in interpreting evidence.

Assessment systems need redesign. Traditional assessment measures outputs. It cannot detect shallow AI use, equity gaps, or whether the triangle is producing learning. Patterns matter more than single artifacts. A notebook, reflection, peer review, oral defense, draft history, and final product together produce stronger evidence than one polished submission.

Technology designers need to build for human judgment. AI may propose skills, identify evidence, or rate confidence, but the educator should accept, override, or adjust. The teacher who knows the student and context remains the interpretive layer.

Policy makers need procurement and accountability rules. The people selecting AI tools for schools must understand the technology, the teacher role, the equity risks, and the preparation required before tools enter classrooms.

Round Table Voices

Danielle Eisenberg argued that AI in education is already role-fluid. It can act as content, tutor, evaluator, and collaborator. Her key warning was that without design, AI will produce unintended consequences.

Jan Morrison connected this to the larger need for system transformation and equity.

Dr. Tzur Karelitz asked why the teacher should mediate between AI and students in evaluation, which sharpened the question of trust and interpretation.

Danielle Eisenberg answered that AI systems may become effective in assessing competencies such as collaboration or diligence, but trust remains limited and bias remains a serious concern. AI systems may be trained on datasets that do not fully represent diverse or neurodivergent students. Human oversight is therefore necessary to ensure the tools help rather than harm.

Dr. Revital Duek asked how students' thinking can be made visible when AI is involved. Danielle Eisenberg described using AI chatbots to interview students about portfolios, assignments, papers, or group projects. Such tools can prompt reflection about the student's role, peer contribution, self-assessment, and learning process.

Prof. Sigal Tifferet raised a possible future in which AI takes over much assessment work, allowing teachers to focus more on guidance, relationships, communication, and psychological support. This is an important possibility, but it depends on careful design. If AI takes over assessment without human interpretation, the system may become efficient but blind.

Jan Morrison emphasized that students focus on what educational systems value and measure. If oral communication, defense of ideas, and critical thinking matter, they must become central to assessment.

Avi Salmon warned that teachers must prevent students from "downloading their brains" into AI without real thinking or understanding. This phrase captures the chapter's core educational risk.

Synthesis

AI policy is not enough. Education systems need learning designs that make student thinking visible.

Danielle Eisenberg's key contribution is the distinction between a learning triangle and a measurement triangle. The learning triangle involves student-AI interaction, teacher-student relationship, and teacher-AI governance. The measurement triangle involves AI proposing evidence, the teacher interpreting context, and the system aggregating patterns over time.

This architecture keeps humans in the loop where context matters most. It does not ask teachers to inspect every detail manually. It uses AI to reduce the cognitive load of evidence review so teachers can invest judgment where it matters.

The central principle is: AI proposes, teachers decide.

Default AI use	Designed AI use
Hidden student process	Visible process evidence
Output judged alone	Drafts, prompts, defense, and reflection reviewed together
Equity risk appears late	Equity guardrails built in from the start
AI use shaped by convenience	AI use shaped by educational purpose
Teacher reacts after submission	Teacher governs the student-AI layer

Figure 6. Default versus design in AI-supported learning.

Recommendation

Design AI use around process evidence: prompts, drafts, explanations, comparisons, critique, reflection, teacher interpretation, and student ownership.

Education systems should implement the following design requirements:

1. Make student thinking visible, not only student output.

2. Require students to demonstrate and defend their reasoning.
3. Use multiple artifacts rather than single submissions.
4. Build human interpretation into AI-supported assessment.
5. Train teachers in evidence design, not only tool operation.
6. Treat equity as a design requirement, not an afterthought.
7. Track patterns over time to distinguish learning from performance.
8. Build procurement rules that require teacher preparation and bias review.

The practical lesson is direct: if education does not design the AI layer, the AI layer will design education by default.

Practical Recommendation for Teachers

Design the student-AI layer instead of only reacting to it. The teacher should govern how AI enters learning by making process evidence visible.

In practice:

1. Ask students to submit prompts, drafts, and reflections for one AI-supported task.
2. Require a short explanation of what the AI did and what the student did.
3. Use multiple artifacts, not one final product: notes, draft, peer feedback, revision, and final explanation.
4. Watch for equity patterns: who uses AI to deepen thinking, and who uses it to avoid thinking?
5. Use teacher judgment to interpret AI-supported evidence, not to rubber-stamp it.

Classroom activity examples:

1. Process evidence notebook. During an AI-supported project, students keep a simple log of prompts, drafts, decisions, checks, and reflections.
2. AI proposes, student decides. Give students an AI-generated set of suggestions. They must accept, reject, or modify each suggestion and explain why.

3. Pattern review. Instead of grading one artifact, students assemble three pieces of evidence, such as draft, peer feedback, and final explanation, to show learning over time.

Evaluation method:

Evaluate success through patterns across evidence. The teacher should look for consistency between prompts, drafts, decisions, checks, and final work. A successful student makes the learning process visible and can explain why AI suggestions were accepted, rejected, or changed.

For your own development, learn evidence design, not only AI tool operation. Take one assignment and redesign it so the thinking around AI becomes visible. This follows Danielle Eisenberg's principle: AI proposes, teachers decide, and patterns across artifacts are stronger than a single polished submission.

SOURCES

Danielle Eisenberg / Ignite Ed presentation, Meeting 4 presentation.

CHAPTER 12

The Futures Lens

The Challenge

Education systems often plan from the past. Students are entering a volatile, uncertain, complex, and ambiguous world.

Dr. Iris Pinto's Meeting 4 lecture, "Future Trends Shaping Reality," placed this challenge at the center of the Round Table. The point of futures thinking is not prophecy. Futures research is not an attempt to know what has not yet happened. Its practical purpose is to read signals in the present, identify trends, imagine several possible futures, and act now in ways that shape the preferred future.

This is a major shift for STEM education. Traditional education often assumes a relatively stable world: teach known knowledge, prepare for known professions, certify known achievement. Iris Pinto's lecture argued that this assumption no longer holds. The pace of change is accelerating, decision-making is becoming more complex, and human thinking remains linear and biased while reality is increasingly interconnected.

The challenge is therefore not only technological. It is cognitive, ethical, institutional, and human. Schools and universities must teach learners how to live with uncertainty without surrendering judgment.

The Dilemma

Should education prepare students for known labor-market needs, or for multiple possible futures that cannot be fully predicted?

If education prepares only for current labor-market needs, it risks training students for work patterns that may disappear. If education prepares only for general adaptability, it may become too abstract and disconnected from real knowledge, tools, and professions.

The futures lens asks for a third path: prepare students through deep disciplinary grounding, cross-boundary thinking, ethical imagination, and the capacity to keep learning as reality changes.

This dilemma is especially sharp for universities. Their traditional model treats undergraduate and graduate study as a concentrated early-life phase followed by professional practice. The Round Table discussion questioned whether this model can survive in a world of rapid technological and professional change.

Main Positions

One position focuses on current market needs. Another focuses on general adaptability. A third uses futures thinking to help students and institutions navigate uncertainty.

The current-market position is practical. It asks what employers need now and designs curricula accordingly. This can strengthen employability, but it may narrow education to the demands of the present.

The general-adaptability position emphasizes creativity, critical thinking, problem-solving, and lifelong learning. This is closer to the Round Table's spirit, but it still needs concrete pedagogy and assessment.

The futures position combines both. It does not abandon current skills, but it teaches students to analyze change. It uses time horizons, scenarios, wild cards, preferred futures, and STEEP dimensions: social, technological, economic, environmental, and political factors.

In this view, futures thinking is not an elective activity. It is a literacy for modern STEM learning.

Pros and Cons

Current-market preparation is concrete, but it can become obsolete. General adaptability is valuable, but can become abstract. Futures thinking offers tools for uncertainty, but requires teacher preparation and institutional culture.

Futures thinking has a strong advantage: it helps learners move beyond passive prediction. Students learn that the future is partly shaped by present decisions. This supports agency, responsibility, and moral imagination.

It also has risks. Poorly taught futures work can become speculation, slogans, or anxiety. Students may be overwhelmed by possible futures without gaining tools to act. Teachers may lack preparation in scenario thinking, systems thinking, and interdisciplinary analysis.

The solution is not to turn every teacher into a futurist. It is to embed futures habits into STEM learning: ask what trends are visible, what assumptions may fail, what systems are connected, who benefits, who is harmed, and what future should be preferred.

Point of View Map

Policy makers need long-term planning. They must prepare systems for futures that cannot be reduced to a single forecast.

Universities need to rethink their relationship with graduates. If knowledge and professional practice change continuously, the university cannot only be a gate through which young adults pass once. It may need to become a long-term learning partner.

Industry sees rapid skill change directly. Iris Pinto connected this to her experience in industry and noted the disconnect between education systems and industry practices. When education and industry solve similar problems separately, both lose time and relevance.

Teachers need preparation to lead students through complexity. Dr. Noa Ragonis raised the teacher development gap, and Iris Pinto strongly supported the importance of STEM teacher leadership. Without prepared teachers, meaningful transformation in competencies, skills, and AI integration will not happen.

Students need agency and identity in a changing world. In a post-work scenario, where automation reduces the centrality of traditional work, education must help students ask not only what job they will perform, but what kind of person and contributor they will become.

Round Table Voices

Dr. Iris Pinto presented the futures lens. She described a VUCA world shaped by technological convergence, human-machine integration, bio-convergence, post-work scenarios, and accelerating change.

Her presentation described several meta-trends changing the nature of humanity. Convergence brings together AI, biotechnology, quantum technologies, nanotechnology, robotics, and other domains to create new capabilities. Human-machine interfaces blur boundaries through brain-computer interfaces, neural implants, subdermal devices, smart devices, and human-machine symbiosis. Human enhancement and gene editing raise the possibility of a bio-cognitive divide between enhanced and non-enhanced people. Automation, AI, robotics, abundant energy, biomanufacturing, and radical life extension raise questions about work, meaning, learning, identity, and equity.

Iris Pinto also described AI as an enabler, but not the endpoint. Human-centric AI may interpret emotion, cognition, senses, and body. Action and creation AI may plan, act, create, and collaborate. Multimodal and phygital AI may expand learning through hybrid physical and digital environments. These trends make STEM broader than technical knowledge.

Prof. Arnon Bentur connected the discussion to lifelong learning and the idea of a long-term relationship between universities and graduates.

He referred to models in which universities maintain learning relationships across a career, sometimes described as a long curriculum. This reframes higher education from a one-time credentialing institution into a continuing knowledge partner.

Iris Pinto gave a similar example. After graduation, universities often lose contact with graduates except when requesting donations. She proposed that universities could provide regular professional updates in fields such as engineering, similar to software or smartphone updates. This would help graduates remain current and would keep universities connected to professional reality.

Dr. Revital Duek noted that industry may force workers into continuous learning as the price of remaining relevant.

This comment exposed the harder side of lifelong learning. Continuous learning can be empowering, but it can also become pressure. If workers must update themselves constantly without institutional support, lifelong learning becomes an individual burden rather than a social design.

Synthesis

The future cannot be predicted, but learners can be prepared to navigate it.

The futures lens changes the definition of STEM education. STEM is not only science, technology, engineering, and mathematics as separate domains. It becomes a way to understand complex socio-technological systems. It asks students to connect technical knowledge with human values, uncertainty, ethics, identity, and responsibility.

This is why the Round Table repeatedly returned to STEAM rather than only STEM. Art, ethics, humanities, philosophy, social understanding, and communication are not decorations around technical knowledge. They are necessary for judging what technology means and how it should be used.

The teacher's role becomes even more important under this lens. AI can provide information and simulation, but teachers help students frame questions, compare futures, examine values, and resist simplistic answers.

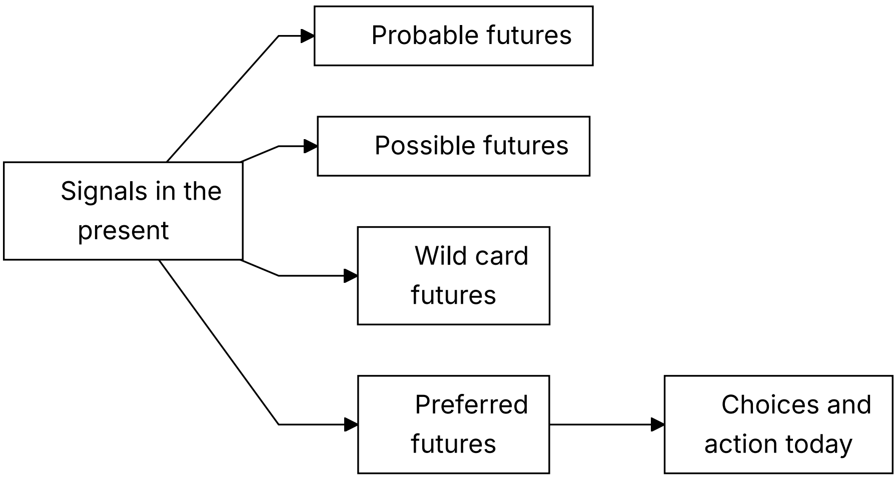

Figure 7. Futures lens for STEM education.

Recommendation

Make futures thinking part of STEM education. Use scenarios, systems thinking, ethical reasoning, lifelong learning, and cross-sector learning to prepare students for uncertainty.

Every STEM program should include structured futures practices:

1. Identify trends in the present and connect them to possible futures.
2. Distinguish probable, possible, wild card, and preferred futures.
3. Use STEEP analysis to connect social, technological, economic, environmental, and political forces.
4. Examine technological convergence and its human implications.
5. Discuss post-work scenarios as questions of identity, meaning, equity, and purpose.
6. Treat lifelong learning as a designed relationship, not only a personal responsibility.
7. Build partnerships among schools, universities, industry, community, and policy.
8. Develop socio-technological leaders, not only competent users of technology.

The Round Table's recommendation is to move STEM beyond technical training. Future-ready STEM education should prepare learners to understand complex problems, shape preferred futures, and carry moral responsibility inside technological change.

Practical Recommendation for Teachers

Add a futures lens to ordinary STEM topics. Students should learn not only how a technology works, but what future it may create and what choices humans still have.

In practice:

1. Begin a unit by asking which trends are visible now.
2. Use STEEP questions: social, technological, economic, environmental, and political effects.
3. Ask students to imagine probable, possible, wild card, and preferred futures.
4. Connect technical learning to human consequences, equity, identity, and responsibility.
5. Ask students what action today would move toward the preferred future.

Classroom activity examples:

1. Four futures map. Students take one technology and describe probable, possible, wild card, and preferred futures, then identify what choices could shape the preferred one.
2. STEEP scan. Groups analyze a STEM development through social, technological, economic, environmental, and political lenses.
3. Future backcasting. Students choose a preferred future, then work backward to identify what learners, teachers, industry, and policy should do now.

Evaluation method:

Evaluate success by assessing the quality of futures reasoning. The student should identify signals, consider more than one possible future, explain assumptions, and connect preferred futures to present choices. Strong work avoids prediction as certainty and shows thoughtful links between technology, society, values, and action.

For your own development, keep a small futures notebook. Once a month, record one emerging technology or social trend, one classroom implication, and one question for students. This follows Dr. Iris Pinto's futures approach: do not predict the future as prophecy, identify signals in the present and act toward a preferred future.

SOURCES

Dr. Iris Pinto presentation, Meeting 4 presentation, OECD Learning Compass 2030.

CHAPTER 13

Risks and Design Principles for AI in Learning

The Challenge

AI can personalize learning and provide feedback, but it can also weaken thinking, hide inequality, and erode trust.

The Round Table did not treat AI as either salvation or threat. It treated AI as a powerful condition that must be governed by educational purpose. The same tool can support deep learning or replace it. It can help teachers see student progress or create a new layer of invisible dependence. It can make learning more equitable or widen existing gaps.

The challenge is that AI's strongest features are also its educational risks. It is fast, articulate, patient, adaptive, and always available. These qualities can support students who need explanation and practice. They can also make students stop struggling, stop questioning, and stop developing the very competencies education is meant to cultivate.

The Dilemma

How can AI enhance learning without replacing the human cognitive and ethical work that education is meant to develop?

The central principle from the Meeting 3 discussion document is keeping the human above the machine. The goal is to maintain human judgment, creativity, and moral responsibility as the final authority over any tool or system.

This creates a practical dilemma. If AI is too restricted, students may lose access to useful support and may use tools secretly. If AI is too open, students may outsource thinking, writing, verification, and even moral responsibility. If AI is designed into learning, teachers must have time, training, assessment infrastructure, and institutional authority.

Main Positions

AI can be used as an efficiency tool, a thinking partner, or a controlled support inside human-centered pedagogy.

The efficiency position uses AI to save time: summarizing, generating examples, grading, tutoring, and producing feedback. It is attractive because education systems are overloaded. Its danger is that efficiency can become the main value, pushing aside reflection, effort, and relationship.

The thinking-partner position asks students to use AI to compare ideas, test assumptions, receive critique, and explore alternatives. This can be powerful when it is structured. It becomes weak when students accept AI output as authority.

The human-centered design position treats AI as a bounded support inside teacher-led learning. It requires disclosure, verification, explanation, oral defense, reflection, and teacher interpretation. This position best matches the Round Table's repeated claim that teachers and lecturers must remain above AI.

Pros and Cons

Efficiency gives speed and access, but may encourage cognitive offloading. A thinking partner can prompt exploration, but may mislead. Controlled support preserves human judgment, but requires careful design.

The benefits are real. AI can offer individualized explanation, practice, simulation, translation, feedback, and access to resources. It can help students rehearse ideas and help teachers review evidence more efficiently.

The risks are also real. The Meeting 4 discussion document identified shallow learning, misinformation, cognitive offloading, weakened teacher-student relationships, inequality, and accountability gaps. Research cited in the materials suggests that AI can boost performance when scaffolded by teachers, while independent or passive use risks dependency and reduced critical thinking.

The danger is not that students use AI. The danger is that AI use becomes invisible, unexamined, and detached from human judgment.

Point of View Map

Teachers need to govern AI use. Students need to learn critical use. Institutions need integrity and equity mechanisms. Society needs citizens who do not delegate judgment to machines.

Students may see AI as helpful, nonjudgmental, and easier to approach than people. That can reduce anxiety, but it can also weaken the human relationships through which confidence, motivation, and identity develop.

Teachers may benefit from AI support, but they also face new burdens: verifying authenticity, interpreting AI-assisted work, protecting equity, and helping students understand the limits of tools.

Institutions must decide what counts as evidence of learning. A polished AI-supported product cannot be enough. Schools and universities need prompts, drafts, comparisons, oral explanations, process logs, peer feedback, reflection, and defense of reasoning.

Society needs graduates who can use AI without surrendering agency. In an AI-rich world, technical fluency is not enough. Learners must develop critical thinking, ethical judgment, collaboration, communication, and the ability to take responsibility.

Round Table Voices

Emlen Metz emphasized that students must use AI thoughtfully and critically. Responsibility cannot be delegated to machines.

Her warning matters because it rejects a shallow definition of AI literacy. AI literacy is not only knowing how to prompt a tool. It includes knowing when not to accept an answer, how to verify, how to decide, and how to remain responsible for one's own judgment.

Prof. Russell Tytler warned that AI may reduce deep thinking if students surrender control over their reasoning.

He emphasized the danger of students relinquishing control over their thinking processes. As AI becomes more sophisticated, students may avoid deep, sustained thinking and shorten cognitive processes. This is not only a learning problem. It is a formation problem, because habits of independent and reflective reasoning take time to build.

Avi Salmon argued that the challenge is not whether to use AI, but how to use it wisely. Teachers and lecturers must remain above AI.

He later sharpened this argument by warning that teachers must prevent students from downloading their brains into AI without real thinking or understanding. This is a precise description of cognitive offloading as an educational failure.

Dr. Gabi Shafat described using AI on challenging problems where tools may fail or show bias, so that students learn critical evaluation.

This is an important design principle. Students should not only use AI where it works smoothly. They should also encounter cases where AI gives biased, incomplete, or incorrect responses. Comparing different tools and discussing their failures turns AI from an answer machine into an object of inquiry.

Dr. Marwa Maklada stated that AI should not design learning or control knowledge. The teacher must remain the architect, guiding learning, preserving deep thinking, and ensuring values-based education.

Prof. Gil Noam warned that AI may increase isolation and encourage students to seek quick answers rather than genuine learning. He emphasized curiosity, critical thinking, teamwork, and human relationships.

Dr. Olena Bekh argued that critical thinking remains the most essential human competency. If society loses it, it risks losing an important part of humanity. She also connected excessive AI dependence to possible reductions in curiosity, creativity, empathy, and emotional engagement.

Dr. Eli Eisenberg raised a concern about pace. AI should be integrated, but society may need time to study ethical, moral, social, and cognitive implications before moving too quickly.

Synthesis

Good AI use increases reflection, judgment, and agency. Bad AI use replaces them.

The Round Table's design logic can be summarized in five principles.

First, AI use must be visible. Students should disclose how they used AI and show the process behind their work.

Second, AI use must be contestable. Students should compare, verify, criticize, and improve AI outputs rather than accept them.

Third, AI use must be relational. It should strengthen teacher guidance, peer discussion, and human explanation rather than isolate students.

Fourth, AI use must be equitable. Systems must prevent a pattern where privileged students augment their thinking while disadvantaged students replace their thinking.

Fifth, AI use must preserve accountability. The student remains responsible for learning, the teacher remains responsible for educational judgment, and the institution remains responsible for tool adoption and safeguards.

AI learning risk	Classroom sign	Design response
Overreliance	Student accepts output without explanation	Require comparison and verification
Bias or error	Output omits context or harms fairness	Use bias checks and source critique
Shallow thinking	Final answer is polished but undefended	Add oral defense and reasoning trail
Inequality	Students use AI in very different ways	Teach responsible use explicitly
Lost accountability	Student cannot say what they own	Require disclosure and responsibility statement

Figure 8. AI learning risks and design responses.

Recommendation

Create learning designs that require students to question, compare, explain, justify, and reflect. Do not treat polished AI output as evidence of learning.

Every AI-supported learning task should include at least four elements:

1. A declaration of AI use.
2. A comparison between AI output and human reasoning.
3. A student explanation of what was accepted, rejected, changed, and why.
4. A teacher-mediated or peer-mediated discussion that tests understanding.

For higher-stakes work, add oral defense, draft history, source verification, and reflection on tool limitations.

The Round Table recommendation is not to slow learning by rejecting AI. It is to slow down the parts that matter: judgment, verification, meaning, and responsibility. AI may accelerate access to information, but education must protect the time needed for thinking.

Practical Recommendation for Teachers

Build friction into AI-supported learning. Students should not move directly from AI output to final submission. They should pause to test, question, and own the result.

In practice:

1. Use AI on a problem where the tool may be incomplete, biased, or wrong.
2. Ask students to compare outputs from different tools or prompts.
3. Require students to identify errors, missing assumptions, or weak reasoning.
4. Use peer discussion to test understanding before submission.
5. Add a final reflection: what did AI help with, and what did it not replace?

Classroom activity examples:

1. AI failure lab. Students investigate an AI response that is incomplete, biased, or wrong and identify the exact point where human judgment is needed.

2. Slow thinking checkpoint. During a task, pause before students use AI and ask them to write their own first explanation, prediction, or strategy.
3. Verify before trust. Students must verify an AI answer using a second method, such as calculation, experiment, source check, peer critique, or teacher conference.

Evaluation method:

Evaluate success through risk awareness and verification behavior. The student should recognize failure modes, slow down before accepting output, and use at least one independent method to check claims. Strong performance includes a correct answer and a documented reason for trusting it.

For your own development, collect a bank of AI failure cases from the discipline. Use them to teach critical evaluation. This follows Dr. Gabi Shafat's example of using challenging AI problems, Emlen Metz's warning about responsibility, Prof. Russell Tytler's concern about losing deep thinking, and Avi Salmon's warning against students downloading their brains into AI.

SOURCES

Meeting 2 presentation, Meeting 3 discussion document, Meeting 4 discussion document.

PART V

Professional Development as System Infrastructure

CHAPTER 14

Who Is Responsible for Teacher Development?

The Challenge

Professional development is essential, but responsibility is scattered. Teachers, institutions, ministries, teaching centers, professional bodies, and funders all play roles, but no single actor clearly owns the whole process.

The steering-team summary of 23 June 2026 stated the problem directly: there is currently no single body that carries overall responsibility for the continuous professional development of teachers and lecturers. Responsibility is distributed across teaching and learning centers, educational institutions, management teams, government ministries, and professional bodies, but it is unclear who ensures that professional development happens systematically over time.

This is the bottleneck in the entire Round Table agenda, not a small administrative problem. Competency-based STEM education, AI literacy, assessment redesign, teacher-student-AI relationships, and future-ready learning environments all depend on educators who can learn continuously. If professional development is weak, every other recommendation remains fragile.

The Dilemma

Who owns sustained professional development: individual teachers, institutions, teaching centers, ministries, professional bodies, or a national infrastructure?

The dilemma is difficult because each answer is partly correct and partly insufficient.

Teachers and lecturers must take responsibility for their own learning. But individual motivation cannot carry a national or institutional transformation.

Institutions must build local conditions for change. But institutions vary widely in resources, priorities, leadership, and incentives.

Teaching and learning centers have expertise. But they often lack authority over promotion, workload, budgeting, curriculum, and departmental culture.

National bodies can create coherence. But they may be too distant from practice and may produce formal requirements without deep implementation.

The Round Table materials therefore point toward shared responsibility, but shared responsibility can easily become no responsibility unless governance is explicit.

Main Positions

Individual responsibility emphasizes teacher agency. Institutional responsibility emphasizes local implementation. National responsibility creates coherence. Shared responsibility combines levels.

The individual responsibility position says that teachers and lecturers are professionals, and professionals must keep learning. This position protects agency and dignity. It also aligns with the OECD Teaching Compass, which emphasizes teacher agency, professional identity, and continuous learning.

The institutional responsibility position says that educators cannot transform practice alone. Leaders must provide time, recognition, resources, peer structures, and incentives. Without institutional involvement, professional learning becomes voluntary enrichment for the already motivated.

The national responsibility position says that systemic change requires shared language, standards, and durable infrastructure. This may include teacher education institutions, national centers for skills development, policy frameworks, funding mechanisms, and evaluation systems.

The shared responsibility position says that professional development must be organized as an ecosystem. Individual learning, institutional support, professional communities, national strategy, and evidence-based evaluation must reinforce one another.

Pros and Cons

Individual responsibility is motivating, but inequitable. Institutional responsibility is close to practice, but uneven. National responsibility creates scale, but can become bureaucratic. Shared responsibility is strongest, but requires governance.

The steering discussion before Meeting 5 highlighted major differences between schools and higher education. In schools, teaching is the core profession. In academia, many lecturers are promoted mainly through research rather than teaching quality. A single professional development model may therefore fail unless it is adapted to different incentive systems.

There are also large differences within higher education itself. Strong institutions may have active teaching centers and strategic leadership. Smaller colleges may have only a limited function for teaching improvement. Professional development policy that assumes equal capacity will widen inequality between institutions.

There is also a motivation problem. Some educators lead change willingly. Some sit on the fence. Some resist. The Meeting 5 steering-team materials raised the possibility of different development pathways and of using leading groups as change agents within the system.

This means responsibility cannot be defined only as attendance at training. Responsibility must include authority, incentives, support, evidence, and follow-up.

Point of View Map

Teachers cannot carry the burden alone. Institutions need structures and incentives. Policy makers must create durable mechanisms. Professional bodies can provide standards and communities.

Teachers need protected time, practical relevance, peer learning, coaching, and opportunities to experience new pedagogy as learners before applying it as teachers.

Lecturers need a development model that recognizes academic freedom and research incentives while still making teaching improvement a serious professional responsibility.

Teaching and learning centers need status and reach. They should not be peripheral service units that work only with volunteers. They need connection to institutional strategy, promotion systems, curriculum design, and assessment reform.

Teacher education institutions need to prepare future educators for AI-rich, competency-centered teaching from the start, rather than treating AI and skills as later add-ons.

National or sector-level bodies need to create coherence across fragmented systems. This may include shared language for competencies, quality expectations, data, and evaluation.

Funders and foundations can support innovation, but the Round Table logic implies that projects must be connected to durable systems, not isolated initiatives.

Round Table Voices

The Meeting 5 steering-team materials reported Prof. Russell Tytler's presentation on teacher professional development and Dr. Yael Granot-Bein's presentation on academic faculty development.

No separate presentation decks for Meeting 5 were found in the collected materials. The available evidence comes from the steering summaries before and after the meeting. Those summaries report that Prof. Russell Tytler addressed professional development for teachers and the question of who is responsible for advancing it and sustaining it over time. Dr. Yael Granot-Bein addressed professional development of lecturers in academia, focusing on activities of the Edmond de Rothschild Bridge for Higher Education and Employment and insights emerging from them.

The steering-team summary reported that the Academia 360 program's aim, as explained by Prof. Arnon Bentur, was to integrate different initiatives under a comprehensive concept of skills and competency development at the undergraduate level, not simply to add new budgets.

Avi Salmon supported a data-based examination and proposed a professional evaluation of the program, similar to evaluations conducted in other projects. This links professional development to institutional learning rather than anecdotal success.

The steering team concluded that no single actor currently carries clear overall responsibility for sustained professional development.

It also concluded that professional development cannot be reduced to one-time training. The question is not only who delivers a workshop, but how educators receive guidance, support, and development across years.

The Meeting 5 steering-team materials emphasized institutional involvement. Effective professional development requires infrastructure: teaching and learning centers, leadership, organizational policy, and resources. Reliance on personal motivation alone is not enough to create broad, sustained change.

Synthesis

Professional development must be treated as infrastructure, not optional enrichment.

The Round Table's deeper insight is that teacher development is not an activity added after reform. It is the mechanism through which reform becomes real. A system cannot ask teachers to develop students' competencies, use AI wisely, redesign assessment, build teamwork, and create future-ready learning if it does not also create a serious system for teacher learning.

Professional development therefore has to be continuous, differentiated, evidence-informed, and connected to implementation. It must support both school teachers and higher education lecturers, while recognizing that their conditions and incentives differ.

The question "who is responsible?" should not be answered by naming one actor. It should be answered by designing a responsibility chain.

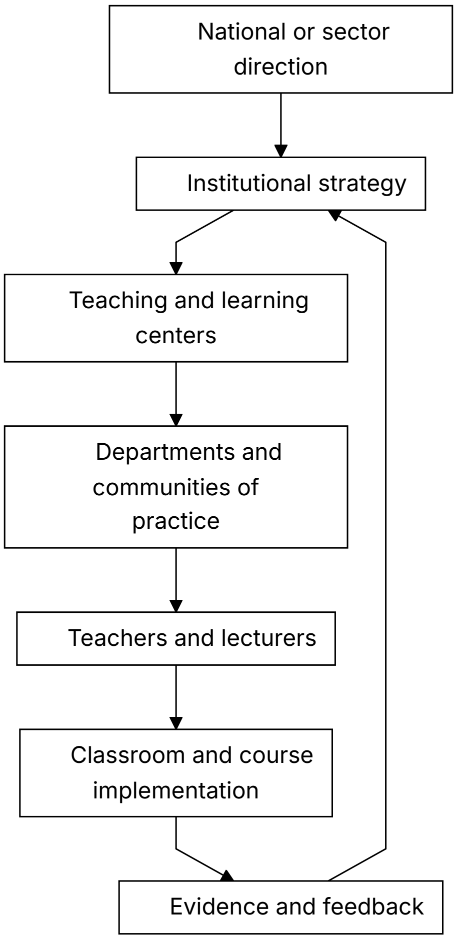

Figure 9. Shared responsibility for sustained teacher development.

Recommendation

Build a shared-responsibility model: national direction, institutional implementation, local communities of practice, teaching and learning centers, and individual professional agency.

The model should assign clear responsibilities:

1. National or sector bodies define direction, language, expectations, and funding mechanisms.

2. Institutions translate those expectations into strategy, time, resources, incentives, and evaluation.
3. Teaching and learning centers provide expertise, coaching, design support, and evidence review.
4. Departments and professional communities create peer learning and shared practice.
5. Teachers and lecturers take active responsibility for growth, experimentation, and reflection.
6. Funders support innovation only when it connects to sustainability and evaluation.

The Round Table recommendation is to stop treating professional development as a sequence of courses. It should become a permanent infrastructure for educational adaptation.

Practical Recommendation for Teachers

Treat professional development as a shared system, while still taking personal responsibility for growth. Do not wait for the perfect institution before beginning, but do not accept isolation as normal.

In practice:

1. Write a one-year personal learning goal linked to AI-era teaching.
2. Identify what support you need: peer group, teaching center, mentor, time, tool access, or leadership backing.
3. Join or create a small professional learning group.
4. Ask the institution to connect professional development to real implementation, not only attendance.
5. Document what changed in your teaching after the development activity.

Classroom activity examples:

1. Teacher learning disclosure. Tell students about one teaching skill you are currently improving and show how it changes the class design.

2. Professional growth inquiry. Ask students what teacher supports would help them learn better, then use one answer to plan a professional learning goal.
3. Shared responsibility map. With older students, map who supports learning: teacher, student, peers, school, family, tools, industry, and policy.

Evaluation method:

Evaluate student success by looking at how well students understand learning as a shared responsibility. The student should be able to identify what they own, what the teacher supports, and what the wider environment contributes. In older classes, success includes proposing a practical support or improvement, not only describing a problem.

For your own development, build a responsibility map: what I own, what my department owns, what the institution owns, and what policy or national infrastructure should own. This follows the Meeting 5 conclusion that no single actor currently carries clear responsibility, and that professional development must become systematic and continuous.

SOURCES

OECD Teaching Compass, Envisioning Educator Roles for Transformation, steering-team summaries for Meeting 5.

CHAPTER 15

Beyond One-Off Training

The Challenge

Workshops are easy to count. Practice change is harder. Teachers and lecturers need support while they implement new pedagogy, AI use, assessment, and collaboration.

The Meeting 5 steering-team materials were explicit: professional development cannot be reduced to one-time training. The question is not only who delivers a course, but also how a system creates guidance, support, and growth over years.

This distinction matters because AI-era teaching is not a technique that can be explained once and then applied mechanically. It changes the teacher's role, the student's role, assessment evidence, classroom dialogue, curriculum design, and institutional expectations. A short workshop may create awareness. It rarely changes practice by itself.

The Dilemma

Should professional development focus on training events, or on long-term implementation support?

Training events are attractive because they are visible, schedulable, and measurable. Leaders can count hours, certificates, attendance, and satisfaction surveys. But these measures do not prove that teaching changed.

Implementation support is harder to count. It requires coaching, peer observation, design time, leadership backing, experimentation, feedback, and reflection. It also requires tolerance for partial success and revision. Yet this is exactly the kind of support needed when educators are asked to move from knowledge transmission to competency formation.

Main Positions

Training events provide exposure. Continuous professional learning supports growth. Implementation accompaniment helps teachers move from knowing to doing.

The exposure position says educators need initial access to new ideas, research, tools, and examples. This is true. Without exposure, teachers may not know what is possible.

The continuous learning position says that professional competence develops over time through cycles of practice, reflection, collaboration, and adjustment. This aligns with the OECD Teaching Compass view of teaching as a lifelong professional journey rather than a one-time qualification.

The implementation accompaniment position goes further. It argues that professional development must be tied to actual classroom or course redesign. Teachers need help while planning, trying, observing, interpreting evidence, and improving. In this model, support follows the educator into practice.

Pros and Cons

Training events are visible and manageable, but transfer weakly into practice. Continuous learning supports change, but requires time and budget. Accompaniment is powerful, but needs mentors and institutional commitment.

One-off training also tends to favor the already motivated. Those who attend are often the educators already inclined toward innovation. Those most in need of change may remain outside the process.

Long-term professional learning can address this, but only when institutions protect time and create recognition. If development is always added on top of existing workload, it becomes a burden rather than a source of professional renewal.

Implementation accompaniment is strongest when it is close to real problems. It can help a lecturer redesign an assessment, help a teacher structure AI use, help a department embed teamwork across courses, or help a school build project-based learning. But it cannot succeed if it is disconnected from leadership, scheduling, and evaluation.

Point of View Map

Teachers need time, confidence, and peer support. Leaders must protect time and resources. Students benefit only when professional learning changes practice. Policy must fund implementation, not only courses.

Teachers need to experience new learning processes themselves. Dr. Einat Sprinzak argued in Meeting 2 that teachers must first experience these processes as learners, gaining a firsthand understanding of the student journey. This is especially important for AI use and competency-based learning. Educators cannot meaningfully guide what they have never experienced.

Lecturers need a path that respects academic identity. Many university lecturers are experts in research and disciplinary knowledge, but may not have been trained in competency development, active learning, or AI-supported assessment. Professional development for lecturers must therefore connect pedagogy to disciplinary depth, not present it as generic teaching advice.

Leaders need to understand that professional development is an implementation system. If leaders ask for new pedagogy but leave class sizes, timetables, promotion criteria, and assessment rules unchanged, the training will not travel into practice.

Students are the final test. A professional development system is successful only when students experience deeper learning, stronger feedback, more meaningful assessment, and better support.

Round Table Voices

The Meeting 5 steering-team materials emphasized the need to move from training to support during implementation.

It stated that the central challenge is not the absence of initiatives or training opportunities, but the absence of a clear, coordinated, binding system for continuous professional development. This is a crucial distinction. The problem is not activity. The problem is system design.

Dr. Einat Sprinzak argued that teachers must experience learning processes themselves as learners.

Her argument gives professional development a pedagogical principle: teachers should not only hear about active learning, teamwork, AI use, or reflection. They should participate in such learning, feel its difficulty, and understand what students experience.

Avi Salmon emphasized that professors must become involved by experiencing new pedagogical and technological realities.

He argued that immersing educators in emerging practices increases the chance that they will internalize change and adapt their teaching. This places experience before formal curriculum decisions. People change practice more deeply when they have felt the new reality, not only read about it.

Jan Morrison's Meeting 2 contribution adds another warning. Problem-based and transdisciplinary learning are often criticized when they are implemented inside systems not designed to support them. Without time for faculty collaboration, cross-sector training, and structured teamwork, strong pedagogical ideas can fail for organizational reasons.

Dr. Gabi Shafat's Afeka material offered a concrete institutional orientation: course skills are defined, assessed, evaluated, and continuously improved. This connects professional development to curriculum and evidence rather than isolated inspiration.

Synthesis

Professional development should be a cycle: experience, design, implement, reflect, evaluate, and improve.

The cycle begins with experience. Educators encounter the kind of learning they are expected to create for students.

It continues with design. Educators adapt the ideas to their discipline, age group, institutional context, and students.

It moves into implementation. Teachers try the new practice with real learners, real constraints, and real uncertainty.

It requires reflection. Educators examine what happened, what students actually did, and what evidence was produced.

It includes evaluation. Institutions ask what changed, for whom, and under what conditions.

It ends by beginning again. Improvement is iterative.

Recommendation

Design professional development as supported practice over time. Include mentoring, peer observation, collaborative design, communities of practice, implementation coaching, and evidence review.

A serious professional development model should include:

1. Initial exposure to concepts, tools, and research.
2. Educator experience as a learner.
3. Collaborative lesson or course design.
4. Classroom or course implementation with support.
5. Peer observation and professional dialogue.
6. Review of student evidence and teacher reflection.
7. Iteration across semesters or school years.
8. Institutional recognition of participation and improvement.

The Round Table recommendation is to stop asking whether professional development occurred and start asking whether professional practice changed. The unit of success is not the workshop. The unit of success is sustained improvement in teaching and learning.

Practical Recommendation for Teachers

Turn every professional development event into an implementation cycle. A workshop should lead to a changed lesson, a trial, evidence, reflection, and revision.

In practice:

1. Before a training event, choose one classroom or course problem you want to solve.
2. During the event, identify one idea you can test within two weeks.
3. After the event, implement a small change with students.
4. Collect evidence: student work, student feedback, peer observation, or your own reflection.
5. Revise the practice and share the result with colleagues.

Classroom activity examples:

1. Try, reflect, revise. Introduce one new learning routine, such as peer critique or AI-supported drafting. Students try it, reflect on its usefulness, and help improve the next version.
2. Teacher-as-learner demonstration. The teacher completes a small version of the student task in front of the class and discusses what was difficult.
3. Implementation journal. Students keep a brief record of how a new classroom method affects their learning across three lessons.

Evaluation method:

Evaluate success through adaptation and reflection. The student should show willingness to try a new learning routine, give useful feedback, and adjust behavior after reflection. Evidence can include journal entries, revised work, and a short explanation of what changed between attempts.

For your own development, experience the new pedagogy as a learner before teaching it. Try the AI tool, teamwork method, reflection routine, or project structure personally. This follows Dr. Einat Sprinzak's point that teachers must experience learning processes themselves, and Avi Salmon's point that educators internalize change through direct experience.

SOURCES

Meeting 2 presentation, OECD Teaching Compass, Envisioning Educator Roles for Transformation.

CHAPTER 16

Evidence, Evaluation, and Institutional Learning

The Challenge

Education innovation often sounds persuasive before it is proven. Institutions need to know what is working, for whom, under what conditions, and why.

The Round Table repeatedly returned to evidence because AI-era education is too complex for slogans. Competency-based learning, project-based learning, AI-supported assessment, teacher development, and industry-connected education all sound promising. But without evidence, institutions cannot distinguish deep change from attractive language.

The challenge is to evaluate without flattening. If evaluation is reduced to simple metrics, it may miss the very competencies the Round Table wants to develop: critical thinking, collaboration, curiosity, ethical judgment, and communication. If evaluation is avoided because the work is complex, systems cannot learn or scale responsibly.

The Dilemma

How can education systems evaluate AI-era teaching reforms without reducing them to shallow metrics?

The dilemma is between accountability and learning. Accountability asks whether public money, institutional time, and professional effort are producing results. Learning asks what is changing, why, where it works, where it fails, and how to improve.

Education systems often confuse the two. They collect data for reporting rather than improvement, or they avoid evaluation because meaningful learning is hard to measure. The Round Table materials point toward a different stance: measurement should support continuous improvement and iteration, not only external accountability.

Main Positions

Quantitative outcomes support accountability. Qualitative evidence captures context. Mixed evidence combines both. Developmental evaluation fits complex change.

Quantitative evidence includes participation rates, course completion, assessment results, retention, progression, and other comparable indicators. It helps leaders see scale and patterns.

Qualitative evidence includes student artifacts, teacher reflections, observations, interviews, oral defenses, peer discussion, and examples of changed practice. It shows meaning and context.

Mixed evidence combines the two. It allows institutions to see both whether change is happening and how it is happening.

Developmental evaluation is especially relevant when reforms are still evolving. Instead of judging a fixed program after implementation, it helps teams learn while designing and adapting.

Pros and Cons

Quantitative evidence is comparable, but may miss deeper learning. Qualitative evidence captures process, but can be harder to scale. Mixed models are richer, but require expertise. Developmental evaluation fits innovation, but may be less familiar to policy systems.

The main risk of quantitative evaluation is false precision. A number may look rigorous while hiding whether students actually became more capable.

The main risk of qualitative evaluation is inconsistency. Rich evidence can become anecdotal unless it is structured, reviewed, and connected to clear questions.

The main risk of mixed evaluation is complexity. It requires institutions to build capacity for data interpretation, evidence review, and improvement cycles.

The main risk of developmental evaluation is discomfort. Leaders may prefer simple answers, while complex reforms require learning from partial and changing evidence.

Point of View Map

Policy needs evidence for funding and scaling. Institutions need learning loops. Teachers need evaluation that improves practice. Industry values evidence of real capability, not grades alone.

Students need assessment that recognizes process and capability, not only final products.

Teachers need evaluation that supports professional judgment. If evidence becomes surveillance, it will reduce trust and experimentation. If evidence becomes feedback, it can strengthen practice.

Institutions need to know whether their initiatives are reaching more than the already motivated. They need to ask who participates, who benefits, and where gaps remain.

Policy makers and funders need evidence of sustainability, not only activity. A program may run workshops, publish reports, and produce enthusiasm without changing practice.

Industry needs graduates who can perform in authentic contexts. This requires evidence of applied capability, teamwork, problem-solving, communication, and adaptation.

Round Table Voices

Avi Salmon proposed professional, data-based evaluation of programs.

This point appeared in the Meeting 5 steering-team materials regarding Academia 360. Avi Salmon supported a data-based examination and proposed a professional evaluation of the program, similar to evaluations conducted in other projects. The significance is that large educational initiatives should not rely on impressions alone. They should be studied seriously.

Dr. Gabi Shafat described continuous feedback and evaluation in courses.

In Dr. Gabi Shafat's Meeting 2 Afeka material, Afeka's approach included defining skills and competencies for each course, asking each lecturer to identify what students should acquire, and examining how those skills are assessed and tested. The question was not only whether content was taught, but whether intended capabilities were achieved.

Dr. Marwa Maklada described competency-based tasks and validation processes.

Her contribution connects evaluation to national school assessment. Competency-based tasks within matriculation and student assessment processes require validation, not only enthusiasm. This is essential if competency language is to become part of formal education rather than remain a parallel discourse.

Jan Morrison described TIES as using measurement for continuous improvement and iteration rather than accountability alone. She also pointed to systems where sustained implementation is supported by long-term data and continuous iteration.

Danielle Eisenberg's Meeting 4 presentation added the idea of evidence design. AI may propose skill evidence and confidence ratings, but the teacher interprets. Patterns across artifacts are stronger than single submissions. This is a practical evaluation principle for AI-rich learning.

Dr. Eli Eisenberg emphasized that meaningful learning requires reflection and evaluation, and that establishing effective feedback, evaluation, and assessment mechanisms is a major challenge across higher education, schools, and the education system as a whole.

Danielle Eisenberg made the hardest evaluation point in the whole Round Table, and she made it at the last meeting. The obstacle to educational innovation is not the absence of successful programs. It is the inability to scale them, because the systems that decide what counts – assessment regimes, university admissions, and employer expectations – still reward traditional academic achievement rather than the competencies innovative programs are designed to build. This turns the evaluation question around. The problem is not that competency-based learning lacks evidence. It is that the measurement systems with real consequences attached are not measuring it, so a school that succeeds at it is rewarded for nothing and a school that ignores it loses nothing.

Synthesis

Evaluation should be part of the learning system, not an external audit after the fact.

The Round Table's approach suggests three levels of evidence.

At the student level, evidence should show process, reasoning, collaboration, reflection, and growth, not only final answers.

At the teacher level, evidence should show changes in practice: new assessment designs, AI governance, collaborative teaching, feedback quality, and student engagement.

At the institutional level, evidence should show adoption, sustainability, equity, and improvement over time.

The key is to treat evidence as a feedback loop. Institutions should ask: What did we intend? What changed? Who benefited? Who was left out? What should we adjust next?

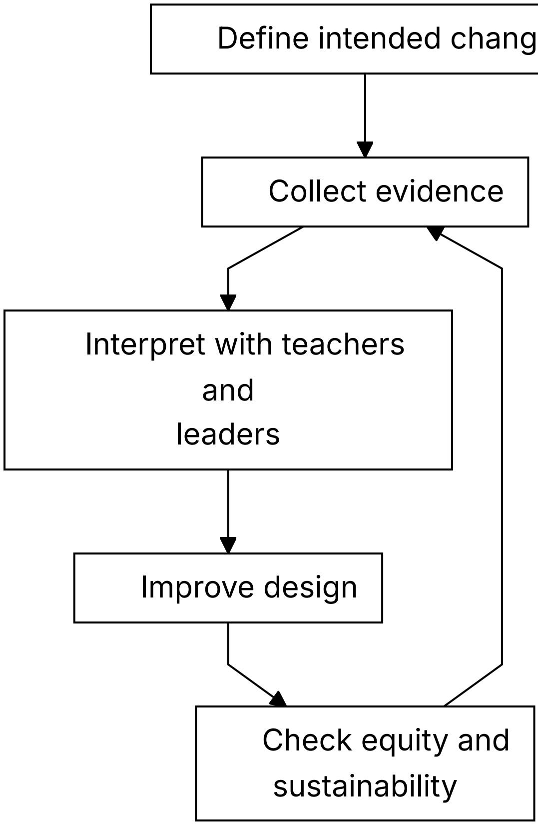

Figure 10. Evidence loop for institutional learning.

Recommendation

Use mixed evidence: student artifacts, process evidence, teacher practice, competency progression, institutional adoption, and longitudinal outcomes.

A serious evaluation model should include:

1. Baseline description of current practice.
2. Clear definition of intended competencies and outcomes.
3. Evidence from student work, drafts, reflection, oral explanation, and collaboration.
4. Evidence from teacher practice and professional learning participation.

5. Data on participation, equity, persistence, and progression.
6. Review cycles in which teachers and leaders interpret evidence together.
7. Program evaluation for large initiatives such as Academia 360 or national competency programs.
8. Longitudinal follow-up where possible, especially in higher education and workforce preparation.

The Round Table recommendation is that evidence should not be used only to judge. It should be used to learn, improve, and protect educational integrity while systems change.

Practical Recommendation for Teachers

Use evidence as a feedback loop for your own teaching. The goal is not to collect data for display. The goal is to learn what changed and what should improve next.

In practice:

1. Define one intended learning change before the unit begins.
2. Collect at least two kinds of evidence, such as student artifacts and reflections.
3. Look for patterns, not only isolated examples.
4. Ask who benefited and who did not.
5. Use the evidence to make one concrete improvement in the next cycle.

Classroom activity examples:

1. Evidence wall. Students post examples of learning evidence, such as drafts, diagrams, questions, corrections, and reflections, then classify what each evidence type shows.
2. What changed review. After a unit, students compare their first and final attempts and identify one change in reasoning, skill, or confidence.
3. Improve the lesson feedback. Students answer three focused questions: what helped learning, what blocked learning, and what should be changed next time.

Evaluation method:

Evaluate success with evidence of change over time. The teacher should compare early and later artifacts, looking for stronger reasoning, better questions, improved accuracy, and clearer reflection. Student feedback about the learning design can also become evidence, but it should be interpreted with the teacher's professional judgment.

For your own development, conduct a small practice inquiry each semester. Choose a question about your teaching, collect evidence, interpret it with a colleague, and revise the design. This follows Jan Morrison's measurement-for-improvement logic, Dr. Gabi Shafat's continuous course evaluation, Danielle Eisenberg's evidence design, and Avi Salmon's call for professional, data-based evaluation.

SOURCES

Meeting 2 presentation, Dr. Gabi Shafat STEM material, OECD Teaching Compass.

PART VI

Environments, Industry, and Experiential Learning

CHAPTER 17

The Classroom Was Built for a Different Problem

The Challenge

Many classrooms were designed for frontal teaching. But creativity, teamwork, making, AI-supported inquiry, and complex problem solving require different environments.

The preparation for Meeting 6 framed the issue clearly: what physical and organizational environment is required to develop twenty-first-century skills in the age of AI? The question was not only what competencies are needed or how teachers should be trained. It was how the learning, teaching, and organizational environment must be designed so those competencies can actually develop.

This matters because space, time, tools, schedules, roles, and partnerships quietly define what kind of learning is possible. A room organized for one teacher speaking to rows of students carries a theory of learning inside its furniture. It assumes attention, reception, and individual performance. But skills such as creativity, teamwork, multidisciplinary problem-solving, making, critique, and AI-supported inquiry require movement, dialogue, experimentation, iteration, and access to tools.

The Dilemma

Can existing physical and organizational environments support AI-era STEM skills, or must they be redesigned?

The dilemma is not only architectural. It is organizational. A school may have a makerspace but no timetable that allows extended projects. A university may have advanced labs but no incentive for cross-disciplinary teaching. A classroom may have digital tools but remain pedagogically frontal. An institution may speak about collaboration while organizing teachers as isolated individuals.

The Round Table question was sharper: can creativity, teamwork, and complex problem solving be developed in classrooms and organizations built for frontal learning?

Main Positions

Existing classrooms can be adapted. Specialized spaces such as makerspaces can support active learning. Broader ecosystem redesign can align space, time, roles, tools, and partnerships.

The adaptation position says that systems cannot wait for new buildings. Teachers can rearrange rooms, use digital tools, create project cycles, invite external partners, and redesign assignments inside existing constraints.

The specialized-space position says that certain forms of learning need dedicated environments. Makerspaces, labs, studios, simulation rooms, and collaborative learning spaces make certain actions possible: building, testing, prototyping, debugging, observing, and revising.

The ecosystem position says that space alone is not enough. The environment includes the physical layout, organizational structure, teacher collaboration, assessment design, AI tools, industry links, community involvement, and leadership mechanisms that sustain change.

Pros and Cons

Adapting existing classrooms is practical, but may limit change. Specialized spaces enable making and experimentation, but can become isolated showcases. Ecosystem redesign is strongest, but complex.

Adaptation is important because most learning still happens in ordinary classrooms. But adaptation can leave the deeper organizational model untouched.

Specialized spaces can signal seriousness and give students access to richer experiences. Yet they can become symbolic islands if only a few students or teachers use them, or if they are disconnected from curriculum and assessment.

Ecosystem redesign is more powerful because it aligns the environment with the intended learning. It asks whether schedules allow project work, whether teachers can co-plan, whether assessment rewards process, whether industry partners are involved, and whether AI is used as a working tool rather than a novelty.

Point of View Map

Teachers need spaces that support intended pedagogy. Students need active and collaborative learning. Institutions need investment and scheduling models. Industry can help connect spaces to real practice.

Teachers need environments that reduce friction. If every collaborative activity requires fighting the room, the timetable, and the assessment system, innovation will remain exceptional.

Students need environments where they can act, not only listen. They need to build, test, present, defend, fail, revise, and collaborate.

Institutions need to decide whether they are organized around knowledge transfer or competency development. This question affects spaces, partnerships, evaluation, AI use, teacher roles, and external relations.

Industry can help connect learning environments to real practice. It can provide problems, tools, mentors, visits, and standards of professional work. But industry involvement must serve learning, not reduce education to short-term labor demand.

Round Table Voices

The Meeting 6 steering-team materials identified physical and organizational environment as a central issue.

The preparation summary listed questions that should guide this chapter: Does real change require organizational redesign, not only teacher training? Are new models of collaborative work among teachers and lecturers needed? Does the current organizational structure support or block innovation? Is responsibility for skill development placed on the individual teacher, or is a broader ecosystem of support required?

Dr. Revital Duek opened the Meeting 6 discussion with the sharpest version of the question: would learning environments look different if education were designed around skills and competencies rather than around knowledge? The rest of the session was an attempt to answer it.

Avi Salmon's Meeting 6 lecture answered from industry. He described hands-on environments – makerspaces, hackathons, technical challenges, and innovation hubs – as places where people learn by designing, building, testing, and improving real solutions, and reported that in engineering training the participants who explored independently and used AI as a support tool were considerably more successful than those who waited for instructions or predefined solutions. He named three qualities that decided the difference: curiosity, determination, and humility. He also described programs in which students teach younger peers or work with their parents on technological challenges, and argued that talented students need room to pursue projects outside the constraints of the curriculum.

Prof. Arnon Bentur put the scaling question to him directly: should experiential learning become an integral part of formal education, or remain an external program? Avi Salmon answered that it must become part of the formal system, and that the obstacle is teacher confidence rather than student interest. Student leadership, peer teaching, and authentic hands-on work are what bridge that gap.

Dr. Gabi Shafat gave the higher-education version. Afeka runs roughly twenty-four maker clubs in fields ranging from electronics to accessibility to music, one of them designing and building musical instruments with 3D printing. Students also work on applied engineering projects, including a fuel-free electric vehicle designed to travel from Dan to Eilat and an autonomous marine vessel entered in an international competition. The decisive feature is not the equipment. It is that faculty members take part as mentors, and that the college funds the space, the equipment, and the academic supervision.

Prof. Arnon Bentur then named the constraint that blocks most of this in universities: many classes are extremely large, and interactive learning cannot become central until the student-to-faculty ratio changes. This is the environment question in its least glamorous and most decisive form.

Danielle Eisenberg reframed the whole problem. The obstacle to educational innovation, she argued, is not a shortage of successful programs. It is the inability to scale them, because assessment systems, university admissions, and workforce expectations continue to reward traditional academic achievement rather than the competencies these environments are built to develop. Chapter 16 returns to this point and Chapter 24 treats it as an objection the book cannot dismiss.

Dr. Rinat Itzhaki argued that the target audience should be teacher education, and that the model and its objectives must be defined before implementation begins, so that everyone involved shares the same understanding of the intended outcome.

Dr. Avigdor Zonnenshain closed the meeting by reaffirming learning by doing as the foundation. Makerspaces, Fab Labs, and project-based programs work because they let learners tackle authentic challenges, collaborate, and develop technical and leadership capability together, with mentors and teachers acting as facilitators who guide rather than supply solutions.

Jan Morrison's TIES material argued for career-connected learning infused with real-world experiences and expertise. It also warned that transdisciplinary and problem-based learning fail when placed inside systems not designed to support them. This directly supports the Meeting 6 environment question.

The sixth meeting treated environment as a way to connect teacher profile, AI, competencies, and institutional design, rather than as a side topic.

Synthesis

Spaces teach. If environments are built for passive learning, they resist skill formation.

A learning environment is a curriculum made visible. It tells students what matters: listening or doing, compliance or inquiry, individual completion or shared problem solving, memorization or design, tool use or tool critique.

The key Round Table insight is that teacher development and learning environments must be designed together. A teacher trained for project-based, AI-supported, competency-centered learning will struggle in a system that provides only frontal classrooms, fixed schedules, isolated courses, and product-only assessment.

The problem is not the classroom as a physical room. The problem is the assumption that changing the teacher is enough while the environment remains unchanged.

Learning environment	What it supports	Main risk
Frontal classroom	Explanation and shared attention	Passive learning and limited collaboration
Flexible classroom	Discussion, teamwork, critique	Requires teacher design and routines
Makerspace or studio	Building, testing, iteration	Can become isolated from curriculum
Workplace-linked environment	Authentic constraints and professional context	Needs safeguards and educational purpose

Figure 11. From frontal classroom to competency-rich learning environment.

Recommendation

Redesign learning environments around collaboration, making, inquiry, critique, AI use, reflection, and real problems.

Every institution should audit its learning environments with the following questions:

1. What kind of learning does this space make easy?
2. What kind of learning does it make difficult?
3. Can students collaborate, build, test, revise, and present?
4. Can teachers work together, not only individually?
5. Is AI used as a tool for inquiry, design, feedback, and reflection?

6. Are industry and community connections built into the learning environment?
7. Does assessment reward process and capability, not only final answers?
8. Can the environment support lifelong learning for teachers as well as students?

The Round Table recommendation is to treat the physical and organizational environment as part of pedagogy. If the goal is competency development, the environment must be designed for competence, not only content delivery.

Practical Recommendation for Teachers

Audit the learning environment before blaming the lesson or the students. Ask whether the space, time, tools, and routines support the kind of learning you are asking students to do.

In practice:

1. Choose one activity that requires teamwork, making, inquiry, or critique.
2. Rearrange the physical or digital space to support that activity.
3. Give students roles that require movement, dialogue, building, testing, or presenting.
4. Use AI as a tool for inquiry or feedback, not as a replacement for student work.
5. After the lesson, ask what the environment made easier and what it blocked.

Classroom activity examples:

1. Room redesign challenge. Students help rearrange the room for collaboration, making, or critique, then explain how the layout changed participation.
2. Makerspace without a makerspace. Use simple materials, paper prototypes, digital sketches, or simulations to let students build and test a model before discussing constraints.

3. Environment audit walk. Students inspect the classroom or school environment and identify what supports or blocks creativity, teamwork, and problem solving.

Evaluation method:

Evaluate success by observing how the environment changes student action. The student should participate actively, use space or materials purposefully, collaborate with others, and explain how the setting affected thinking. Evidence can include prototypes, observation notes, peer feedback, and a short environment reflection.

For your own development, keep an environment log for one month. Record which arrangements supported active learning and which pushed the class back toward passive listening. This follows Meeting 6's question about whether classrooms built for frontal learning can develop creativity, teamwork, and complex problem solving.

SOURCES

Meeting 6 summary, Avi Salmon's Meeting 6 presentation, TIES Jan Morrison presentation, OECD Learning Compass 2030.

CHAPTER 18

Virtual, XR, VR, and AI-Supported Learning Spaces

The Challenge

Virtual, XR, VR, and AI-supported environments create possibilities for simulation, access, and repetition. But they may also detach students from physical, social, and ethical realities.

The Meeting 6 preparation material asked what kinds of learning environments allow students and teachers to work alongside AI. It also asked how digital tools and AI can be integrated without damaging the development of human skills. This is the right framing for XR, VR, virtual learning, and AI-supported spaces.

The question is not whether new technologies are impressive. The question is whether they deepen the learner's encounter with reality, complexity, evidence, collaboration, and judgment. A virtual environment can make an invisible process visible. It can also become a polished substitute for authentic practice.

The Dilemma

What is the right balance between physical environments and virtual ones?

Physical learning offers material resistance. Tools break, prototypes fail, measurements vary, teammates disagree, and real constraints appear. These experiences are educational because they force students to connect ideas to consequences.

Virtual learning offers access, safety, repeatability, and scale. Students can simulate rare events, explore complex systems, practice procedures, or visualize processes that are too small, large, expensive, dangerous, or slow to observe directly.

The dilemma is that both are necessary. A purely physical model may be limited, expensive, and unequal. A purely virtual model may weaken embodied experience, social negotiation, and contact with real-world complexity.

Main Positions

Physical-first learning values embodied experience. Virtual-first learning values access and simulation. Blended experiential learning combines both.

The physical-first position says students should learn through real materials, real tools, real collaboration, and real consequences. This is especially important in engineering, science, design, and makerspace learning.

The virtual-first position says students need access to simulations, AI tutors, immersive environments, and adaptive practice. This can widen opportunity and help students repeat difficult learning experiences.

The blended experiential position says that virtual and physical environments should be designed together. XR, VR, AI, simulation, laboratories, makerspaces, peer discussion, and industry-connected problems each serve different learning functions.

Pros and Cons

Physical-first learning preserves material and social reality, but may be expensive and limited. Virtual learning is scalable and safe, but can flatten complexity. Blended learning is strongest, but demands design skill.

Virtual and XR environments are strongest when they do one of three things: make hidden processes visible, allow safe practice of difficult situations, or widen access to experiences students could not otherwise have.

They are weakest when they are used as novelty. A simulated lab that merely entertains does not build scientific judgment. A VR tour that replaces real inquiry may reduce learning to passive experience. An AI-supported environment that gives answers without requiring explanation may weaken thinking.

Blended learning is strongest when the teacher intentionally connects the virtual and the physical. Students simulate, test, build, compare, revise, explain, and reflect. The virtual environment becomes a bridge into deeper understanding, not an escape from effort.

Point of View Map

Students need both access and authenticity. Teachers need tools that serve pedagogy, not novelty. Institutions need infrastructure and evaluation. Industry can provide realistic scenarios and tools.

Students need opportunities to practice in multiple modes: physical, digital, collaborative, reflective, and AI-supported. They also need to understand the limits of each mode.

Teachers need professional development in learning design. They should not be handed XR, VR, or AI tools and expected to invent pedagogy alone.

Institutions need procurement discipline. New learning technologies should be adopted because they serve defined learning goals, not because they look modern.

Industry can help provide authentic scenarios, tools, and constraints. This is especially valuable when virtual environments model real professional practice.

Round Table Voices

Meeting 6 directly raised the balance between physical environments and virtual ones such as XR and VR.

The preparation summary asked what environments would allow work alongside AI, how digital tools and AI can be integrated without harming human skill development, and which human competencies require changes in the organization of learning.

Emlen Metz's Meeting 6 lecture answered the question at the level of classroom practice rather than technology. She separated productive uses of AI from unproductive ones. Productive use is dialogic: asking the system to generate counterarguments, challenge an assumption, or explain a concept at a chosen level, rather than requesting a finished answer. Translation into another language or reading level widens access. Active evaluation means checking what AI asserts, and she gave lateral reading as the working example – the skill is difficult when no sources are offered, so the student should ask the system for its sources and then examine those sources for reliability and relevance. Think-first means brainstorming independently before consulting AI, because consulting it early narrows the range of ideas a student will consider.

She also proposed a category the environment discussion needs: AI-robust tasks. These are activities whose value does not collapse when a language model is available – laboratory work, drawing or building models, card sorts, handwriting, pairwise and group discussion, jigsaw grouping, collaborative work with a defined role for each student. Alongside them she argued for deliberate no-AI practice: students must still practice reading, writing, and mathematics with pen and paper, oral examination, or locked machines, because difficulty and effort are where learning happens and students should not outsource a skill they have not yet mastered. Her term for the failure mode is cognitive surrender, the point at which a student accepts an output because checking it is harder than believing it.

Danielle Eisenberg asked how these environments should serve neurodiverse students and what the design trade-offs are. Emlen Metz answered that AI offers real flexibility and personalization for neurodiverse learners but should complement rather than replace face-to-face interaction, because structured collaborative activity and hands-on experience are what build social confidence and meaningful participation. Dr. Olena Bekh asked the system-scale version: how can education systems, given their size and inertia, equip teachers with the competencies this requires? The answer was accessible professional learning, and more importantly communities of practice in which educators learn from one another rather than from a single training event.

Dr. Noa Ragonis brought the question back to a case the field has already run. During the COVID-19 pandemic, whether virtual laboratories could replace physical ones became a live issue, and the conclusion was that the two must be combined, because each carries distinct educational value. Virtual laboratories, AI, and digital tools improve access and flexibility, but physical experimentation, prototyping, and engineering design remain essential. She pointed to at-home making kits and to teacher training that uses 3D printers, laser cutters, and Arduino as evidence that hands-on learning has not been superseded. Dr. Einat Sprinzak made the parallel point for simulation: it has genuine educational value, and it complements rather than replaces hands-on work and active student participation.

Prof. Gil Noam argued that the debate is framed too narrowly. The choice is not AI or hands-on learning. The more interesting question is how AI can act as a collaborative learning partner – systems designed to ask questions rather than supply answers, students collaborating with each other while interacting with AI, and the teacher strengthened in the role of facilitating collective reflection. Emlen Metz extended this into classroom design: have students compare AI-generated arguments, evaluate the reliability of sources, and analyze together why two AI responses differ. Avi Salmon noted that current systems can already be prompted to guide a student through questions rather than answer them, and that the teacher's job is to create the discussion and independent thinking that happen before a student turns to the tool.

Dr. Yael Granot-Bein widened the frame to the educator's own position. When students increasingly rely on recorded lectures and AI rather than attending class, teachers and professors have to redefine what their added value is beyond delivering content. Avi Salmon offered the flipped classroom as one answer: content studied independently, and contact time spent on discussion, experiential learning, mentorship, and the practical work that neither a recording nor a model can replicate.

Dr. Iris Pinto's Meeting 4 lecture adds useful conceptual language. She described multimodal AI, which integrates text, image, audio, and video, and phygital AI, which connects physical and digital realities through AR and VR. She also described AI-enabled learning through simulations, data, and immersive environments. These concepts help define why the virtual and physical should be connected rather than separated.

Synthesis

Virtual environments should extend learning, not replace the human, social, and material dimensions of learning.

The best virtual learning spaces do not remove reality. They prepare students to engage reality more intelligently. They let students see a process, rehearse an action, test a model, compare alternatives, or receive feedback before returning to physical, social, and ethical practice.

The weakest virtual learning spaces replace friction with smoothness. They create the feeling of learning without the burden of making decisions, facing consequences, collaborating with others, or defending reasoning.

AI-supported environments must therefore be judged by what they ask students to do. Do students only consume? Or do they question, build, explain, compare, decide, and reflect?

Recommendation

Use XR, VR, and AI when they deepen experience, widen access, or make invisible processes visible. Avoid using them as decoration or as a substitute for authentic practice.

Institutions should apply five design tests before adopting virtual or AI-supported learning spaces:

1. Does the tool serve a clearly defined learning goal?
2. Does it require active student reasoning rather than passive viewing?
3. Does it connect to physical, social, or professional reality?
4. Does it strengthen human competencies such as collaboration, communication, judgment, and ethical responsibility?
5. Does it produce evidence teachers can interpret and use?

The Round Table recommendation is not physical instead of virtual, or virtual instead of physical. It is intentional integration. The learning environment should be rich enough to include simulation and material practice, AI support and human judgment, access and authenticity.

Practical Recommendation for Teachers

Use virtual, XR, VR, and AI-supported tools only when they serve a learning function that ordinary instruction cannot serve as well.

In practice:

1. Define the learning purpose before choosing the tool.
2. Use simulation to reveal hidden processes, rehearse difficult actions, or widen access.
3. Connect every virtual experience to discussion, explanation, or physical practice.
4. Ask students to name what the virtual model shows and what it leaves out.
5. Collect evidence of reasoning, not only evidence that students completed the experience.

Classroom activity examples:

1. Simulation plus reality. Students explore a virtual simulation, then compare it with a physical demonstration, data set, or real-world case.
2. What the model hides. After using an XR, VR, digital, or AI tool, students list what the tool made visible and what it simplified or excluded.
3. Design test for tools. Students evaluate a learning technology using five criteria: purpose, active reasoning, connection to reality, human skills, and evidence for the teacher.

Evaluation method:

Evaluate success through the student's ability to connect virtual or AI-supported experience to real understanding. The student should explain what the tool revealed, what it hid, and how the experience changed their reasoning. Strong work includes critique of the tool, not only enjoyment or technical use.

For your own development, test one digital or AI-supported environment as a learner, then write a short design note: what it made visible, what it hid, what student action it required, and how it should be assessed. This follows Dr. Iris Pinto's phygital and multimodal AI framing, Emlen Metz's distinction between productive and passive AI use, and Dr. Noa Ragonis's conclusion that virtual and physical laboratories should complement one another.

SOURCES

Meeting 6 summary, Emlen Metz's Meeting 6 presentation, Dr. Iris Pinto presentation, OECD Learning Compass 2030.

CHAPTER 19

Industry as a Learning Partner

The Challenge

STEM learning is often disconnected from the contexts where STEM is practiced. Students may master school tasks without seeing how knowledge operates in real work.

The disconnection is not a failure of enthusiasm. It is structural. School problems are bounded, single-answer, and scheduled to fit a lesson. Professional problems are underspecified, constrained by cost and time, contested between stakeholders, and solved by teams over months. A student can be excellent at the first kind and unprepared for the second. AI widens this gap, because the bounded school problem is exactly the kind of problem a language model solves best.

Dr. Iris Pinto described this gap from both sides. Having moved from Intel into the Ministry of Education, she saw how differently the two systems define a good outcome, how differently they treat failure, and how rarely they share a vocabulary. Industry treats a failed prototype as information. Education often treats a failed attempt as a lower grade.

The Dilemma

Should industry be an occasional guest in education, or an ongoing partner in designing and supporting STEM learning?

Behind this sits a sharper question about purpose. Education serves the learner across a whole life. Industry serves a market that changes in a few years. A partnership that lets the second define the first will produce narrow training that ages badly. A partnership that keeps industry at arm's length will produce learning that never meets a real constraint.

Main Positions

The guest model brings industry speakers and site visits. The mentorship model creates continuing relationships. The deep partnership model supports projects, externships, real problems, and feedback loops.

A fourth position, raised through Jan Morrison's TIES material, treats the question as one of system design rather than individual relationships. In a STEM ecosystem, industry is not a visitor to a school. Schools, colleges, employers, museums, community organizations, and out-of-school programs form a connected learning landscape, and a student moves through it. On this view, asking whether industry should visit is the wrong question. The question is whether the region has an ecosystem at all.

Pros and Cons

Guest visits are easy but limited. Mentorship creates human connection but requires continuity. Deep partnership gives authenticity, but must protect educational purpose and equity.

Equity is the risk most often underestimated. Partnerships form where the partners already are. Schools near a technology corridor accumulate mentors, equipment, and site visits. Schools in peripheral regions accumulate none of it. A partnership strategy that grows by enthusiasm alone will widen the gap it was meant to close, and only deliberate allocation, remote formats, and shared regional programs will prevent that.

Continuity is the second risk. Partnerships are usually built on one committed engineer and one committed teacher. When either moves on, the partnership ends. Partnerships survive when they are attached to a role and a calendar rather than to two individuals.

Point of View Map

Industry can provide real problems, tools, mentors, and professional context. Teachers need support translating industry experience into pedagogy. Students need exposure without narrow vocational tracking. Policy must define partnership models and safeguards.

For industry partners, the honest constraint is time. Engineers volunteer, and volunteering competes with delivery schedules. Partnerships that ask for a full day per month fail. Partnerships that ask for two hours per term, with the teacher doing the pedagogical translation, survive.

For school leaders, the constraint is liability, scheduling, and transport. These are unglamorous and decisive. A partnership that no one has timetabled does not exist.

For students, the risk is that a visit becomes recruitment. The purpose is to see how knowledge behaves under real constraints, not to be sorted into a career track at fifteen.

Round Table Voices

Jan Morrison emphasized externships, industry collaboration, and career-connected learning. Her argument was that teacher externships matter more than student visits, because a teacher who has spent two weeks inside a working engineering team redesigns every unit afterwards, while a student who visits for an afternoon remembers the building.

Avi Salmon brought the industry perspective on future skills and AI as a working tool. His point was that industry no longer needs graduates who can perform routine technical production, because AI performs it. Industry needs people who understand a problem deeply enough to know when the automated answer is wrong.

His Meeting 6 lecture made the partnership concrete rather than aspirational. The programs he described run in both directions. Inside the company, engineering upskilling happens through hands-on formats – a hardware challenge, a processor teardown workshop, a crash-course technical competition, an autonomous-driving contest – built on the finding that people who tolerate uncertainty and explore independently learn more than people who wait for the specification. Outside it, the same logic produces young technology leadership programs, students teaching younger students, and challenges that parents and children build together. The common element is that the learner produces something under real constraints, which is precisely what a site visit cannot deliver.

Jan Morrison pressed the point that matters most for scale: how should teacher education institutions prepare future teachers to use makerspaces and experiential learning, rather than concentrating only on the needs of teachers already in post? Avi Salmon answered that teacher educators have to experience it first. Until they have been learners in a makerspace themselves, they cannot integrate the approach into what they teach. Prof. Arnon Bentur asked the companion question – whether such programs should become part of formal education or stay external – and the answer was that they must become formal, with teacher confidence as the binding constraint.

Dr. Iris Pinto described the disconnect she observed between education and industry after moving from Intel into the Ministry of Education.

Dr. Avigdor Zonnenshain connected the same theme to lifelong learning: the professional context students meet in school should teach them that learning does not stop at graduation, because in industry it does not.

Synthesis

Industry should not replace education's broader purpose. But it can make STEM learning more authentic, current, and motivating.

The practical test of a partnership is what changes in student work. If the industry connection changed nothing about what students produced, revised, or understood, it was hospitality rather than education. If a student revised a design because of a real constraint, asked a better question because of a professional conversation, or understood why a method matters because they saw it used, the partnership did its work.

Recommendation

Build sustained education-industry partnerships around mentoring, real problems, teacher externships, student projects, site visits, and feedback loops.

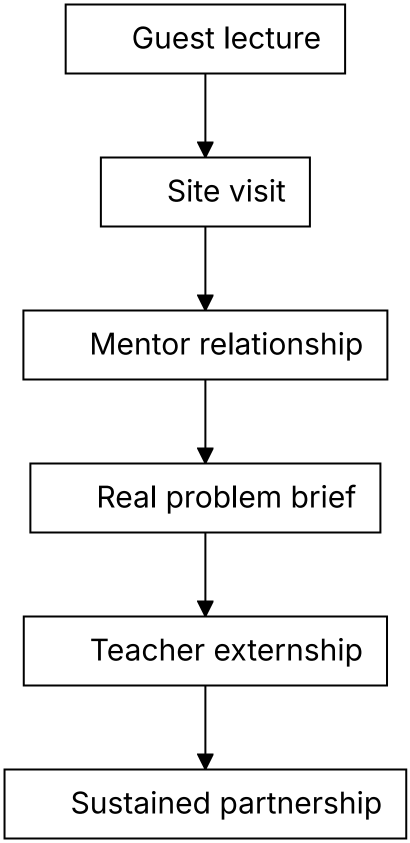

Figure 12. The partnership ladder, from one-off visit to sustained collaboration.

Practical Recommendation for Teachers

Bring industry into learning as context, not decoration. A guest lecture is useful only if it connects to student work before and after the visit.

In practice:

1. Identify one unit where real professional context would deepen learning.

2. Invite an industry partner to provide a problem, constraint, example, or feedback.
3. Prepare students with disciplinary background before the interaction.
4. Ask students to produce something after the interaction: revised design, reflection, question set, or presentation.
5. Debrief what students learned about knowledge, teamwork, tools, uncertainty, and professional judgment.

Classroom activity examples:

1. Real constraint brief. An industry partner or teacher provides a realistic constraint, such as cost, reliability, user need, safety, or time, and students revise their solution accordingly.
2. Professional question clinic. Students prepare questions for a guest from industry, then turn the answers into design criteria or career-learning reflections.
3. Feedback from practice. Students present a project to an external mentor or professional and revise one element based on feedback.

Evaluation method:

Evaluate success by looking at how students use professional context to improve their work. The student should ask relevant questions, respond to constraints, revise based on feedback, and explain what changed because of the industry connection. The assessment should protect learning depth, not reward contact with industry as a symbolic event.

For your own development, build one sustained professional connection per year: an externship, site visit, mentor relationship, or recurring project partner. This follows Jan Morrison's emphasis on externships and career-connected learning, Avi Salmon's industry perspective, and Dr. Iris Pinto's warning about the disconnect between education and industry.

SOURCES

Meeting 2 presentation, TIES Jan Morrison presentation, Meeting 4 presentation, Meeting 6 summary, Avi Salmon's Meeting 6 presentation.

CHAPTER 20

Ecosystem Around the Teacher

The Challenge

Expecting individual teachers to transform STEM education alone is unrealistic.

Almost every recommendation in this book adds something to a teacher's workload. Design tasks so thinking is visible. Learn the tools before setting policy. Collect evidence. Build partnerships. Reflect each semester. Taken one at a time, each is reasonable. Taken together, they describe a job no one has time to do. This is the point at which reform proposals usually fail, and it fails quietly: teachers do not refuse, they simply run out of hours, and the reform survives as language in a document.

The TALIS material named the same problem from the teacher's side. Workload, pressure, behavior, diverse learning needs, and constant change are job demands. Learning opportunities, agency, leadership, support, and professional status are resources. When demands rise and resources do not, the system does not get transformation. It gets exhaustion.

The Dilemma

Is skill development mainly the teacher's responsibility, or does it require a full ecosystem around the teacher?

Main Positions

Teacher-centered responsibility honors professional agency. Institution-centered responsibility creates structure. Ecosystem responsibility distributes support across peers, leaders, technology, policy, industry, and community.

The strongest objection to the ecosystem position deserves stating plainly: when responsibility is shared among everyone, it can end up held by no one. "The ecosystem should support teachers" is a sentence with no subject who can be asked what they did about it. An ecosystem is only real if each part of it has a name, a budget line, and something it can be held to.

Pros and Cons

Teacher-centered responsibility empowers educators, but can overload them. Institution-centered responsibility creates support, but can become top-down. Ecosystem responsibility is strongest, but requires coordination.

Coordination is not free. Every additional partner adds meetings, alignment work, and points of failure. An ecosystem with seven weak links is worse than three strong ones. The practical advice is to build the ecosystem one link at a time and to strengthen a link before adding another.

Point of View Map

Teachers need support, not only expectations. Leaders must create enabling conditions. Industry and community can contribute context. Policy must support continuity beyond individual champions.

The Envisioning Educator Roles for Transformation material adds a practical design discipline to this ecosystem view. New educator roles should not be invented as slogans. They should be specified through concrete questions: what is the role called, where does it sit in the system, what responsibilities does it carry, what is it not responsible for, what qualifications are required or preferred, what compensation and working conditions make it sustainable, and what happens if the role is or is not filled. This turns the idea of an ecosystem into an organizational design task.

The question about what a role is *not* responsible for is the one most often skipped, and it is the one that protects teachers. An ecosystem that only adds responsibilities is not a support system. It is a longer list.

Round Table Voices

Meeting 6 asked whether the teacher alone can carry the responsibility or whether a full ecosystem is needed.

Dr. Avigdor Zonnenshain emphasized education that fosters lifelong learning and team-driven practice, and brought a systems-engineering habit to the discussion: a system that depends on the exceptional performance of its individual components is a badly designed system.

Dr. Eli Eisenberg and Prof. Arnon Bentur framed the overall Phase III process around the future educator profile and its system conditions. Their sequence of dilemmas was deliberate: profile, competencies, AI, professional development, environments. Each earlier answer creates a system requirement that a later one has to meet.

Jan Morrison's ecosystem argument from Meeting 2 returns here in a different form. She argued that transdisciplinary and problem-based learning fail when inserted into systems built for isolated individual teaching. The ecosystem is not an enhancement to the reform. It is the precondition for it. At the closing session she extended the same logic to the Round Table itself, urging it to connect with other international round tables and professional networks so that expertise is exchanged, effort is not duplicated, and the group's work reaches practice. A group that argues for ecosystems should not operate as an island.

Dr. Eli Eisenberg's closing remarks drew the six meetings together around the same claim. Preparing learners for the future means rethinking not only which competencies students should develop, but the roles of teachers, higher education, industry, and the learning environment at the same time. Three findings recurred: there is no single ideal educator profile, and effective education may need diverse and complementary roles; teachers need to develop the competencies they are expected to cultivate, particularly around AI and experiential learning; and learning should increasingly incorporate authentic workplace experience, industry mentors, and hands-on activity. He also insisted that the outcome be a product rather than a conversation – a draft report circulated for comment, finalized collectively, and published.

Synthesis

The teacher remains central, but not solitary.

Central means the teacher still holds the judgment that cannot be delegated: what this student needs, whether this evidence is trustworthy, what learning is for. Not solitary means the system carries the load that does not require that judgment: materials, tools, time, structure, evaluation capacity, and partnership infrastructure. Reform fails when these are reversed, when the system dictates the judgment and leaves the teacher to find the time.

Recommendation

Define the AI-era educator as the center of a designed support ecosystem: peers, leaders, mentors, AI tools, teaching centers, industry, policy, and community.

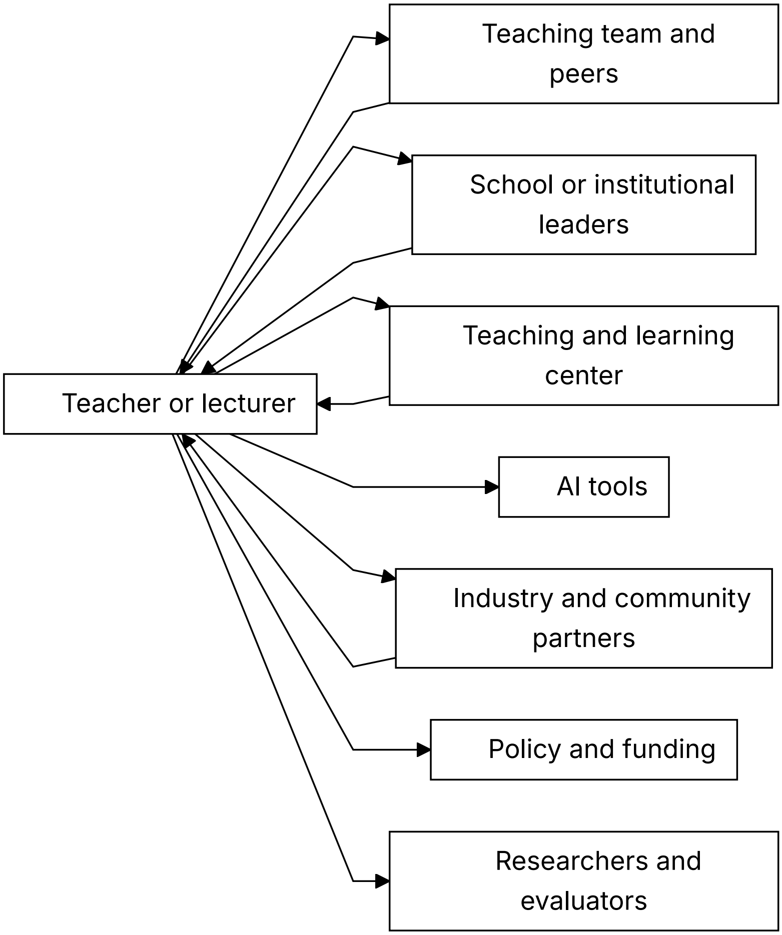

Figure 13. The support ecosystem around the AI-era educator. Arrows in both directions mark the relationships that must be reciprocal.

Practical Recommendation for Teachers

Build a support map around your teaching. You are central, but not solitary.

In practice:

1. List the people and resources that currently support your teaching.
2. Identify one missing support: peer planning, leadership time, AI guidance, assessment help, industry contact, community partner, or teaching center support.

3. Ask for or create one concrete support structure.
4. Define the missing support as a real role: responsibility, boundaries, qualifications, time, and working conditions.
5. Share one teaching challenge with the ecosystem instead of solving it alone.
6. Revisit the map each semester and strengthen weak links.

Classroom activity examples:

1. Learning ecosystem map. Students map who and what supports their learning: teacher, peers, family, AI tools, mentors, community, school systems, and industry.
2. Peer support protocol. Students rotate roles as explainer, checker, questioner, and reflector, showing that learning is distributed across a group.
3. Support request exercise. Students identify one support they need to learn better and propose who in the ecosystem could provide it.

Evaluation method:

Evaluate success through the student's ability to use support responsibly. The student should identify available supports, seek help appropriately, contribute to peers, and explain how the ecosystem improved learning. Strong performance includes both independence and wise use of others.

For your own development, move from self-improvement to ecosystem-building. This means developing personal competence while also creating conditions that help colleagues and students. It follows Dr. Avigdor Zonnenshain's system perspective and the Meeting 6 question of whether skill development belongs to the teacher alone or to a full support system.

SOURCES

OECD Teaching Compass, Envisioning Educator Roles for Transformation, Meeting 6 summary.

PART VII

A Compass for STEM Education in the AI Era

CHAPTER 21

Insights Across the Six Meetings

The six meetings point to a coherent set of insights.

First, the future educator profile should be common in foundation but diverse in expression. Every educator needs a baseline of AI literacy, ethical judgment, pedagogical responsibility, and commitment to student agency. But not every educator needs the same strengths or roles.

Second, competencies must be embedded in disciplinary learning. Skills become real through practice, reflection, feedback, and repeated use in meaningful contexts.

Third, teachers need lived experience with the competencies they cultivate, but they do not need to be perfect models of everything. The system should distribute expertise.

Fourth, AI must be designed into learning. Ignoring it or banning it will not work. The question is how to make thinking visible and keep responsibility human.

Fifth, professional development must become infrastructure. Workshops are not enough. Teachers and lecturers need long-term support.

Sixth, environments matter. Physical space, virtual tools, industry connections, mentoring, organizational structures, and assessment systems all shape what learning is possible.

Seventh, equity must be designed from the start. AI can widen gaps if advantaged students use it to amplify thinking while disadvantaged students use it to replace thinking.

PHASE III ROUND TABLE JOURNEY

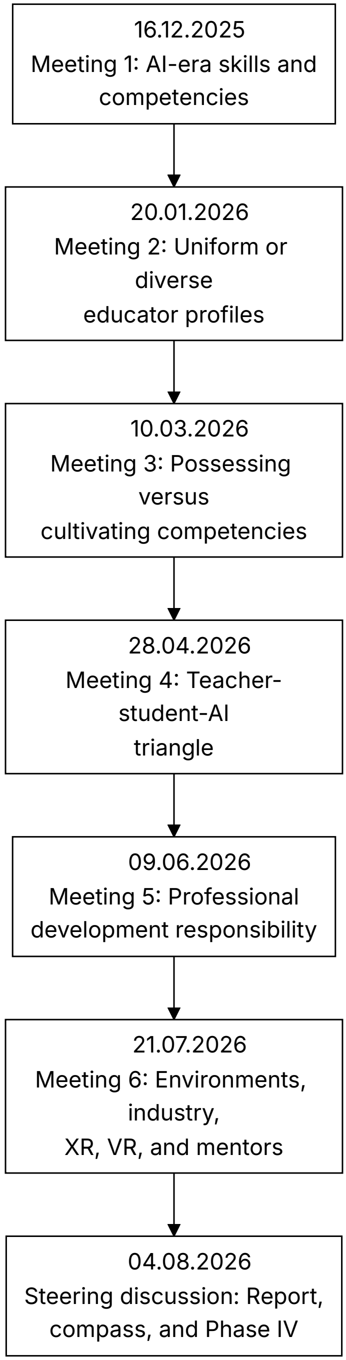

Figure 14. The six-meeting journey of the Round Table, and the question each meeting opened.

Practical Recommendation for Teachers

Use the seven insights as a personal professional compass. Do not try to act on all of them at once. Choose one insight each term and translate it into a visible classroom practice.

In practice:

1. Select one insight that is most urgent for your students.
2. Define one teaching behavior that would make the insight real.
3. Design one student task that reflects it.
4. Collect evidence that the practice changed learning.
5. Share the result with colleagues and revise.

Classroom activity examples:

1. Six insights reflection. Students choose one insight, such as visible thinking or equity by design, and explain how the class practiced it during the unit.
2. Human above machine review. After an AI-supported task, students identify where human judgment mattered most.
3. Equity check discussion. Students examine whether a learning tool or activity helped all learners equally, and propose one adjustment.

Evaluation method:

Evaluate success through synthesis. The student should connect classroom experience to the larger Round Table insights: human judgment, competencies, evidence, equity, AI responsibility, and support systems. Strong work shows that the student can move from one activity to a broader principle.

For your own development, review the six-meeting insights at the end of each semester and ask: where did I keep human judgment central, where did I make thinking visible, where did I support equity, and where did I still work alone? This turns the Round Table synthesis into a recurring teacher reflection tool.

CHAPTER 22

Guiding Principles

The following principles express the spirit of the Round Table:

1. Keep the human above the machine.
2. Treat AI as part of the learning ecology, not as a replacement for teachers.
3. Preserve disciplinary rigor while embedding human competencies.
4. Design for visible thinking, not polished outputs.
5. Build diverse educator ecosystems, not one impossible ideal profile.
6. Give teachers and lecturers lived experience with new learning models.
7. Make professional development continuous and supported.
8. Connect STEM learning to real-world practice and industry.
9. Use physical, virtual, and AI environments purposefully.
10. Evaluate reforms with evidence and feedback loops.
11. Design for equity from the start.
12. Build institutions for lifelong learning.

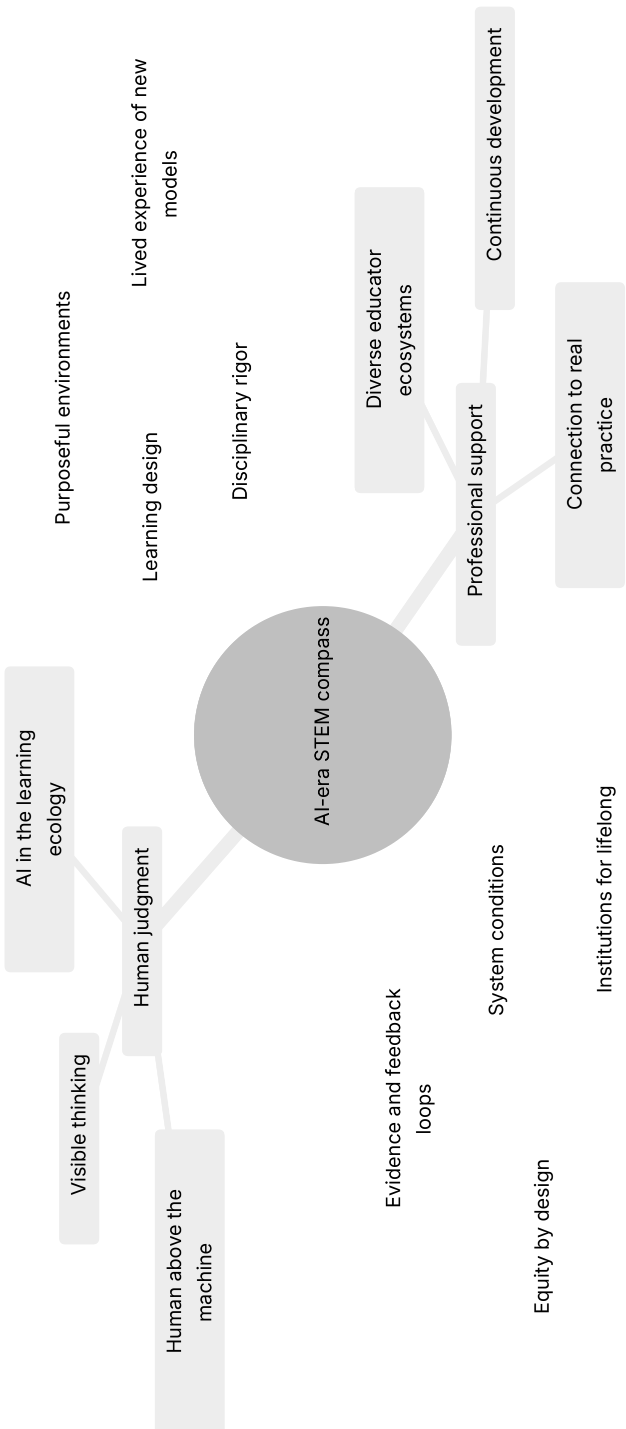

Figure 15. The compass: all twelve guiding principles, grouped into four bearings.

Chapter 23 turns these principles into a single teachable sequence.

Practical Recommendation for Teachers

Turn the guiding principles into a short design checklist before launching a major task, project, or AI-supported activity.

In practice:

1. Ask whether the task keeps human judgment above AI.
2. Check whether disciplinary rigor is preserved.
3. Identify where student thinking will be visible.
4. Decide how equity risks will be reduced.
5. Plan how evidence from the task will improve the next version.

Classroom activity examples:

1. Principle-to-task checklist. Before a project, students and teacher check which guiding principles are built into the task and which are missing.
2. Visible thinking checkpoint. Students pause mid-task to show evidence of reasoning, uncertainty, tool use, and next steps.
3. Principle evidence gallery. At the end of a unit, students display artifacts that show one guiding principle in action.

Evaluation method:

Evaluate success by asking students to produce evidence for a principle, not only repeat the principle. A student succeeds when they can show an artifact, decision, revision, or reflection that proves visible thinking, responsible AI use, collaboration, equity, or another guiding principle in action.

For your own development, choose three principles as annual growth priorities. For example: visible thinking, lived experience with AI, and continuous professional learning. At the end of the year, gather evidence of progress for each. This makes the book's compass operational rather than decorative.

CHAPTER 23

The Human-First AI Learning Cycle

The guiding principles become practical when teachers design a learning sequence in which students meet the world before they meet AI. AI should enter after observation, curiosity, early reasoning, measurement, and prediction. It should help students expand, compare, model, and validate understanding, not replace the human work of learning.

This is the book's single most transferable proposal. Everything else in these pages describes what should be true of AI-era STEM education. This chapter describes an order of operations that a teacher can use on Monday.

Why Order Matters

The sequence is the whole argument. The same ten activities in a different order produce a different education.

If AI comes first, it supplies the frame. The student receives a vocabulary, a model, and a set of expected answers before they have looked at anything themselves. Every later observation is then read through that frame. The student is not wrong, exactly, but the thinking has been outsourced at the only point where it was formative.

If AI comes after the student's own explanation, its role inverts. It becomes something to compare against, argue with, and check. The student now has a position, which means they have something to defend or revise. This is the difference between a tool that thinks for you and a tool that gives you something to think about.

A Worked Example

This cycle can be used in many STEM topics. A simple example is a swing in a park. Students first observe the swing, form explanations with the teacher, measure the motion, make predictions, and only then use AI to deepen the model, build a simulator, and test whether the simulator fits reality.

The example is deliberately humble. A swing needs no laboratory, no budget, and no specialist knowledge to observe. Students will notice that pushing harder does not change the timing much, which contradicts almost everyone's first intuition. That contradiction, discovered before AI enters, is the engine of the whole sequence. A student who has been surprised by a swing will interrogate an AI explanation of a pendulum. A student who was told about pendulums first will not.

The same structure works for a falling object, a circuit that dims, a plant that grows toward light, a bridge made of paper, a queue at the cafeteria, or a set of local weather records.

The Ten Steps

1. Observe the phenomenon. Begin with the real world. Students watch, sketch, describe, photograph, or record what happens before naming formulas or asking AI.
2. Form a human thesis. Students explain what they think is happening and why, with teacher guidance and peer discussion. The first explanation may be incomplete, but it must be theirs.
3. Experiment and measure. Students collect evidence through measurement, comparison, controlled change, or repeated observation.
4. Predict before AI. Students use their own explanation and data to predict what will happen if one condition changes. They also state what evidence would prove them wrong.
5. Use AI as a learning partner. Students ask AI to explain the topic, identify variables, suggest common mistakes, or propose a simple model. They compare the AI answer with their own evidence.

6. Explain verbally what was learned. Students explain the phenomenon in their own words before building or submitting anything AI-assisted.
7. Build a simulator with AI. Students use AI to help create a model, spreadsheet, code simulation, Scratch activity, GeoGebra model, or other representation of the phenomenon.
8. Validate the simulator against reality. Students compare the simulator with their observations and measurements. They identify where the model fits, where it fails, and which assumptions are too simple.
9. Present verbally and defend the learning. Students present the full path: observation, thesis, experiment, prediction, AI learning, simulator, validation, and conclusion.
10. Reflect on human responsibility. Students explain how they stayed above AI: what they understood before AI, what AI helped clarify, what they checked themselves, and where they made the final judgment.

The short teacher version is:

1. Observe reality.
2. Form a human explanation.
3. Measure or experiment.
4. Predict before AI.
5. Use AI to learn and compare.
6. Explain verbally.
7. Build a simulator with AI.
8. Validate against reality.
9. Present and defend.
10. Reflect on human responsibility.

What Can Go Wrong

Three failure modes are common enough to name.

The first is the ceremonial thesis. Students are asked for their own explanation, give one in a sentence, and move on. Nothing has been committed to, so nothing can be revised. The fix is to make the thesis specific and falsifiable: not "it depends on the weight" but "if we double the weight, the time will halve."

The second is the validation that never happens. Steps seven and eight are the easiest to build and the easiest to skip. A simulator that is never compared against the measurements is a decorative artifact. The comparison, including the places where the model fails, is the learning.

The third is compression. Ten steps look like ten lessons, and no timetable has room. In practice the cycle runs in two to four lessons, because steps one to four are fast when the phenomenon is simple. If the cycle needs a month, the phenomenon was too ambitious.

Where It Fits in a Course

The cycle is not a replacement for direct instruction, and it should not be used for everything. Novices learning a new formalism need explicit teaching, worked examples, and practice. The evidence on this is strong and Prof. Sigal Tifferet's challenge in Chapter 24 applies here directly.

The realistic pattern is one full cycle per unit, positioned where the discipline has a phenomenon worth being surprised by, surrounded by ordinary instruction and practice. Two or three per term is enough to change what students believe learning is for.

Practical Recommendation for Teachers

Run the cycle once, on a topic you already teach, before redesigning anything else.

In practice:

1. Choose a phenomenon students can observe directly, in the room or nearby.
2. Ask for a written prediction with a number in it, before any tool is opened.
3. Measure, and let the measurement disagree with the prediction.
4. Only then allow AI, and require students to state where it agrees and disagrees with their data.

5. Ask each student to explain the phenomenon aloud, without notes, before any AI-assisted artifact is submitted.
6. End with the reflection: what did I understand before AI, what did AI clarify, what did I verify myself.

Classroom activity examples:

1. Prediction envelope. Students seal a written prediction before the experiment and open it after the measurement. The gap between the two becomes the lesson.
2. Disagreement log. While using AI, students record every point where the AI explanation and their own evidence do not match, then resolve each one.
3. Model autopsy. After building a simulator, students list three ways the model is wrong and explain why each simplification was acceptable or not.

Evaluation method:

Evaluate the path, not the artifact. A strong performance shows a specific initial thesis, evidence that was actually collected, an identified disagreement between the student's understanding and the AI output, a resolution of that disagreement, and a clear statement of what the student verified personally. A polished simulator with no disagreement log is weak work, however impressive it looks.

For your own development, run the cycle on yourself first, on a topic you know well. Notice how strongly the AI explanation pulls you toward its framing, even when you have taught the topic for years. That pull is what the sequence is designed to resist.

CHAPTER 24

The Case Against This Book

A book that asks students to state a position before checking it against a stronger one should do the same. This chapter sets out the strongest arguments against the case made here, without softening them.

The Competency Claims Are Under-Evidenced

The central claim of Part I is that STEM education should reorganize around a set of transferable human competencies. Prof. Sigal Tifferet asked the question that this book has not fully answered: is that evidence-based?

The honest answer is partly. There is good evidence that critical thinking instruction works when it is domain-specific and explicit. There is much weaker evidence that critical thinking taught inside a discipline transfers to other disciplines, and transfer is precisely what the competency argument assumes. Half a century of research on transfer of learning is, at best, mixed. A recommendation to redesign national curricula around transferable competencies is a large bet placed on a contested literature.

This book proceeds anyway, on the grounds that the alternative, assessing students only on outputs AI can produce, is clearly failing. But "the current approach is failing" is an argument for change, not evidence that this particular change works.

Explicit Instruction Usually Beats Discovery for Novices

The book repeatedly recommends inquiry, projects, problem-based learning, and student-led investigation. The research on novice learners points the other way. For students who lack domain knowledge, minimally guided instruction is generally less effective than explicit instruction with worked examples, because novices lack the schemas needed to direct their own search and their working memory is consumed by the search itself.

The Human-First Learning Cycle in Chapter 23 is vulnerable to this objection. Asking a student to form a thesis about a phenomenon they have no conceptual vocabulary for may produce confident nonsense rather than productive struggle. The defense is that the cycle is bounded, guided, and followed by instruction, not that discovery works.

A reader who takes this objection seriously should use the cycle sparingly, on phenomena that are directly observable, and should not conclude that projects can replace teaching.

"Ecosystem" Can Mean Nobody Is Responsible

Chapter 20 states this objection and then argues past it. It deserves to be stated once more, harder.

Distributed responsibility is the standard way institutions avoid accountability. Every actor in an ecosystem can point to another actor. Policy blames funding, funding blames priorities, leadership blames policy, and the teacher is still alone in the classroom at eight in the morning. The ecosystem framing is attractive to everyone except the person it is supposed to help, because for everyone else it converts a demand into a diagram.

The book's answer, that each part of the ecosystem needs a name, a budget line, and something it can be held to, is correct and almost never implemented.

The Workload Arithmetic Does Not Work

This book makes something close to a hundred practical recommendations. Each is small. Collectively they describe a teacher who redesigns assessment, learns new tools, collects evidence, builds partnerships, mentors colleagues, runs inquiry cycles, and reflects each semester, on top of teaching a full load.

No serious costing accompanies any of it. Neither does any account of what should be removed to make room. A reform program that only adds is not a program. It is a wish, and teachers have learned to recognize the difference.

The most useful thing a reader can do with this book is to pick two recommendations and ignore the rest.

Nothing Here Scales While the Gatekeepers Reward Something Else

Danielle Eisenberg put this to the final meeting and the book has no adequate answer to it. Innovation in education does not fail for lack of good programs. It fails because it cannot scale, and it cannot scale because assessment systems, university admissions, and employer hiring practices continue to reward traditional academic achievement.

Every recommendation in this book asks a teacher, a lecturer, or an institution to invest in capabilities that the gatekeeping systems do not measure. A student who spends a term building judgment, collaboration, and the habit of checking a tool still faces an admissions process that reads a grade. A school that redesigns assessment around visible thinking still reports the same examination results. The incentive runs against the argument at every point where the argument would have to become general.

This book's answer is that the measurement systems should change, which is true and is not a plan. Until matriculation, admissions, and hiring recognize the competencies described here, the model in these pages will keep working in individual classrooms and keep failing to become the system.

So the question stays open, and it is better stated plainly than buried. What would make an examination board, an admissions officer, or an employer actually value the competencies described in this book? Not what should make them, which is easy to assert and costs nothing. What would. Who moves first, what evidence would persuade them, and what does a school do in the years before any of that happens.

The Round Table did not answer that, and neither does this book. It is left here as an open question, and it is the most important one this work hands on.

The Evidence Base Is Uneven and Partly Retrospective

Meetings three and five left no primary record of the session itself. Part III rests largely on a discussion document written before Meeting 3, which means it reports the questions the organizers posed rather than the answers the room gave. Part V rests on steering-team summaries written after Meeting 5. Several attributed positions in both parts are therefore reported at second hand. The book says so in its Editorial Note, but a reader should hold the attributions accordingly: these are reported contributions, filtered through a summarizer, then synthesized by an editor who was himself a participant.

AI Capability May Outrun the Argument

The book's core move is to identify what remains human when machines can produce fluent answers: judgment, responsibility, meaning, and the decision about when not to trust a tool. That boundary is drawn at a moment in time, and it has moved before.

If systems become substantially better at calibrated uncertainty, at knowing what they do not know, and at explaining their reasoning in ways that survive checking, some of the tasks reserved here for humans will not stay reserved. The claim that responsibility cannot be delegated to a machine is a claim about accountability, which is a social and legal fact rather than a technical one. Social and legal facts also change.

What Survives

Taking all of this seriously, a smaller claim survives, and this book stands behind it:

Producing a correct-looking answer no longer demonstrates learning. Educators therefore need new ways to see whether a student understands. Making thinking visible, requiring explanation, and asking students to check tools rather than trust them are reasonable responses to that specific problem, whatever else turns out to be true.

That is a narrower thesis than the one in the Preface. It is also the part most likely to still be right in ten years.

CHAPTER 25

Recommendations by Audience

For Teachers

Use AI to deepen questioning, critique, and reflection. Do not use it only to produce better-looking answers. Ask students to explain their reasoning, compare alternatives, identify errors, and reflect on what changed in their understanding.

For Lecturers

Redesign courses around graduate profiles and competencies. Connect foundations to application. Use AI-aware assessment that asks for process evidence, not only final submissions.

For School Leaders

Create time and structures for teacher collaboration. Support project-based and experiential learning. Build AI policies around learning design.

For Higher-Education Leaders

Strengthen teaching and learning centers. Map competencies across programs. Support faculty development as a long-term institutional priority.

For Policy Makers

Define shared goals and infrastructure. Fund implementation support, not only training events. Support mechanisms for continuous professional development.

For Industry Partners

Provide mentors, real problems, externships, site visits, and feedback. Support education without narrowing it to short-term labor needs.

For Researchers

Study AI-era learning processes, teacher development, assessment, equity, and long-term outcomes.

Audience	Main action	Evidence to collect
Teachers	Make thinking visible and guide AI use	Student reasoning, drafts, reflection
Lecturers	Redesign courses around graduate profiles	Competency maps and course artifacts
School leaders	Create time and structures for collaboration	Participation, implementation, student work
Higher-education leaders	Strengthen teaching centers and incentives	Program evidence and faculty development records
Policy makers	Fund infrastructure and continuity	System adoption and equity indicators
Industry partners	Provide real problems, mentors, and feedback	Student revisions and partnership continuity
Researchers	Study process, outcomes, and equity	Mixed evidence and long-term follow-up

Figure 16. What each audience should do next, and what evidence would show it worked.

Practical Recommendation for Teachers

Translate audience recommendations into a local action plan. A teacher does not control every audience, but can initiate the first step with each one.

In practice:

1. For students, redesign one task so AI deepens questioning rather than replaces thinking.
2. For colleagues, propose one shared planning or peer review session.
3. For leaders, ask for one enabling condition: time, space, tool access, or assessment support.
4. For industry partners, request one authentic problem, visit, mentor, or feedback opportunity.
5. For researchers or evaluators, frame one question about what is working in your classroom.

Classroom activity examples:

1. Audience redesign. Students adapt one explanation of their work for three audiences: peer, teacher, and industry or community partner.
2. Stakeholder feedback round. Students identify who would care about their project outcome and seek feedback from at least one relevant audience.
3. Recommendation workshop. Students write practical recommendations from their project for teachers, leaders, policy makers, or industry partners.

Evaluation method:

Evaluate success through audience awareness and practical recommendation quality. The student should adapt language, evidence, and emphasis for different stakeholders. Strong work includes a recommendation that is specific, realistic, and grounded in what was learned.

For your own development, become a bridge among audiences. Keep a small stakeholder map and update it each semester. This reflects the Round Table's repeated message that AI-era STEM education is not a private classroom matter. It is a coordinated professional, institutional, and social effort.

CHAPTER 26

Open Questions and Phase IV

The Round Table did not close every question. It opened the next layer.

Dr. Eli Eisenberg set out the direction in his closing remarks at the sixth meeting, and invited participants to shape it collectively rather than receive it from the organizers. Three themes were proposed. The first is how rising global uncertainty – technological disruption, geopolitical instability, and social change – affects not only cognitive competencies but the social and emotional ones that learners and professionals depend on. The second is the relationship between top-down policy and bottom-up innovation: rather than waiting for ministries to define change, the next phase should study how initiatives that already work in schools, universities, industry, and local communities can inform policy. The third is STEM leadership – moving past the preparation of technically expert engineers and scientists to the leadership capability needed to guide organizations, education systems, and innovation ecosystems. Jan Morrison added a fourth direction that is procedural rather than thematic: connect the Round Table to other international round tables and professional networks, so that expertise is shared and effort is not duplicated.

Future work should therefore examine STEM leadership in the AI era, uncertainty and resilience, the connection between bottom-up innovation and top-down policy, systems thinking and multidisciplinarity, and the role of values and attitudes alongside skills.

The next phase should move from defining the AI-era educator profile to designing the AI-era STEM education system: leadership, institutions, policy, practice, tools, environments, and evidence.

Practical Recommendation for Teachers

Treat the open questions as an invitation to inquiry. Do not wait for all answers before improving practice, but do not pretend the questions are settled either.

In practice:

1. Choose one open question that matters in your context: STEM leadership, resilience, bottom-up innovation, systems thinking, values, or attitudes.
2. Turn it into a classroom or course experiment.
3. Define what evidence would show progress.
4. Involve students in reflecting on the question.
5. Share the result with a professional community so practice can inform the next phase.

Classroom activity examples:

1. Open question inquiry. Students choose one unresolved question, such as resilience, STEM leadership, values, or systems thinking, and run a small inquiry project.
2. Bottom-up innovation lab. Students identify a problem in their learning environment and prototype a practical improvement.
3. Values and AI scenario. Students analyze a future STEM scenario and identify the skills, values, and attitudes needed to act responsibly.

Evaluation method:

Evaluate success through inquiry quality and responsible imagination. The student should frame a meaningful question, gather evidence, test or prototype an idea, and reflect on values and consequences. Strong work does not claim final certainty. It shows disciplined curiosity and a practical next step.

For your own development, adopt an inquiry stance: learn, test, document, share, and revise. This fits the Phase IV direction suggested by the steering discussion, where the next challenge is not only to define the educator profile but to design the wider STEM education system around leadership, practice, tools, environments, values, and evidence.

Appendices

APPENDIX A

Contributors and Participants

This appendix records who took part in the International STEM Skills Round Table Phase III. Entries describe each person's role in the discussion and, where a verified public profile exists, their wider work. Where no public profile was available, the entry describes the Round Table contribution only and makes no claim about publications or career.

Dr. Eli Eisenberg

Role in the Round Table: Initiator, organizer, opening speaker, moderator, and co-author of the Meeting 3 and Meeting 4 discussion documents. He opened Meeting 2, Meeting 4, and Meeting 6, introduced later meeting dilemmas, delivered the closing remarks that set the Phase IV agendaing dilemmas, delivered the closing remarks that set the Phase IV agendaing dilemmas, delivered the closing remarks that set the Phase IV agenda, and helped frame the Round Table as an active professional inquiry rather than only a documentation project. He is not related to Danielle Eisenberg, the Meeting 4 speaker listed later in this appendix.

Bio: The Samuel Neaman Institute profile identifies Dr. Eisenberg as a Senior Research Fellow specializing in technological and vocational education and training. His education includes a PhD and MSc in Education in Technology and Science from the Technion and a BSc in Electrical Engineering from the Technion.

Main achievements, publications, and interests: His public profile describes him as a leading pedagogue in TVET, consultant to governments and international organizations, former Senior Deputy Director General of ORT Israel and Head of its Administration for R&D and Training, and author of many publications, research reports, and books in technology education and ICT in education and training. Relevant listed publications and projects include the National Skills Interface Program in STEM Subjects, Faculty Attitudes Toward Integration of Core Skills and Competencies into Academic Courses, the International Round Table for Advancing Skills in STEM Education Phase I and Phase II, Academy Interface, active online education, and engineering education for Industry 4.0.

Prof. Arnon Bentur

Role in the Round Table: Senior conceptual framer and moderator. The Meeting 2 program assigns him the presentation of the key dilemmas and moderation of the discussion, and he moderated both question sessions at Meeting 6, and he moderated both question sessions at Meeting 6, and he moderated both question sessions at Meeting 6. He is also named as a co-author of the Meeting 3 and Meeting 4 discussion documents.

Bio: Prof. Bentur is one of the Round Table's senior academic leaders. Samuel Neaman Institute publication listings name him as a co-author on education, engineering education, academy-school interface, skills, competencies, lifelong learning, and STEM excellence work.

Main achievements, publications, and interests: Neaman publication listings connect him to the International Round Table Phase I and Phase II, the Skills Interface Program in STEM, faculty attitudes toward integrating competencies into academic courses, academy-school interface work, engineering education in the 21st century, digital engineering, lifelong learning, and university-industry relations. His contribution to the book is the systemic framing of educator profile, competencies, professional development, and learning environments as connected dilemmas rather than isolated topics.

Dr. Avigdor Zonnenshain

Role in the Round Table: Senior systems and engineering-education voice. The Meeting 2, Meeting 4, and Meeting 6 programs assign him the meeting summary role, and this book uses his comments on lifelong learning, teamwork, industry realities, system design, evidence, and learning by doingevidence, and learning by doingevidence, and learning by doing.

Bio: The Samuel Neaman Institute profile identifies Dr. Zonnenshain as a Senior Research Fellow specializing in quality, reliability and safety management, systems engineering and management, systems thinking, advanced manufacturing, engineering education, regional strategic development, and social responsibility. He holds a PhD in Systems Engineering from the University of Arizona, an MSc in Operations Research from the Technion, and a BSc in Industrial Engineering from the Technion.

Main achievements, publications, and interests: His public profile describes senior roles in quality, reliability, project management, and systems engineering at Rafael, leadership of the Center for Quality and Excellence in the Prime Minister's Office, work at the Technion Gordon Center for Systems Engineering, chairing the Standardization Committee for Management and Quality at the Israel Standards Institute, and recognition as an INCOSE Fellow in 2010. Relevant listed publications and projects include AI Integration in Engineering and Systems Engineering, Engineering Leadership: Skills and Competencies, engineering education in the 21st century, digital engineering, lifelong learning, advanced manufacturing, and the methodology of generating information from round-table discussions.

Tamar Dayan

Role in the Round Table: Research, information, and source-infrastructure contributor. The Neaman public pages list her as a member of the International Round Table for Advancing STEM Excellence Skills and as a co-author on Phase I, Phase II, and related skills publications.

Bio: The Samuel Neaman Institute profile identifies Tamar Dayan as a researcher and information specialist at the Samuel Neaman Institute, with specializations in information science. Her education includes an MLS from the University of Haifa and a BA in Education and General Studies from the University of Haifa.

Main achievements, publications, and interests: The public profile notes her participation in projects on Intel Israel's contribution to the national economy, engineering education in the 21st century, economic infrastructure in Northern Israel, continuing education for STEM graduates, the Forum for Engineering Education in the 21st Century, the Industry Forum of the Center for Industrial Excellence, and the International Round Table for Advancing STEM Excellence Skills. Relevant listed publications include Faculty Attitudes Toward Integration of Core Skills and Competencies into Academic Courses, the International Round Table Phase I and Phase II reports, Digital Engineering and its Assimilation in Engineering Education, and Academy Interface.

Avi Salmon

Role in the Round Table: Industry perspective contributor, Meeting 6 speaker, and co-author of the Meeting 3 and Meeting 4 discussion documents. This book uses his arguments about deep understanding, curiosity, authentic professional engagement, AI as a working tool, makerspaces, evidence-based program evaluation, and the principle that teachers and lecturers must remain above AI. He is also the editor of this book, which means his own contributions appear here more often than a neutral count would justify; readers should weigh them accordingly.

Bio: Avi Salmon took part as an industry voice from the Innovation Department of Intel Israel.

Main achievements, publications, and interests: His Meeting 6 presentation, Training by Experience, describes engineering upskilling through hands-on formats and school-facing programs in which students teach younger students or build alongside their parents, and names curiosity, determination, and humility as the qualities that decide who learns in an AI-rich environment. His Meeting 6 presentation, Training by Experience, describes engineering upskilling through

hands-on formats and school-facing programs in which students teach younger students or build alongside their parents, and names curiosity, determination, and humility as the qualities that decide who learns in an AI-rich environment. Within the Round Table, his areas of interest are AI-era STEM learning, industry-relevant skills, deep conceptual understanding, curiosity, program evaluation, experiential learning environments, makerspaces, and practical AI use.

Dr. Marwa Maklada

Role in the Round Table: Assessment and policy contributor. This book uses her contributions on competency-based tasks, formal assessment, matriculation processes, and the claim that AI should not design learning or control knowledge.

Bio: Dr. Maklada contributed from the Ministry of Education and from the assessment and education-system perspective.

Main achievements, publications, and interests: The Dr. Eli Eisenberg profile lists Marwa Maklada as a co-author of The National Skills Interface Program in STEM Subjects summary report and the International Round Table Phase II report. In this book, her areas of interest are STEM competency assessment, validation of competency-based tasks, and preserving teacher authority in AI-rich learning.

Dr. Revital Duek

Role in the Round Table: Discussion moderator and contributor on teamwork, visible thinking, and what must remain fundamentally human in the teacher-student-AI triangle. The Meeting 4 program lists her as a discussion moderator together with Dr. Yael Granot-Bein, and the two moderated the Meeting 6 discussion as well.

Bio: Dr. Duek took part as a policy advisor on STEM.

Main achievements, publications, and interests: In this book, her contribution centers on teamwork as a foundational competency, making students' thinking visible when AI is involved, and clarifying the non-delegable human role in learning. Her opening question at Meeting 6, whether learning environments would look different if education were designed around skills and competencies rather than knowledge, frames Chapter 17. Her opening question at Meeting 6, whether learning environments would look different if education were designed around skills and competencies rather than knowledge, frames Chapter 17.

Dr. Yael Granot-Bein

Role in the Round Table: Discussion moderator and contributor on competency development, repeated practice, reflection, and faculty development. The Meeting 4 program lists her as a discussion moderator, the Meeting 5 steering-team materials report her presentation on academic faculty development, and she co-moderated the Meeting 6 discussion.

Bio: Dr. Granot-Bein works in higher-education faculty development at the Edmond de Rothschild Bridge for Higher Education and Employment.

Main achievements, publications, and interests: Her Round Table contribution emphasizes that competencies are internalized when linked to a discipline, reinforced over time, and supported by reflection. In the book, she anchors the professional-development question for academic lecturers. At Meeting 6 she asked which educational paradigm most needs breaking, and raised the question of how teachers and professors should redefine their added value when students rely on recorded lectures and AI instead of attending.

Dr. Noa Ragonis

Role in the Round Table: Participant and contributor on teacher-development gaps and on the balance between physical and virtual laboratories. This book uses her point that teachers need preparation to lead students through complexity, and her Meeting 6 conclusion, drawn from the pandemic experience, that virtual and physical laboratories must be combined rather than substituted.

Bio: Dr. Ragonis took part from Beit Berl Academic College.

Main achievements, publications, and interests: In the book, her areas of contribution are teacher preparation for AI-era STEM complexity, the development gap that appears when schools expect transformation without preparing educators, and the continuing educational value of physical experimentation, prototyping, and engineering design alongside digital tools. She pointed to at-home making kits and to teacher training with 3D printers, laser cutters, and Arduino as evidence that hands-on learning has not been superseded.

Prof. Sigal Tifferet

Role in the Round Table: Critical discussant on evidence and pedagogy. This book uses her questions about whether embedding competencies inside disciplinary courses is evidence-based and whether transdisciplinary instruction or problem-based learning can weaken explicit instruction.

Bio: Prof. Tifferet took part from Ruppin Academic Center.

Main achievements, publications, and interests: Her Round Table contribution is methodological and evaluative: do not assume a fashionable pedagogy works merely because it sounds progressive. In the book, she strengthens the demand for evidence, explicit instruction, and careful design.

Dr. Gabi Shafat

Role in the Round Table: Higher-education speaker in Meeting 2. His Afeka presentation is one of the main sources for the book's treatment of engineering education, graduate profiles, the future lecturer, AI use, project-based learning, continuous feedback, and team models.

Bio: Dr. Shafat represented Afeka Academic College of Engineering and presented The Challenge in Engineering Higher Education.

Main achievements, publications, and interests: His presentation describes Afeka's BSc and MSc engineering programs, the graduate profile, the Afeka learning experience, adaptive and dynamic course design, pre-assessment, flipped classroom practice, personalized learning, students teaching, AI as a learning-support tool, project/problem-based learning, continuous improvement, and fair assessment of individual contributions in team projects. At Meeting 6 he described roughly twenty-four maker clubs at Afeka, applied projects including a fuel-free electric vehicle designed to travel from Dan to Eilat and an autonomous marine vessel entered in an international competition, and the institutional funding, space, equipment, and faculty mentorship that make them work. At

Meeting 6 he described roughly twenty-four maker clubs at Afeka, applied projects including a fuel-free electric vehicle designed to travel from Dan to Eilat and an autonomous marine vessel entered in an international competition, and the institutional funding, space, equipment, and faculty mentorship that make them work.

Dr. Tzur Karelitz

Role in the Round Table: Assessment and trust discussant. This book uses his Meeting 4 question about why the teacher should mediate between AI and students in evaluation.

Bio: Dr. Karelitz took part from NITE, the National Institute for Testing and Evaluation.

Main achievements, publications, and interests: In the book, his contribution sharpens the trust issue: AI may propose evidence, but student context, bias, relationships, and interpretation still require human judgment.

Dr. Einat Sprinzak

Role in the Round Table: Contributor on educator professional learning. This book uses her point that teachers must experience new learning processes themselves before they can mediate them for students.

Bio: Dr. Sprinzak took part from the Davidson Institute at the Weizmann Institute of Science.

Main achievements, publications, and interests: Her Round Table contribution concerns teacher development, experiential professional learning, and the need for educators to internalize AI-era pedagogy by experiencing it as learners. At Meeting 6 she argued for the educational value of simulations and technology-enhanced environments while insisting that technology complement rather than replace hands-on learning and active participation. At Meeting 6 she argued for the educational value of simulations and technology-enhanced environments while insisting that technology complement rather than replace hands-on learning and active participation.

Prof. Russell Tytler

Role in the Round Table: International expert contributor on science education and teacher professional development. The Meeting 5 steering-team materials report his presentation on teacher professional development, and this book uses his warning that AI can reduce deep, sustained thinking if students surrender control of reasoning.

Bio: Prof. Tytler is a science education researcher at Deakin University, Australia. No detailed public biography is claimed here beyond his Round Table contribution.

Main achievements, publications, and interests: His areas of interest are science education, teacher professional development, deep thinking, AI literacy, critical reasoning, and the responsibility of teachers to cultivate long-term intellectual habits.

Emlen Metz

Role in the Round Table: Meeting 6 speaker and international perspective contributor. Her presentation, Balancing In-Person vs. Computer-Based and AI Experiences in Education, is a main source for the book's treatment of productive AI use, AI-robust tasks, and deliberate no-AI practice. This book also uses her warnings about thoughtful and critical AI use, independent thinking, decision-making, responsibility, and the learning environment.

Bio: S. Emlen Metz took part from the OECD and the University of California, Berkeley.

Main achievements, publications, and interests: Her presentation sets out four productive uses of AI, namely dialogic learning, translation into level and language, active evaluation of claims and sources, and thinking first before consulting the tool. It also proposes a category of AI-robust tasks that keep their value when a language model is available, including laboratory work, model building, card sorts, handwriting, and structured group discussion. It argues that difficulty and effort are where learning happens, that students should not outsource a skill they have not mastered, and that systems must protect offline practice against what she calls cognitive surrender.

Jan Morrison

Role in the Round Table: Meeting 2 education-system speaker. The TIES presentation is a major source for the book's chapters on team-based teaching, transdisciplinary learning, externships, diverse educator pathways, career-connected learning, and STEM ecosystems.

Bio: The TIES About page identifies Jan Morrison as Founder of TIES and describes the organization as emerging from the vision of a 35-year veteran teacher and principal. The TIES Our Work page describes TIES as working across STEM ecosystems, emerging technologies, digital fabrication, school design, workforce development, government engagement, and STEM investment strategy.

Main achievements, publications, and interests: The public TIES pages describe a 22-year organizational history, a mission to advance education and employment outcomes, STEM ecosystem work across almost 120 national and global communities, and more than 40 million learners connected to ecosystem spaces. In the Round Table, Morrison's main contribution is the argument that STEM teaching must become team-based, cross-sector, career-connected, and designed as a system rather than left to isolated teachers. At Meeting 6 she pressed the question of how teacher education institutions should prepare future teachers to work in makerspaces, and proposed that the Round Table itself connect with other international round tables and professional networks to exchange expertise and avoid duplicated effort. At Meeting 6 she pressed the question of how teacher education institutions should prepare future teachers to work in makerspaces, and proposed that the Round Table itself connect with other international round tables and professional networks to exchange expertise and avoid duplicated effort.

Prof. Gil Noam

Role in the Round Table: Contributor on the distinction between school teachers and university lecturers, and on risks of AI increasing isolation and quick-answer behavior.

Bio: The PEAR profile identifies Gil Noam, Ed.D., Dr. Habil, as an associate professor at Harvard Medical School focusing on prevention and resilience, leader of the Institute for the Study of Resilience in Youth at McLean Hospital, former director of the Risk and Prevention Program, and founder of the RALLY Prevention Program.

Main achievements, publications, and interests: The PEAR profile states that he has published over 200 papers, articles, and books in child and adolescent development, risk and resiliency in clinical, school, and afterschool settings, and served as editor-in-chief of New Directions for Youth Development. In the book, his contribution keeps the educator-profile discussion from flattening school and university contexts and highlights the social and relational risks of AI-rich learning. At Meeting 6 he reframed the debate away from AI versus hands-on learning and toward AI as a collaborative learning partner: systems designed to ask questions rather than give answers, students collaborating with one another while using AI, and teachers strengthened in the role of facilitating collective reflection. At Meeting 6 he reframed the debate away from AI versus hands-on learning and toward AI as a collaborative learning partner: systems designed to ask questions rather than give answers, students collaborating with one another while using AI, and teachers strengthened in the role of facilitating collective reflection.

Danielle Eisenberg

Role in the Round Table: Meeting 4 speaker on AI, assessment, and system design. Her Default or Design presentation is the main source for the book's discussion of role-fluid AI, evidence design, the learning triangle, and the measurement triangle.

Bio: Danielle Eisenberg is Founder of Ignite/Ed Consulting and presented the Ignite Ed material Default or Design. Her presentation also identifies her as Executive Director, Product Strategy: Skills for the Future, at ETS. She is not related to Dr. Eli Eisenberg.

Main achievements, publications, and interests: Her Round Table contribution argues that AI is already acting as content, tutor, evaluator, and collaborator; that AI may propose evidence but teachers decide; that assessment must see the learning process and not only the products; that patterns across artifacts are stronger than one polished submission; and that equity and bias risks require human oversight. At Meeting 6 she named the constraint the book cannot dismiss: the obstacle to educational innovation is not a shortage of successful programs but the inability to scale them, because assessment systems, university admissions, and workforce expectations still reward traditional academic achievement.

Dr. Iris Pinto

Role in the Round Table: Meeting 4 speaker on futures and trends. Her Future Trends Shaping Reality presentation supports the book's treatment of VUCA, futures thinking, STEEP analysis, convergence, human-machine interfaces, multimodal and phygital AI, post-work questions, and the education-industry gap.

Bio: Dr. Pinto works at the Institute for Futures Research in Education at the Ministry of Education and gave the Meeting 4 future-trends presentation.

Main achievements, publications, and interests: Her presentation contributes a futures-literacy lens: education should prepare learners not only for today's tools but for unstable, converging, AI-mediated realities.

Dr. Olena Bekh

Role in the Round Table: Contributor on critical thinking and the human risk of over-dependence on AI. This book uses her argument that critical thinking remains an essential human competency and that excessive AI dependence may reduce curiosity, creativity, empathy, and emotional engagement, together with her Meeting 6 question about how education systems of that size and inertia can equip teachers with the competencies AI integration demands.

Bio: Dr. Bekh took part from the European Training Foundation.

Main achievements, publications, and interests: Her Round Table contribution centers on critical thinking, curiosity, creativity, empathy, emotional engagement, the human consequences of relying too heavily on AI, and the system-scale problem of building teacher capacity quickly enough to matter.

Dr. Rinat Itzhaki

Role in the Round Table: Teacher-education contributor. This book uses her Meeting 6 argument that teacher education should be the target audience for work on learning environments, and that the intended model and its objectives must be defined before implementation so that everyone involved shares the same understanding of the goal.

Bio: Dr. Itzhaki took part from the Henrietta Szold Institute.

Main achievements, publications, and interests: Her Round Table contribution concerns teacher education, clarity of purpose in educational design, and the gap between adopting a practice and agreeing on what it is for.

Dr. Yair Noam

Role in the Round Table: Contributor from the defense sector. Lt. Col. (res.) Dr. Noam took part from the Behavioral Sciences Department of the Israel Defense Forces and joined the Phase III sessions from the first meeting onward.

Bio: Dr. Noam represents the Behavioral Sciences Center of the IDF.

Main achievements, publications, and interests: His stated interest in the Round Table was the retention of professional personnel and the strengthening of competencies among commanders and leaders, and the translation of wider educational thinking into structured development processes inside a large institution that is not a school.

Oren Baratz

Role in the Round Table: Contributor on ethics and values in STEM education, taking part from the first meeting.

Bio: Oren Baratz is a former Senior Deputy at the Jewish Federation of Cleveland and led the Federation's STEM program in Israel.

Main achievements, publications, and interests: His stated interest is how ethics and values can be integrated into STEM education so that technological advancement rests on moral and social foundations rather than capability alone.

Additional Participants

The following people took part in Phase III sessions. This book does not attribute a specific quoted position to them, and their inclusion here records attendance rather than authorship of any argument above.

- Inna Zertser, Samuel Neaman Institute. An organizational consultant specializing in skills development and human capital, and a member of the Phase III steering team.
- Golan Tamir, Samuel Neaman Institute.
- Orly Rauch, Social Finance Israel. Her work concerns the measurement of skills development and the generation of actionable insight for STEM initiatives run with several municipalities in partnership with the Cleveland Federation.
- Ruchie Avital, English-Hebrew interpreter for the sessions.

APPENDIX B

Meeting Timeline

- Meeting 1: 16.12.2025, introduction and AI-era skills. A full meeting summary and a published recording exist for this meeting. The invitation circulated in November 2025 announced 02.12.2025; the meeting summary is headed 16.12.2025, and this book follows the summary.
- Meeting 2: 20.01.2026, uniform or diverse teacher and lecturer profiles. A full meeting summary, both presentation decks, and a published recording exist for this meeting.
- Meeting 3: 10.03.2026, possessing versus cultivating competencies. No meeting summary was found in the collected materials. The record for this session is the discussion document circulated on 22.02.2026, before the meeting.
- Meeting 4: 28.04.2026, teacher-student-AI triangle. A full meeting summary, presentation files, and a published recording exist for this meeting.
- Meeting 5: 09.06.2026, professional development responsibility. No meeting summary and no presentation files were found. The steering summaries written before and after the session are the only record.
- Meeting 6: 21.07.2026, learning environments, industry, XR/VR, mentors. A full meeting summary, both presentation decks, and a published recording exist for this meeting. The summary was still circulating as a draft for participant comment when this edition was prepared, so its attributions are reported rather than finally approved.

- Steering-committee discussions: 19.10.2025, 16.12.2025, 03.02.2026, 24.03.2026, 12.05.2026, 23.06.2026, and 04.08.2026. Where no full meeting summary or presentation file was available, these steering summaries are the primary record for the synthesis.

APPENDIX C

Presentation Inventory

Presentation files used in Phase III:

- Presentation 2nd meeting, 20.01.2026.
- TIES Jan Morrison International Round Table Presentation, Future of Teaching, 20.01.2026.
- Default or Design, Ignite Ed STEM Roundtable Presentation, 04.2026.
- Presentation 4th meeting, 28.04.2026.
- Presentation 6th meeting, 21.07.2026.
- Balancing In-Person vs. Computer-Based and AI Experiences in Education, Emlen Metz, 21.07.2026.
- Training by Experience, Avi Salmon, 21.07.2026.

Additional presentation or background materials:

- STEM, Dr. Gabi Shafat.
- Future Trends Shaping Reality, Dr. Iris Pinto.
- The Teacher in the AI Era discussion document.
- Meeting 4 discussion document.
- Meeting 1 draft summary, 16.12.2025.
- Meeting 2 draft summary, 20.01.2026.
- Meeting 4 draft summary, 28.04.2026.
- Meeting 6 draft summary, 21.07.2026.

- International Round Table for Advancing Skills in STEM Education, Phase I report.
- International Round Table for Advancing Skills in STEM Education, Phase II report, March 2025.
- Steering-team summaries, 19.10.2025 through 04.08.2026, held as internal Round Table documents.

APPENDIX D

External Sources and Frameworks

External references appearing in the collected materials include:

- OECD Teaching Compass: Reimagining Teachers as Agents of Curriculum Change.
- OECD Learning Compass 2030.
- TALIS 2024 material.
- UNESCO AI Competency Framework for Teachers.
- UNESCO AI competency framework overview.
- AI Literacy Framework.
- What Should Teachers Teach in an AI Future, listed in the Meeting 3 discussion document.
- Teachers' Professional Learning for the AI Era, listed in the Meeting 3 discussion document.
- ISTE Standards for Educators and ISTE Standards for Students.
- Collins, Brown, and Holum on cognitive apprenticeship, listed in the Meeting 3 discussion document.
- Bandura on self-efficacy, listed in the Meeting 3 discussion document.
- Perkins and Salomon on transfer of learning, listed in the Meeting 3 discussion document.
- Rest and Narvaez on moral development in the professions, listed in the Meeting 3 discussion document.

- Abrami and colleagues on critical thinking instruction, listed in the Meeting 3 discussion document.

Closing Note

The Round Table did not settle the question it opened. It sharpened it.

What the six meetings established is an architecture rather than an answer: AI-era STEM education requires a renewed educator profile, a stronger professional support system, and a practical compass that keeps human judgment at the center. What they did not establish is how any education system pays for it, staffs it, or sustains it past the enthusiasm of the people who started it.

That is the work of the next phase, and of the readers of this book. If a single unit is redesigned so that a student has to think before the tool answers, the discussion recorded here will have done more than most discussions do.